Best-Arm Identification under One-Bit Threshold Feedback

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Abstract

We study fixed-confidence best-arm identification when each reward is observed only through a one-bit threshold comparison. We consider two observation models. In Model E, rewards belong to a regular one-parameter exponential family and the learner adaptively chooses the threshold. In Model U, each arm has an arbitrary distribution supported on $[0, B]$, while an independent uniform threshold is revealed together with the comparison bit.

For both models, we derive instance-dependent information-theoretic lower bounds and construct fixed-β Top-Two procedures whose leading expected sample complexity matches these bounds. Here, fixed-β means that asymptotically a prescribed fraction $\beta \in (0, 1)$ of samples is allocated to the best arm, while the remaining samples are allocated to challengers. For Model E, the optimal information allocation is a single feasible arm–threshold design shared across all challengers. For Model U, the identity $Y \sim \mathrm{Ber}(\mu/B)$ provides a distribution-free statistic for identifying the leader, while the revealed thresholds retain additional information that is used for challenger selection and stopping.

Beyond first-order asymptotic optimality, we establish finite-confidence guarantees. A separate Model-E companion procedure admits an expected stopping-time bound with leading term proportional to $\log(1/\delta)$ and an explicit $O\Big(\log^{2/3}(1/\delta)\sqrt{\log(e + \log(1/\delta))}\Big)$ correction, up to lower-order terms. For the exact certified Model-U rule, we obtain a leading $\log(1/\delta)$ term with an explicit $O\Big(\log^{3/4}(1/\delta)\log^{3/2}(e + \log(1/\delta))\Big)$ correction for arbitrary $[0, B]$-supported laws with interior means, without full-support or global-slope assumptions. Experiments compare one-bit and full-feedback sample complexities and evaluate the finite-confidence behavior of the proposed procedures.

1 INTRODUCTION

Best-arm identification (BAI) asks a learner to identify the arm with the largest mean while controlling the probability of a wrong recommendation. In the fixed-confidence setting, the learner is given a risk level $\delta \in (0, 1)$ and chooses a data-dependent stopping time τ_δ such that the recommended arm is incorrect with probability at most δ. Under full feedback, the optimal first-order sample complexity is characterized by change-of-distribution information, and asymptotically optimal procedures are known for broad classes of bandit models [1, 2].

We study a more restrictive feedback model. After sampling an arm with reward X, the learner does not observe X itself, but only the comparison bit

$$Y = \mathbf{1}\{X \geq q\},$$

where q is a threshold. The choice, or generation, of q is therefore part of the statistical experiment.

We consider two models. In *Model E*, each arm belongs to a regular one-parameter exponential family and the learner adaptively chooses the threshold q. In *Model U*, an arm may have any probability law supported on $[0, B]$; after the arm is selected, an independent threshold $Q \sim \mathrm{Unif}[0, B]$ is generated and both Q and the bit $Y = \mathbf{1}\{X \geq Q\}$ are revealed. Model U therefore provides a *current-status observation*: rather than observing the value of X, we learn only whether X lies above or below a revealed inspection threshold.

A key identity in Model U is

$$\Pr(Y = 1) = \frac{1}{B}\int_0^B \Pr(X \geq q)\, dq = \frac{\mathbb{E}[X]}{B}.$$

Hence marginally $Y \sim \mathrm{Ber}(\mu/B)$, so the bits alone provide a distribution-free estimator of the arm mean. Conditioning on the revealed threshold Q, however, retains more information about the survival function $q \mapsto \Pr(X \geq q)$. Our algorithms use these two views for different purposes: the marginal bits identify the empirical leader, whereas the richer threshold-conditioned likelihood is used to compare the leader with its challengers and to construct stopping evidence.

When $K > 2$, the pairwise comparisons are statistically coupled: all challengers reuse the same observations collected from the best arm. Thus, optimizing a different best-arm threshold distribution separately for each challenger can produce mutually incompatible pairwise designs. For Model E we therefore optimize over a single feasible arm–threshold occupation measure shared by all challengers. Its optimal worst-case information rate is the *characteristic value*; its reciprocal is the corresponding *characteristic time*, which determines the leading asymptotic number of samples.

Our algorithms follow the Top-Two principle: at each round, one arm acts as a current leader and another as its most informative challenger, and sampling is split between them. In the fixed-β version, a prescribed fraction $\beta \in (0, 1)$ is asymptotically assigned to the best arm and the remaining fraction $1 - \beta$ to challengers.

Contributions. *Matching instance-dependent lower and upper bounds.* For both observation models we derive K-arm change-of-distribution lower bounds and construct fixed-β Top-Two procedures whose leading expected stopping times match the corresponding characteristic times. This yields an explicit measure of the information loss caused by one-bit feedback through the ratio

$$\frac{T^\star_{\beta,\mathrm{1b}}}{T^\star_{\beta,\mathrm{full}}}.$$

For Model E, the optimization uses one feasible threshold design shared by all challengers. For Model U, we obtain an exact reduction of the population pairwise transportation problem to a one-dimensional mean-separation problem.

Finite-confidence guarantees. We provide δ-PAC stopping rules and non-asymptotic probability and expected-time bounds. A separate Model-E global-design companion admits an explicit correction of order

$$L_\delta^{2/3}\sqrt{\log(e+L_\delta)}, \qquad L_\delta := \log(1/\delta),$$

up to lower-order terms. For the exact certified Model-U procedure, the corresponding correction is

$$L_\delta^{3/4}\log^{3/2}(e+L_\delta)$$

for arbitrary probability laws supported on $[0, B]$ with interior means, without density, full-support, or global-slope assumptions.

Empirical comparison. Paired simulations compare full and one-bit feedback and examine practical fixed-point predictors. The practical simulations are separated from the paper-certified Model-U finite-sample calibration, whose sieve penalty is deliberately conservative at moderate sample sizes.

Related work. Our analysis builds on BAI lower bounds and Top-Two methods [1, 3, 2, 4]. One-bit mean and quantile estimation, bit-constrained BAI, and classical current-status likelihoods provide related tools but address different statistical experiments [5, 6, 7, 8, 9, 11, 12].

Concurrent one-bit BAI. Concurrent work by Luong et al. [10] studies fixed-confidence BAI under adaptive one-bit set queries with only a finite-variance assumption. It develops a time-uniform one-bit mean-estimation primitive and clipping-based BAI algorithms, together with a gap-adaptive upper bound and a two-arm worst-case lower bound matching the leading gap dependence up to lower-order iterated logarithms. Its lower bound is a worst-case two-arm construction rather than an exact K-arm instance-dependent characteristic; the paper explicitly leaves the exact instance-dependent complexity of one-bit BAI open. Our setting and objective are complementary: under Model E and Model U we characterize the exact fixed-β first-order instance-dependent complexity through observed-law KL information and construct procedures attaining the corresponding characteristic times.

2 PROBLEM FORMULATION

Fixed-confidence objective. There are $K \geq 2$ arms and the true instance has a unique best arm $i^\star$. At each round the learner chooses an arm, receives a one-bit observation according to one of the models below, and may stop at a data-dependent time τ_δ. We call an algorithm δ-PAC if, for every admissible instance,

$$\mathbb{P}(\tau_\delta < \infty) = 1, \qquad \mathbb{P}(\widehat{i}_{\tau_\delta} \neq i^\star) \leq \delta. \tag{1}$$

When only the second inequality is used for an arbitrary sampling rule, we call it an incorrect-stop probability bound. Write

$$L_\delta := \log(1/\delta).$$

Divergence notation. For $p, r \in [0, 1]$, let

$$\mathrm{kl}(p, r) := p\log\frac{p}{r} + (1-p)\log\frac{1-p}{1-r}$$

denote the binary Kullback–Leibler divergence, with the usual extended-value conventions at 0 and 1. For general probability laws P and Q, $\mathrm{KL}(P, Q)$ denotes their Kullback–Leibler divergence. If P and Q admit densities p and q with respect to a common dominating measure ν, then

$$\mathrm{KL}(P, Q) = \int p(x)\log\frac{p(x)}{q(x)}\,d\nu(x),$$

with $\mathrm{KL}(P, Q) = +\infty$ when $P \not\ll Q$. Thus kl is reserved for Bernoulli parameters, whereas KL denotes divergence between general probability laws.

Our first-order complexity is the ratio $\mathbb{E}[\tau_\delta]/L_\delta$. We study its limit as $\delta \downarrow 0$. Suboptimal arms are allowed to have equal means throughout.

2.1 Model E: adaptive thresholds in an exponential family

Arm k has law P_{θ_k} from the canonical one-parameter exponential family

$$f_\theta(x) = \exp\{\theta x - b(\theta)\}h(x), \qquad \theta \in \Theta, \tag{2}$$

where $\Theta = [\theta_-, \theta_+]$ is compact, $b \in C^2(\Theta)$ and $b'' > 0$. Hence the mean $b'(\theta)$ is strictly increasing and $i^\star = \arg\max_k \theta_k$. The learner chooses a threshold q_t in a compact set Q and observes only

$$Y_t = \mathbf{1}\{X_t \geq q_t\}, \qquad X_t \sim P_{\theta_{I_t}}.$$

Write $p_q(u) = P_u(X \geq q)$ and

$$d_q(u, v) = \mathrm{kl}(p_q(u), p_q(v)) \,. \tag{3}$$

Thus $d_q(u, v)$ is the KL divergence between the two Bernoulli observations induced at threshold q under candidate parameters u and v. We assume $(q, u) \mapsto p_q(u)$ is continuous, continuously differentiable in u, bounded away from 0 and 1 on $Q \times \Theta$, and that the one-bit log likelihood is uniformly Lipschitz in u. These compact regularity assumptions are sufficient for the finite-time likelihood control and uniform information separation used below. The uniform bound applies to the entire *admissible parametric family* and the *selected compact query set*; it is not the mere pointwise condition $0 < p_q(u) < 1$.

For $a \in \Theta$, define the armwise one-bit log likelihood

$$\begin{aligned} \ell_{n,k}(a) &:= \sum_{\substack{t \leq n \\ I_t = k}} \big[Y_t \log p_{q_t}(a) + (1 - Y_t) \log(1 - p_{q_t}(a)) \big], \\ L_n(u) &:= \sum_{k=1}^{K} \ell_{n,k}(u_k). \end{aligned} \tag{4}$$

Because the joint likelihood is separable across arms, define the armwise MLE by the measurable deterministic rule

$$\widehat{\theta}_k(n) := \min\left(\arg\max_{a \in \Theta} \ell_{n,k}(a) \right), \qquad \widehat{\theta}(n) = \big(\widehat{\theta}_1(n), \ldots, \widehat{\theta}_K(n)\big).$$

The minimum selects the smallest maximizer whenever the MLE is not unique.

For an ordered pair (i, j) define

$$\begin{aligned} W_n^{\mathrm{E}}(i, j) = &\ell_{n,i}(\widehat{\theta}_i(n)) + \ell_{n,j}(\widehat{\theta}_j(n)) \\ &- \sup_{u_i \leq u_j} \{\ell_{n,i}(u_i) + \ell_{n,j}(u_j)\}. \end{aligned} \tag{5}$$

Thus $W_n^{\mathrm{E}}(i, j)$ is the generalized likelihood-ratio evidence against the alternative that arm j is at least as good as arm i: it compares the unrestricted pairwise maximum likelihood with the best fit satisfying $u_i \leq u_j$. Large values therefore provide evidence that arm i is better than arm j.

The global statistic for recommending arm i is

$$Z_i(n) := \min_{j \neq i} W_n^{\mathrm{E}}(i, j).$$

The minimum appears because arm i can be recommended only after it has accumulated evidence against every challenger; hence $Z_i(n)$ is controlled by the currently least-separated competitor.

2.2 Model U: revealed uniform thresholds

Arm k has an otherwise arbitrary law F_k supported on $[0, B]$, with mean $\mu_k \in (0, B)$. After the learner chooses arm I_t, nature draws $Q_t \sim \mathrm{Unif}[0, B]$ independently of $X_t \sim F_{I_t}$ and reveals

$$(Q_t, Y_t), \qquad Y_t = \mathbf{1}\{X_t \geq Q_t\}.$$

If $S_F(q) = P_F(X \geq q)$, then

$$Y \sim \mathrm{Ber}(\mu(F)/B), \qquad Y \mid Q = q \sim \mathrm{Ber}(S_F(q)). \tag{6}$$

Thus the bit marginal is exactly $\mathrm{Ber}(\mu(F)/B)$. Consequently, B times the empirical bit mean is a distribution-free consistent estimator of the arm mean; at each fixed local pull count it is also unbiased. The full observed pair has KL divergence

$$D_{\mathrm{U}}(F, G) = \frac{1}{B} \int_0^B \mathrm{kl}(S_F(q), S_G(q)) \, dq. \tag{7}$$

The first identity in (6) will define the leader; the richer current-status likelihood associated with (7) will define the challenger and the stopping rule.

Regularity differs between the models. Model E assumes that its finite-dimensional Bernoulli probabilities are uniformly interior on the compact parameter–query domain. Model U retains all $[0, B]$-supported arm laws and optimizes over unrestricted candidate survival functions, which can equal zero or one. Its profile analysis therefore uses a *common, horizon-dependent smoothing map* in the proof, not an extra restriction on the statistical experiment; see Lemma C.15 and the accompanying appendix remark.

3 A K-ARM LOWER BOUND

We state the change-of-measure argument once, for a generic one-bit experiment. When arm k is sampled with action a (a threshold in Model E; only the arm label in Model U), let $P_{\nu_k}^a$ denote the law of the resulting observed one-bit datum. For two candidate arm models ν_k and λ_k, define

$$d_a(\nu_k, \lambda_k) := \mathrm{KL}(P_{\nu_k}^a, P_{\lambda_k}^a),$$

the expected log-likelihood ratio contributed by one observation from arm k under action a. For a probability design ρ over all arm-action pairs and an alternative instance λ, set

$$\mathcal{I}(\nu, \lambda; \rho) = \sum_{k=1}^{K} \int d_a(\nu_k, \lambda_k) \, \rho_k(da). \tag{8}$$

Thus $\mathcal{I}(\nu, \lambda; \rho)$ is the average information gained per sample for discriminating the true instance ν from λ under design ρ. Let $\mathcal{A}(\nu)$ be the set of *strict* alternatives having a unique best arm different from $i^\star$. The lower bound below is stated for this strict set. For the fixed-β designs with positive best-arm mass used to identify the matching characteristics, the supplement separately verifies the strict-alternative/closed-order-reversal reduction. The generic lower bound itself is stated directly for strict alternatives.

Theorem 1 (K-arm action-space lower bound)**.** *For every $\delta \in (0, 1/2)$ and every δ-PAC algorithm with $\mathbb{E}_\nu[\tau_\delta] < \infty$,*

$$\mathbb{E}_\nu[\tau_\delta]\,\Gamma^\star(\nu) \geq \mathrm{kl}(1-\delta, \delta), \qquad \Gamma^\star(\nu) = \sup_{\rho} \inf_{\lambda \in \mathcal{A}(\nu)} \mathcal{I}(\nu, \lambda; \rho). \tag{9}$$

The ratio form follows when $\Gamma^\star(\nu) > 0$; if it is zero, no δ-PAC algorithm has finite expected stopping time on ν.

The quantity $\Gamma^\star(\nu)$ is the largest worst-case information rate achievable by a feasible design, and $1/\Gamma^\star(\nu)$ is the corresponding first-order characteristic time.

The proof takes the expected stopped occupation measure of the actual adaptive experiment and applies the bandit change-of-distribution inequality to each strict alternative. The important point for $K > 2$ is structural: there is a single measure ρ before nature chooses the challenger. Hence the same best-arm threshold stream must serve all pairwise tests.

Most Top-Two algorithms fix the asymptotic fraction $\beta \in (0, 1)$ of best-arm samples. Restricting the design in (9) to mass β on the best arm gives $\Gamma^\star_\beta(\nu)$.

Corollary 2 (Fixed-β lower bound)**.** *Under the compactness and finite-KL continuity conditions verified for Models E and U in the supplement, suppose a family of δ-PAC algorithms satisfies*

$$\frac{\mathbb{E}_\nu[N_{\tau_\delta, i^\star}]}{\mathbb{E}_\nu[\tau_\delta]} \longrightarrow \beta.$$

Then

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu[\tau_\delta]}{L_\delta} \geq \frac{1}{\Gamma^\star_\beta(\nu)}. \tag{10}$$

with the convention $1/0 = +\infty$.

3.1 The two characteristic values

For Model E, let $\mathcal{D}^b_\beta$ collect all nonnegative threshold measures $\rho = (\rho_1, \ldots, \rho_K)$ with $\sum_k \rho_k(Q) = 1$ and $\rho_b(Q) = \beta$. The pairwise scalar reduction proved in the supplement yields

$$\Gamma^{\mathrm{E}}_\beta(\theta) = \max_{\rho \in \mathcal{D}^{i^\star}_\beta} \min_{\substack{j \neq i^\star \\ x \in \Theta}} \Big[\int_Q d_q(\theta_{i^\star}, x) \rho_{i^\star}(dq) + \int_Q d_q(\theta_j, x) \rho_j(dq) \Big]. \tag{11}$$

The best-arm measure $\rho_{i^\star}$ is visibly shared across all challengers.

For Model U, define the one-sided observed transportation costs

$$K^-_{\mathrm{U}}(F, x) = \inf_{G: \mu(G) \leq x} D_{\mathrm{U}}(F, G), \qquad K^+_{\mathrm{U}}(F, x) = \inf_{G: \mu(G) \geq x} D_{\mathrm{U}}(F, G).$$

Let $\mathcal{W}_\beta := \{w_j \geq 0 : \sum_{j \neq i^\star} w_j = 1 - \beta\}$. Then

$$\Gamma^{\mathrm{U}}_\beta(\mathbf{F}) = \max_{w \in \mathcal{W}_\beta} \min_{j \neq i^\star} \inf_{x \in [0, B]} \big[\beta K^-_{\mathrm{U}}(F_{i^\star}, x) + w_j K^+_{\mathrm{U}}(F_j, x)\big]. \tag{12}$$

In (12), $0 \cdot (+\infty)$ is interpreted as zero when a challenger receives zero mass. Corollary 2 specializes exactly to $1/\Gamma^{\mathrm{E}}_\beta$ and $1/\Gamma^{\mathrm{U}}_\beta$.

The Model-E best-arm query measure must be common to all challengers. The theoretical comparison with the best *common fixed threshold* and with the independently optimized pairwise relaxation is simply

$$\Gamma^{\mathrm{fixed}}_\beta \leq \Gamma^{\mathrm{E}}_\beta(\theta) \leq \Gamma^{\mathrm{pair}}_\beta(\theta),$$

These comparators are defined in the supplement; neither inequality is asserted to be strict. No fixed-threshold experiment is included.

4 ALGORITHMS

4.1 A common Top-Two template

Both algorithms use the same arm-selection architecture. They maintain a leader, enforce sublinear arm exploration, choose the least-informed challenger on regular rounds, sample the leader/challenger with probabilities β and $1 - \beta$, and stop when all pairwise GLRs cross an anytime-valid boundary. Model E additionally designs thresholds through a shared oracle; Model U uses Uniform-Observed thresholds and a current-status profile likelihood.

4.2 Model E algorithm

Initialize by sampling every arm once, using a threshold drawn from a fixed full-support probability measure on Q. After n observations, the empirical leader is $\widehat{b}_n = \arg\max_k \widehat{\theta}_k(n)$, with deterministic tie-breaking. The action at round $n + 1$ is chosen from the history through time n.

To choose thresholds, we regularize the shared design problem at the current MLE. For a parameter vector v and a proposed leader b, write

$$F^b_v(\rho) := \inf_{u \in \mathrm{Alt}_b} \mathcal{I}(v, u; \rho),$$

where $\mathrm{Alt}_b := \{u \in \Theta^K : u_b \leq \max_{j \neq b} u_j\}$. Here $F^b_v(\rho)$ evaluates a given *shared* design; maximizing it gives the characteristic value. Let $b(v)$ be the tie-broken best arm and

define

$$\begin{aligned} R_\Theta &:= \operatorname{diam}(\Theta) = \theta_+ - \theta_-, \\ D_{\mathrm{E}}(r) &:= \min_{\substack{q\in Q,\ u,v\in\Theta \\ |u-v|\geq r}} d_q(u,v), \\ r_n &:= \sup\Big\{ r \in [0, R_\Theta] : \\ &\qquad D_{\mathrm{E}}(r) \leq 2n^{-1/8} \Big\}. \end{aligned} \tag{13}$$

Set $\varepsilon_n := \sqrt{r_n + n^{-1}}$ and $b_n = b(\widehat{\theta}(n))$. The supplement establishes eventual MLE localization within r_n, with a finite-moment entrance. The oracle is

$$\begin{aligned} \rho_n = \arg\max_{\rho\in\mathcal{D}_\beta^{b_n}} \big\{ & F^{b_n}_{\widehat{\theta}(n)}(\rho) \\ & - \varepsilon_n \mathcal{R}(\rho) \big\}. \end{aligned} \tag{14}$$

The continuous strictly convex kernel energy $\mathcal{R}$ selects the canonical optimum: after leader stabilization the uniform objective error $\delta_n \leq L_d r_n$, so $\varepsilon_n \to 0$ and $\delta_n/\varepsilon_n \to 0$. Details are in the supplement.

Fix $\gamma \in (1/2, 1)$. If some arm satisfies $N_{n,k} < (n+1)^\gamma$, a least-sampled arm is forced. Otherwise the challenger is

$$C_{n+1} \in \arg\min_{j\neq \widehat{b}_n} \Big\{ W_n^{\mathrm{E}}(\widehat{b}_n, j) + \xi_n(j) \Big\}, \tag{15}$$

where $\xi_n(j) = 0$ gives TC and $\xi_n(j) = \log N_{n,j}$ gives TCI. Conditional on the selected arm k, if $m_{n,k} := \rho_{n,k}(Q) > 0$ the threshold is drawn from $\rho_{n,k}/m_{n,k}$; if $m_{n,k} = 0$, it is drawn from the fixed full-support initialization law. The same rule is used on forced-arm rounds. Thus the guard perturbs arm counts by only $o(n)$ and does not inject an artificial fixed-threshold atom into the limiting query distribution.

The stopping rule is

$$\begin{aligned} \tau_\delta^{\mathrm{E}} := \inf\Big\{ n \geq K : \ & \min_{j\neq\widehat{b}_n} W_n^{\mathrm{E}}(\widehat{b}_n, j) \\ & \geq c_{\mathrm{E}}(n,\delta) \Big\}. \end{aligned} \tag{16}$$

The recommendation is $\widehat{i}_{\tau_\delta^{\mathrm{E}}} := \widehat{b}_{\tau_\delta^{\mathrm{E}}}$. The likelihood-covering boundary satisfies

$$c_{\mathrm{E}}(n,\delta) = L_\delta + O(\log n).$$

4.3 Guarded EB–TCI for Model U

Sample each arm once. By (6), the empirical mean estimator

$$\widehat{\mu}_{n,k} = B\, N_{n,k}^{-1} \sum_{\substack{t\leq n \\ I_t = k}} Y_t$$

is strongly consistent under the guarded sampling rule. The leader after n observations is $\widehat{b}_n = \arg\max_k \widehat{\mu}_{n,k}$. For each arm k, let $(Q_{k,r}, Y_{k,r})$ denote the observation obtained on the rth local pull of that arm. For challenging and stopping we use the current-status likelihood

$$\begin{aligned} \ell_{n,k}(G) = \sum_{r=1}^{N_{n,k}} \big[& Y_{k,r} \log S_G(Q_{k,r}) \\ & + (1 - Y_{k,r}) \log(1 - S_G(Q_{k,r})) \big]. \end{aligned}$$

Let $\mathcal{M}_{n,k} = \sup_G \ell_{n,k}(G)$. The raw profile GLR for the order hypothesis $\mu(G_i) \leq \mu(G_j)$ is

$$\begin{aligned} W_n^{\mathrm{U}}(i,j) = & \mathcal{M}_{n,i} + \mathcal{M}_{n,j} \\ & - \sup_{\mu(G_i)\leq\mu(G_j)} \{ \ell_{n,i}(G_i) + \ell_{n,j}(G_j) \}. \end{aligned} \tag{17}$$

The likelihood is optimized over probability laws; implementation uses isotonic/profile fits (supplement). Interiorized sieves occur only in the calibration proof; true survival values may equal 0 or 1.

If $\min_k N_{n,k} < \sqrt{n/K}$, pull a least-sampled arm. Otherwise choose

$$C_{n+1} \in \arg\min_{j\neq\widehat{b}_n} \{ W_n^{\mathrm{U}}(\widehat{b}_n, j) + \log N_{n,j} \}, \tag{18}$$

and perform the fixed-β Top-Two coin. Nature then draws a fresh $Q \sim \mathrm{Unif}[0, B]$ and only (Q, Y) is retained.

The stopping rule uses the *same raw profile GLR*:

$$\begin{aligned} \tau_\delta^{\mathrm{U}} := \inf\Big\{ n \geq K : \ & \min_{j\neq\widehat{b}_n} W_n^{\mathrm{U}}(\widehat{b}_n, j) \\ & \geq c_n + \log \frac{(K-1)\pi^2 n^2}{6\delta} \Big\}. \end{aligned} \tag{19}$$

The recommendation is $\widehat{i}_{\tau_\delta^{\mathrm{U}}} := \widehat{b}_{\tau_\delta^{\mathrm{U}}}$. Here the proof constructs an explicit deterministic profile-complexity penalty $c_n = O(n^{2/3}\log(n+1)) = o(n)$ by approximating the monotone profile class with a finite sieve. The sieve is used only for calibration and is not part of the sampling algorithm.

5 THEORETICAL RESULTS

We now connect the two algorithms to the lower bound. Detailed concentration, compactness, tracking, and allocation arguments are given in the supplement.

5.1 Model E: validity, shared-oracle geometry, and optimality

For every threshold, $u \mapsto p_q(u)$ is strictly increasing because $\partial_u p_q(u) = \mathrm{Cov}_u(X, \mathbf{1}\{X \geq q\}) > 0$; hence forced *arm* exploration suffices for identification. Let R be the parameter interval diameter and G the uniform Lipschitz constant of one log-likelihood contribution.

Theorem 3 (Explicit adaptive GLR calibration)**.** *For*

$$c_{\mathrm{E}}(n,\delta) = 1+\log\frac{\pi^2}{6\delta}+K\log n+K\log(2+GRn), \quad (20)$$

the stopping rule (16) *has incorrect-stop probability at most δ under arbitrary predictable arm and threshold choices. In particular, the sampling theorem does not need to be used to prove validity of the stopping rule.*

The proof combines a $1/(Gn)$ parameter net with likelihood-ratio martingales and a time-uniform error budget.

Shared-oracle geometry. For a fixed threshold design of a best arm i and challenger j, the pair alternative can be reduced from two parameters to one separator. If $\theta_i > \theta_j$, then

$$\inf_{u_i\le u_j}\Big\{\int d_q(\theta_i,u_i)\rho_i(dq)+\int d_q(\theta_j,u_j)\rho_j(dq)\Big\} = \min_x\Big\{\int d_q(\theta_i,x)\rho_i(dq)+\int d_q(\theta_j,x)\rho_j(dq)\Big\}. \quad (21)$$

Monotonicity of $x\mapsto d_q(a,x)$ places the minimizing separator strictly between the true parameters whenever both masses are positive. This yields strict increase in challenger mass, positivity of all challenger masses, and equalization at a shared optimum. The raw optimum need not be unique (e.g. Bernoulli rewards make all $q\in(0,1)$ equivalent), so the vanishing kernel regularizer canonically selects one stable measurable optimizer.

Theorem 4 (Shared-design structure)**.** *Under the compact regularity assumptions and a unique best arm,* (11) *attains a positive maximum. Every optimum gives positive mass to each challenger and equalizes all active pair information at $\Gamma^{\mathrm{E}}_\beta(\theta)$. The optimal set is compact and convex; the strictly convex kernel regularizer selects a unique $\rho^\dagger$, and the empirical oracle* (14) *converges weakly to it when $\varepsilon_n=\sqrt{r_n+n^{-1}}$ and $\delta_n/\varepsilon_n\to 0$.*

Theorem 5 (Consistency, tracking, and finite-time correctness)**.** *For the forced-arm regularized-shared algorithms β-EB-TC and β-EB-TCI, forced rounds are $o(n)$, the MLE is strongly consistent, the leader stabilizes at $i^\star$, and threshold-information averages track the canonical oracle. Arm proportions converge to $w^\dagger_k=\rho^\dagger_k(Q)$ with finite-mean fixed-accuracy entrance times, with $w^\dagger_{i^\star}=\beta$; the stopping rule* (16) *has incorrect-stop probability at most δ.*

Theorem 6 (Matching fixed-β complexity for Model E)**.** *Under the assumptions above, both forced-arm regularized-shared β-EB-TC and β-EB-TCI stop almost surely and satisfy*

$$\mathbb{P}_\theta(\widehat{i}_{\tau^{\mathrm{E}}_\delta}\neq i^\star)\le\delta, \qquad \lim_{\delta\downarrow 0}\frac{\mathbb{E}_\theta[\tau^{\mathrm{E}}_\delta]}{L_\delta}=\frac{1}{\Gamma^{\mathrm{E}}_\beta(\theta)}. \quad (22)$$

Thus the algorithms are asymptotically β-optimal for the K-arm adaptive-threshold experiment.

The upper bound uses oracle convergence, threshold tracking, and oversampled-challenger rejection; Corollary 2 supplies the matching stopped-fraction lower bound.

5.2 Model U: calibration, profile geometry, and optimality

Data processing applied to $(Q,Y)\mapsto Y$ gives

$$\mathrm{kl}\Big(\frac{\mu(F)}{B},\frac{\mu(G)}{B}\Big)\le D_{\mathrm{U}}(F,G), \quad (23)$$

The empirical profile GLR has the same scalar separator. Set $\mathcal{M}_{n,k}:=\sup_G\ell_{n,k}(G)$ and define

$$\mathcal{K}^-_{n,k}(x):=\mathcal{M}_{n,k}-\sup_{\mu(G)\le x}\ell_{n,k}(G),$$
$$\mathcal{K}^+_{n,k}(x):=\mathcal{M}_{n,k}-\sup_{\mu(G)\ge x}\ell_{n,k}(G).$$

Then

$$W^{\mathrm{U}}_n(i,j)=\inf_{x\in[0,B]}\{\mathcal{K}^-_{n,i}(x)+\mathcal{K}^+_{n,j}(x)\}. \quad (24)$$

This identity is exact: the scalar x separates the null constraint $\mu(G_i)\le\mu(G_j)$ into the two one-sided constraints $\mu(G_i)\le x$ and $\mu(G_j)\ge x$. We use an infimum because the representation does not require the optimal separator x to be attained.

The calligraphic $\mathcal{K}^\pm_{n,k}(x)$ are *empirical profile-likelihood losses*; the earlier $K^\pm_{\mathrm{U}}(F_k,x)$ are their *population KL* analogues. Since the unrestricted profile numerator is not itself a likelihood-ratio martingale, a finite monotone sieve supplies the calibration penalty c_n.

Theorem 7 (Pairwise profile calibration)**.** *Under arbitrary predictable arm sampling respecting the arm-before-threshold protocol, if the null order $\mu_i\le\mu_j$ is true, then for every n and $x>0$,*

$$\mathbb{P}\big(W^{\mathrm{U}}_n(i,j)\ge c_n+x\big)\le e^{-x},$$
$$c_n=O(n^{2/3}\log(n+1)).$$

Hence the boundary in (19)*, obtained by summing $6/(\pi^2n^2)$ over time and $\delta/(K-1)$ over wrong recommendations, is anytime valid.*

Critical-level allocation geometry. Pair information is continuous, concave, and nondecreasing in the challenger mass, but may have flat pieces. We assume neither uniqueness nor equalization. For $0<\gamma<\Gamma^{\mathrm{U}}_\beta(\mathbf{F})$ define

$$h_j(w):=\inf_x\{\beta K^-_U(F_{i^\star},x)+wK^+_U(F_j,x)\},$$
$$c_j(\gamma):=\inf\{w\ge 0: h_j(w)\ge\gamma\}.$$

Thus $h_j(w)$ is the population information rate against challenger j when the best arm has mass β and challenger j

has mass w; $c_j(\gamma)$ is the smallest challenger mass attaining level γ. Only $\gamma < \Gamma^{\mathrm{U}}_\beta$ enters the proof, ensuring that every critical mass is below the physical challenger budget $1-\beta$. Strict total slack,

$$\sum_{j \neq i^\star} c_j(\gamma) < 1 - \beta,$$

and uniform profile approximation imply a deterministic deficit barrier: TCI cannot keep serving an already-safe challenger while another remains below its critical mass. Hence every pair eventually exceeds any fixed $\gamma < \Gamma^{\mathrm{U}}_\beta$.

Theorem 8 (Exact scalar current-status transportation)**.** *For $F \in \mathcal{P}([0,B])$ with $0 < \mu(F) < B$, define $T_\lambda(p)$ as the minimizer of* $\mathrm{kl}(p,r) + \lambda r$ *over $r \in [0,1]$, and let $m_F(\lambda) = \int_0^B T_\lambda(S_F(q))\,dq$. For $x \in (0, \mu(F))$, select $\lambda_x \geq 0$ with $m_F(\lambda_x) = x$. Then*

$$K^-_{\mathrm{U}}(F,x) = \frac{1}{B}\int_0^B \mathrm{kl}\big(S_F(q), T_{\lambda_x}(S_F(q))\big)\,dq, \quad (25)$$

with the analogous K^+_{U} formula above $\mu(F)$. For positive masses a, b, the pairwise alternative cost is

$$\inf_{x \in [0,B]} \{aK^-_{\mathrm{U}}(F_i,x) + bK^+_{\mathrm{U}}(F_j,x)\}.$$

The explicit formula for T_λ, endpoint conventions, and pairwise reduction proof appear in the supplement.

Theorem 9 (Profile likelihood, stabilization, and anytime validity)**.** *For guarded β-EB-TCI, the profile GLR has the exact separator identity,* (19) *is δ-valid under arbitrary predictable arm sampling respecting the arm-before-threshold protocol, and the guard uses $O(\sqrt{Kn})$ rounds. The marginal-bit leader stabilizes after a finite-moment entrance, and for every pair*

$$\frac{W^{\mathrm{U}}_n(i,j)}{n} - V^{\mathrm{U}}_{ij}\left(\frac{N_{n,i}}{n}, \frac{N_{n,j}}{n}\right) \to 0$$

with finite-moment fixed-accuracy entrance times.

Theorem 10 (Matching fixed-β complexity for Model U)**.** *For otherwise arbitrary arm laws supported on $[0,B]$ with $0 < \mu_k < B$ for all k and a unique best mean, guarded β-EB-TCI with* (19) *stops almost surely and satisfies*

$$\mathbb{P}_{\mathbf{F}}(\widehat{i}_{\tau^{\mathrm{U}}_\delta} \neq i^\star) \leq \delta, \qquad \lim_{\delta \downarrow 0} \frac{\mathbb{E}_{\mathbf{F}}[\tau^{\mathrm{U}}_\delta]}{L_\delta} = \frac{1}{\Gamma^{\mathrm{U}}_\beta(\mathbf{F})}. \quad (26)$$

Hence the procedure is δ-PAC. The result requires only a unique best arm; suboptimal means may tie.

The upper bound uses the critical-level deficit barrier and $o(n)$ profile calibration; Corollary 2 supplies the lower bound.

5.3 Finite-confidence bounds

Let $L = L_\delta$, $T^\star_{\mathrm{E}} = [\Gamma^{\mathrm{E}}_\beta(\theta)]^{-1}$, and $T^\star_{\mathrm{U}} = [\Gamma^{\mathrm{U}}_\beta(\mathbf{F})]^{-1}$. These stopping-time bounds are separate from the δ-PAC error calibration in Theorems 6 and 10.

Theorem 11 (Model E: explicit companion bound)**.** *Under the Model-E assumptions and the identifying-threshold condition stated in the supplement, the* separate *global-design companion with forced-identification exponent $\gamma = 2/3$ satisfies, for fixed K,*

$$\mathbb{E}_\theta[\tau^{\mathrm{E,ft}}_\delta] \leq T^\star_{\mathrm{E}} L + O\Big(L^{2/3}\sqrt{\log(e+L)} + \sqrt{L\log(e+L)} + \log(e+L)\Big) \quad (27)$$

The supplement gives the corresponding high-probability statement with explicit instance-dependent constants. The $2/3$ correction is not claimed for the main Model-E Top-Two rule or for the full-feedback comparator.

Theorem 12 (Model U: explicit finite-confidence bound)**.** *Under Theorem 10's assumptions, the* exact *guarded raw-profile rule with certified boundary* (19) *satisfies*

$$\mathbb{E}_{\mathbf{F}}[\tau^{\mathrm{U}}_\delta] \leq T^\star_{\mathrm{U}} L + O\Big(1 + L^{3/4}\log^{3/2}(e+L)\Big). \quad (28)$$

A quantitative high-probability form is given in the supplement. This bound holds for arbitrary $[0,B]$-supported arm laws with interior means and a unique best arm, but it does not apply to the practical approximate GLR used in the finite-sample plots.

The different exponents reflect different proof mechanisms: the Model-E companion balances forced identification against global-oracle transfer, whereas Model U balances subcritical allocation slack, raw-profile approximation, and the certified sieve penalty. Neither term is claimed to be an optimal second-order expansion.

5.4 Comparison with full feedback

Both experiments are garblings of full rewards:

$$d_q(u,v) \leq \mathrm{KL}(P_u, P_v), \qquad D_{\mathrm{U}}(F,G) \leq \mathrm{KL}(F,G).$$

Thus the one-bit characteristic rate cannot exceed its full-feedback counterpart at the same β; the loss is instance dependent. For Bernoulli Model E, thresholds $q \in (0,1)$ reveal the reward itself.

6 EXPERIMENTS

Protocol. We use synthetic Gaussian and bounded instances to test three aspects of the theory: the relative information loss caused by one-bit feedback, direct first-order behavior, and finite-confidence calibration of practical stopping rules. Model E uses Gaussian rewards with known $\sigma = 0.2$ and compares full-feedback Top–Two with adaptive

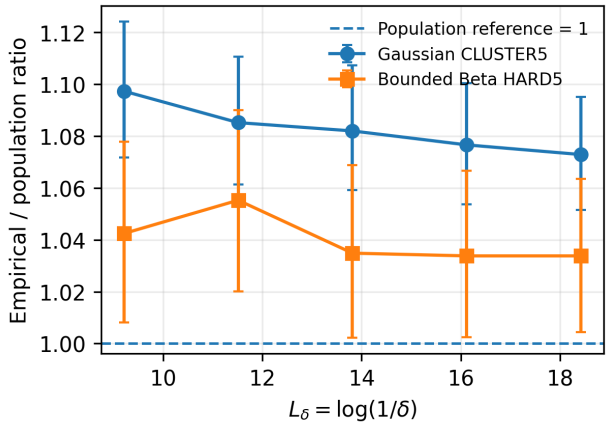


Figure 1: **Relative full-versus-one-bit efficiency.** For Gaussian CLUSTER5 and bounded Beta HARD5 at $\beta = .5$, we plot $R_{\mathrm{rel}}(\delta) = (\overline{\tau}_{1\mathrm{bit}}/\overline{\tau}_{\mathrm{full}})/(T^\star_{1\mathrm{bit}}/T^\star_{\mathrm{full}})$. Intervals are 95% paired-bootstrap confidence intervals over replication IDs. The dashed reference at one represents agreement with the *population relative* characteristic-time prediction; it is not a finite-δ bound.

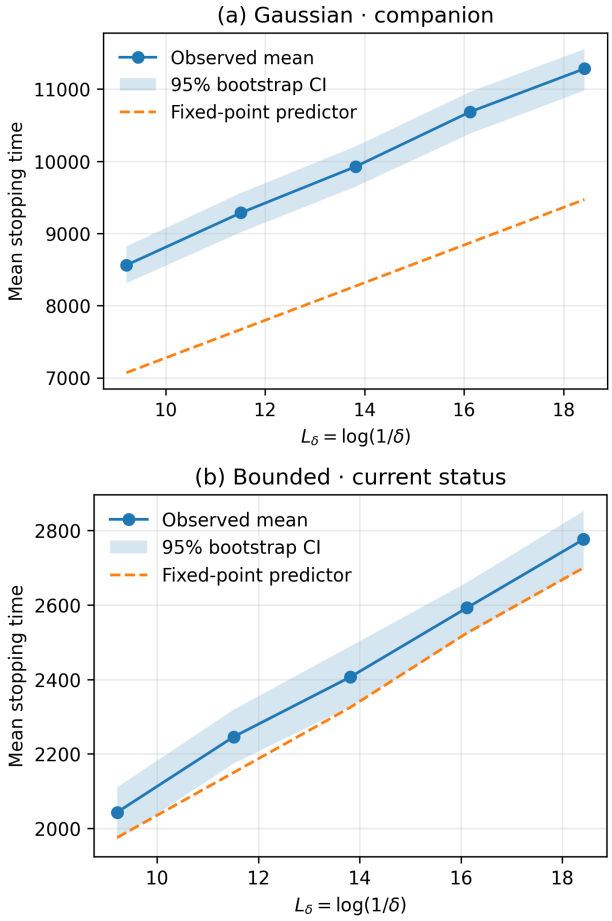


Figure 2: **Practical one-bit finite-confidence calibration.** Top: the separate Gaussian identification-schedule companion. Bottom: bounded raw current-status one-bit. Solid curves show empirical mean stopping times; dashed curves show the fixed-point predictor (29); shaded regions are bootstrap 95% confidence intervals. These use practical implementation boundaries, not the certified non-asymptotic upper bounds.

shared-threshold one-bit Top–Two. Model U uses Beta laws only as bounded reward generators; the full-feedback baseline is nonparametric, while the one-bit method retains the raw current-status observation (Q, Y) with $Q \sim \mathrm{Unif}[0,1]$. Full and one-bit replications are paired by instance, confidence level, and replication identifier for lower-variance ratio estimates. Appendix E gives the complete grids, seeds, caps, correctness checks, and stress runs.

Relative efficiency and first-order behavior. Figure 1 compares the empirical one-bit/full stopping-time ratio with the corresponding ratio of population characteristic times. The normalized ratio is already near one on the representative five-arm instances, indicating that the population characteristic times capture most of the relative information loss. This remains a *relative* diagnostic: finite-time overheads can partially cancel. The supplement therefore also reports the direct normalization $\overline{\tau}_\delta/(T^\star \log(1/\delta))$ separately for full and one-bit feedback and the deeper δ grid used to assess approach to the first-order limit.

Practical finite-confidence calibration. For each one-bit implementation, define

$$n_{\mathrm{pred}}(\delta) := \min\{n \in \mathcal{T} : n\Gamma^{\mathrm{pop}}_{1\mathrm{bit}} \geq b_{\mathrm{prac}}(n,\delta)\}, \quad (29)$$

where $\mathcal{T}$ is the implementation's stopping-check schedule and b_{prac} is the practical simulation boundary. This predicts the scale of the implemented stopping time; it is neither an exact formula for $\mathbb{E}[\tau_\delta]$ nor the numerical upper bound supplied by the certified non-asymptotic theorems. Figure 2 shows that the predictor tracks the raw current-status stopping scale more closely than the Gaussian companion on the archived benchmark. The distinction matters because the Gaussian $2/3$ result belongs to a separate companion policy, whereas the practical raw-U boundary is not the paper-certified sieve boundary.

Takeaway. Population characteristic times describe the instance-dependent information loss, while the fixed-point predictor captures finite-confidence implementation overhead. The fixed/shared/pairwise distinction in Model E remains theoretical: shared is the feasible joint K-arm design, while pairwise is a relaxation.

AI Use Statement

Generative AI was used substantively to develop and discuss mathematical claims, intermediate lemmas and proof strategies; to identify potential gaps; to explore research and numerical-method alternatives; to assist with literature discovery, experimental design, implementation and debugging of simulation and plotting code; and to draft, restructure and edit manuscript text, tables and figure captions. It was also used to interpret preliminary and final synthetic-simulation outputs. The authors remain solely responsible for auditing all generated mathematical arguments, validating numerical calculations and references, inspecting scripts, checking reported results against raw data, and approving the final manuscript. AI tools are not authors.

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Reproducibility Checklist

1. **Claims and assumptions.** The main text states the model-specific regularity, unique-best, fixed-K, compactness/interiority, and fixed-β assumptions used by the theorems. The supplement gives complete proofs and records where stronger finite-moment stabilization is used.
2. **Theoretical reproducibility.** Definitions of the observed-law KLs, characteristic values, sampling rules, stopping boundaries, and finite-confidence entrance quantities are given explicitly. The supplement contains the change-of-measure, oracle, profile-likelihood, allocation, and stopping arguments needed to reproduce the theoretical results.
3. **Experiments and numerical calculations.** Synthetic Gaussian and Beta instances, independent population optimizations, full-reward and threshold-bit algorithms, stopping checks, seeds, paired replication policy, and implemented boundary are specified in Section 6 and the appended experimental supplement (Section E). Continuum oracle calculations use numerical discretization; no quadrature error certificate is claimed.
4. **Code, data, and compute.** No external or human-subject dataset is used. Source code and CSV outputs are required to reproduce the $10{,}800$ method trajectories and their paired bootstrap analysis; the raw-profile five-arm simulations require substantial CPU time. A manuscript png alone does not include these executable artifacts.
5. **Uncertainty and limitations.** The finite-confidence refinements concern exact, certified model-specific algorithms and are not claimed to be sharp second-order expansions. Other limitations include the unique-best assumption and the model-specific compact/interior regularity conditions.
6. **AI-assisted workflow.** Generative-AI use is disclosed in the preceding AI Use Statement; all mathematical claims, citations, and numerical values remain the responsibility of the authors.

Guide to the supplement. The first two numbered parts are consecutive Model-E proofs: (A) asymptotic analysis, beginning with a cross-model proof roadmap and the shared change-of-measure/stopping-inversion tools, followed immediately by (B) the separate Model-E finite-confidence companion. Parts (C)–(D) give the corresponding asymptotic and finite-confidence Model-U proofs; (E) contains numerical and experimental details. A shared argument is stated and proved at its first substantive use rather than duplicated. In particular, Model U reuses the common lower-bound tool from (A) and the marked-CDF/profile estimates from (C) in its finite-confidence proof (D).

A MODEL E: ASYMPTOTIC OPTIMALITY

Proof guide: common milestones and model-specific tools

Let $b = i^\star$, $s = 1 - \beta$, $L = \log(1/\delta)$ and $\Gamma_M = (T_M^\star)^{-1}$ for $M \in \{\mathrm{E}, \mathrm{U}\}$. For both models the proof follows the Top–Two chain

$$\text{change of measure} \;\to\; B_n = b \;\to\; \text{information tracking} \;\to\; \text{challenger control} \;\to\; \mathbb{E}\tau_\delta \sim L/\Gamma_M. \tag{30}$$

Unlike the full-feedback analysis in [2, Section 4], each arrow must respect the observed one-bit experiment. We isolate the seven proof steps and refer to the full arguments below.

1. Lower bound: one feasible experiment, not separate pairwise queries. For every strict wrong-best alternative λ, data processing on the stopped transcript and its expected action occupation $\bar\rho_\delta$ yield

$$\mathrm{kl}(1-\delta, \delta) \le \mathbb{E}_\nu \tau_\delta \sum_k \int d_a(\nu_k, \lambda_k)\, \bar\rho_{\delta,k}(da). \tag{31}$$

In Model E, $d_a = d_q(\theta_k, u_k)$ and $\bar\rho_\delta$ is a single shared arm–threshold design. Its stopped best-arm mass is

$$\bar\rho_{\delta,b}(Q) = \frac{\mathbb{E}_\nu N_{\tau_\delta, b}}{\mathbb{E}_\nu \tau_\delta},$$

which need not equal β at a fixed confidence level. The stopped-β lemma below shows instead that

$$\bar\rho_{\delta,b}(Q) \longrightarrow \beta \qquad (\delta \downarrow 0).$$

In Model U the threshold is exogenous, so only the arm masses are controlled, while the observed KL is

$$D_U(F, G) = B^{-1} \int_0^B \mathrm{kl}(S_F(q), S_G(q))\, dq.$$

Thus Model E must use one feasible shared query design, whereas Model U retains the revealed threshold in its change-of-measure information. Strict/closed alternative equality is proved separately for the two models. After passing to the stopped-β limit, the generic lower bound yields

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu \tau_\delta}{L} \ge \frac{1}{\Gamma_M}.$$

2. Sufficient exploration and eventual correct leader. The full-feedback Top–Two route uses $N_{n,k} \gtrsim \sqrt{n}$ followed by mean concentration. In E, the forced *arm* rule yields

$$\min_k N_{n,k} \gtrsim n^\gamma, \qquad \inf_{q \in Q,\, |u - \theta_k| \ge \varepsilon} d_q(\theta_k, u) > 0 \implies \widehat\theta(n) \to \theta \implies B_n = b.$$

No fixed-threshold force is needed for the main E rule: compact canonical-family monotone likelihood ratio makes every admissible q uniformly identifying. Its *separate finite-time companion* instead forces an identifying q_0 for an explicit concentration rate. For U, uniform Q makes the *marginal bit* exactly Bernoulli:

$$\mathbb{E}[Y \mid I = k] = \mu_k / B, \qquad \min_k N_{n,k} \gtrsim \sqrt{n/K} \Longrightarrow \widehat\mu_k = B\overline{Y}_k \to \mu_k \Longrightarrow B_n = b.$$

The distinct guards give finite-moment leader entrances without requiring distinct suboptimal means.

3. The oracle: E equalizes a shared design; U targets a critical level. For E, with $\rho \in \mathcal{D}_\beta^b$ and shared best-arm component ρ_b, define

$$V_{bj}^{\mathrm{E}}(\theta;\rho_b,\rho_j) = \inf_{u_b \le u_j} \left\{ \int d_q(\theta_b,u_b)\,\rho_b(dq) + \int d_q(\theta_j,u_j)\,\rho_j(dq) \right\},$$
$$\Gamma_\beta^{\mathrm{E}}(\theta) = \max_{\rho\in\mathcal{D}_\beta^b} \min_{j\neq b} V_{bj}^{\mathrm{E}}(\theta;\rho_b,\rho_j).$$

Strict pairwise monotonicity forces positive challenger masses and equalization at every shared optimum. A vanishing strictly convex kernel regularizer chooses a canonical $\rho^\dagger$ even if the unregularized query design is non-unique. For U, scalar separation gives $h_j(w) = V_{bj}^U(\beta,w)$ and

$$\Gamma_\beta^{\mathrm{U}} = \max_{w_j\ge 0,\,\sum_{j\neq b} w_j = s} \min_{j\neq b} h_j(w_j), \quad c_j(g) = \inf\{w : h_j(w) \ge g\}, \quad g < \Gamma_\beta^{\mathrm{U}}.$$

Here $\sum_j c_j(g) < s$, but h_j may have plateaus: U therefore does *not* import E's strict-allocation proof or a unique optimal challenger vector.

4. Empirical information must track the relevant oracle. E tracks *threshold occupation*, not merely arm counts. Uniform MLE concentration, convergence of the regularized shared oracle, and martingale sampling control give, for each $j \neq b$,

$$\rho_n \Rightarrow \rho^\dagger, \qquad n^{-1}W_n^{\mathrm{E}}(b,j) - H_j^\dagger(N_{n,b}/n, N_{n,j}/n) \longrightarrow 0.$$

Here $H_j^\dagger$ is the population pairwise rate computed with the *same canonical threshold marginals* $\lambda_b^\dagger$ and $\lambda_j^\dagger$; the local threshold-sampling martingale is essential for this reduction. The crucial distinction from full-feedback transport is that a challenger cannot be assessed using the marginal arm allocation alone. U instead compares *unrestricted current-status profiles*:

$$\max_{i\neq j} \left| W_n^U(i,j) - nV_{ij}^U(N_{n,i}/n, N_{n,j}/n) \right| \le C\left[1 + \log(en)\sqrt{nH_n(\eta)}\right]. \tag{32}$$

Unlike a Bernoulli effective-sample-size GLR inequality, this follows from all-prefix marked-CDF DKW and BV/Stieltjes integration after *common* law-level smoothing. It remains valid at survival probabilities 0 or 1 and for tied candidate thresholds.

5. Challenger allocation: oversampled rejection versus a deficit barrier. With $B_n = b$, E's TC/TCI nomination minimizes $W_n^{\mathrm{E}}(b,j) + \xi_n(j)$, $\xi_n(j) \in \{0, \log N_{n,j}\}$. Strict geometry and Step 4 imply the familiar Top–Two implication

$$N_{n,j}/n > w_j^\dagger + \varepsilon \implies W_n^{\mathrm{E}}(b,j) - \min_{\ell\neq b} W_n^{\mathrm{E}}(b,\ell) \ge c_\varepsilon n \implies C_{n+1} \neq j.$$

Together with $N_{n,k} - \Psi_{n,k}$ concentration and a *random-prefix-aware* last-crossing argument, this gives $N_n/n \to w^\dagger$. For U, fix $g = \Gamma_\beta^{\mathrm{U}} - \varepsilon$ and let $c_j(g)$ be the challenger mass required to reach level g. If R_n is the number of regular correct-leader rounds, define the virtual allocation debt

$$D_{n,j} = c_j(g)R_n - \Psi_{n,j}^{\mathrm{nom}}, \qquad D_n^+ = \sum_j [D_{n,j}]_+.$$

On a regular round nominating challenger a, $D_{n+1,j} - D_{n,j} = c_j(g) - s\mathbf{1}\{j=a\}$. The subcritical secant and chord inequalities, together with (32), force a sufficiently under-served challenger to be selected. Since $\sum_j c_j(g) \le s$, serving such a challenger cannot increase D_n^+. A pathwise induction (not optional stopping at a last-exit time) therefore yields the target information level g without asserting convergence to a unique Model-U allocation.

6. Calibrated stopping and the matching first-order constant. Wrong-arm recommendations are controlled by a likelihood ratio, not by convergence alone. In E, a deterministic likelihood net gives the uniform boundary $b_E(n,\delta) = L + O(K\log n)$; for the *exact* U raw-profile statistic a conditional finite monotone sieve gives

$$\mathbb{P}\{W_n^U(i,j) \ge c_n + x\} \le e^{-x} \quad (\mu_i \le \mu_j), \qquad c_n = na_n + 2\log M_n = O(n^{2/3}\log n).$$

Unlike a practical U boundary omitting c_n, this one is δ-certified under adaptive arm choice. Steps 4–5 show that, for every fixed $g < \Gamma_M$, the correct evidence eventually exceeds gn after an integrable entrance. Consequently

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}\tau_\delta^M}{L} \le \frac{1}{\Gamma_M}.$$

The reverse inequality uses *expected stopped* sampling fractions: the shared stopped-β lemma establishes $\mathbb{E}N_{\tau,b}/\mathbb{E}\tau \to \beta$ from finite-mean leader entry, sublinear guards and bounded-increment martingales; insert this in (31) to obtain $\mathbb{E}\tau_\delta^M/(T_M^\star L) \to 1$. This conclusion concerns the exact theoretical algorithms.

7. Quantitative refinements belong to different policies. For the *separate E global-design companion*, forced q_0 exploration costs $O(n^\gamma)$ rounds; Hellinger/net localization costs $O(n^{1-\gamma/2}\sqrt{H_n})$ after oracle transfer. Hence

$$n^\gamma + n^{1-\gamma/2}\sqrt{H_n}, \quad \gamma = 2/3 \implies \mathbb{E}\tau_\delta^{\mathrm{E,ft}} \le T_E^\star L + O\Big(L^{2/3}\sqrt{\log(e+L)}\Big).$$

For the *same exact guarded U raw-profile policy* as above, the subcritical envelope combines a $n\varepsilon$ loss with the $\sqrt{n}/\varepsilon$ tracking cost and the sieve penalty:

$$\min_j W_n^U(b,j) \ge n\Gamma_\beta^{\mathrm{U}} - C\left[n\varepsilon + \frac{\log(en)\sqrt{nH_n(\eta)} + T_{\mathrm{lead}} + 1}{\varepsilon}\right],$$
$$\varepsilon_n \asymp n^{-1/4} \implies \mathbb{E}\tau_\delta^U \le T_U^\star L + O\Big(L^{3/4}\log^{3/2}(e+L)\Big).$$

For both displays, the detailed probability bounds follow from uniform trial-time inversion and quantile integration below; only U's exact profile rule carries the $3/4$ claim. Neither correction is claimed as an optimal second-order rate.

A.1 Shared change-of-measure and stopping inversion

We give the change-of-measure argument used by the main paper. The stopped-transcript inequality itself is generic and applies to both one-bit models. The later model-specific sections separately identify the resulting strict-alternative infimum with the closed order-reversal characteristic used in each achievability proof. Here δ-PAC means almost-sure stopping and recommendation error at most δ on every admissible instance. The fixed-β strict/closed identification below is applied with best-arm mass $\beta > 0$; no such equality is invoked for an arbitrary zero-best-mass design.

For arm k and action $a \in \mathcal{A}_k$, let $P^a_{\nu_k}$ be the observed law. Actions are predictable. Let $d_a(\nu_k, \lambda_k) = \mathrm{KL}(P^a_{\nu_k}, P^a_{\lambda_k})$, and let $\mathcal{A}(\nu)$ denote strict unique-best alternatives whose best arm differs from that of ν. For a design ρ define

$$I(\nu, \lambda; \rho) = \sum_k \int d_a(\nu_k, \lambda_k)\rho_k(da).$$

Proof of Theorem 1. Assume first $\mathbb{E}_\nu[\tau_\delta] < \infty$ and fix a strict alternative $\lambda \in \mathcal{A}(\nu)$. Write H_τ for the stopped history, including arm choices, thresholds where applicable, observations, stopping and recommendation. The symbol $\mathcal{L}_\nu(H_\tau)$ means its *probability law* under instance ν; it is not the parameter likelihood L_n used elsewhere. Let E be the event that the algorithm recommends the true best arm of ν. By δ-correctness, $P_\nu(E) \ge 1-\delta$, whereas $P_\lambda(E) \le \delta$. Data processing from the stopped transcript to the binary decision gives

$$\mathrm{kl}(1-\delta, \delta) \le \mathrm{KL}(\mathcal{L}_\nu(H_\tau), \mathcal{L}_\lambda(H_\tau)). \tag{33}$$

Pad the transcript after stopping by a cemetery symbol. The chain rule for relative entropy and predictability of the chosen arm-action pair imply

$$\mathrm{KL}(\mathcal{L}_\nu(H_\tau), \mathcal{L}_\lambda(H_\tau)) = \mathbb{E}_\nu\left[\sum_{t=1}^{\tau} d_{A_t}(\nu_{I_t}, \lambda_{I_t})\right]. \tag{34}$$

Define the expected stopped occupation measure

$$\bar\rho_\delta(C) = \frac{1}{\mathbb{E}_\nu[\tau_\delta]}\mathbb{E}_\nu\left[\sum_{t=1}^{\tau_\delta} \mathbf{1}\{(I_t, A_t) \in C\}\right], \quad C \subseteq \bigsqcup_k \big(\{k\} \times \mathcal{A}_k\big) \text{ measurable.}$$

It is a probability measure because exactly one action is taken per pre-stopping round. By Tonelli,

$$\mathbb{E}_\nu\left[\sum_{t=1}^{\tau_\delta} d_{A_t}(\nu_{I_t}, \lambda_{I_t})\right] = \mathbb{E}_\nu[\tau_\delta] I(\nu, \lambda; \bar{\rho}_\delta).$$

Since the inequality holds for every strict alternative,

$$\mathrm{kl}(1-\delta, \delta) \le \mathbb{E}_\nu[\tau_\delta] \inf_{\lambda \in \mathcal{A}(\nu)} I(\nu, \lambda; \bar{\rho}_\delta) \le \mathbb{E}_\nu[\tau_\delta] \Gamma^\star(\nu).$$

If $\Gamma^\star(\nu) = 0$, this contradicts finiteness of $\mathbb{E}_\nu[\tau_\delta]$. No generic closure argument is used here. In the fixed-β specialization with positive best-arm mass, Model E identifies strict and closed pair alternatives by compactness and continuity; for Model U, Lemma C.24 provides finite-KL strict approximation at those weights. □

Corollary A.1 (Fixed-β asymptotic lower bound)**.** *Suppose*

$$\mathbb{E}_\nu[N_{\tau_\delta, i^\star}] / \mathbb{E}_\nu[\tau_\delta] \to \beta.$$

Assume additionally that the design space is weakly compact and that, for every limiting design ρ of best-arm mass β and every $\varepsilon > 0$, the inner alternative infimum admits a strict ε-optimal alternative λ_ε for which $\rho' \mapsto I(\nu, \lambda_\varepsilon; \rho')$ is finite and continuous at ρ. Let $\Gamma^\star_\beta$ be the previous max–min restricted to designs assigning mass β to the best arm. Then

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu[\tau_\delta]}{\log(1/\delta)} \ge \frac{1}{\Gamma^\star_\beta(\nu)}.$$

For Model E these conditions follow from compact regularity. For Model U the design space is finite-dimensional in the arm masses, while Lemma C.24 provides the required strict finite-KL approximations.

Proof. Let

$$L := \liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu[\tau_\delta]}{\log(1/\delta)}$$

If $L = +\infty$, the claim is immediate. Assume $L < \infty$ and choose a sequence $\delta_m \downarrow 0$ along which the displayed ratio converges to L. Along this sequence $\mathbb{E}_\nu[\tau_{\delta_m}] < \infty$ for all sufficiently large m, so the stopped occupation measures are well defined. Weak compactness of the design space gives a further subsequence, not relabelled, such that $\bar{\rho}_{\delta_m} \Rightarrow \rho$. The stopped-fraction assumption forces the best-arm mass of ρ to be β.

Fix $\varepsilon > 0$. By the definition of the inner infimum, choose one strict alternative $\lambda_\varepsilon \in \mathcal{A}(\nu)$ such that

$$I(\nu, \lambda_\varepsilon; \rho) \le \inf_{\lambda \in \mathcal{A}(\nu)} I(\nu, \lambda; \rho) + \varepsilon \le \Gamma^\star_\beta(\nu) + \varepsilon.$$

For this fixed alternative, the continuity assumption in the corollary gives

$$I(\nu, \lambda_\varepsilon; \bar{\rho}_{\delta_m}) \longrightarrow I(\nu, \lambda_\varepsilon; \rho).$$

The finite-δ change-of-measure inequality therefore yields

$$\frac{\mathbb{E}_\nu[\tau_{\delta_m}]}{\log(1/\delta_m)} \ge \frac{\mathrm{kl}(1-\delta_m, \delta_m)/\log(1/\delta_m)}{I(\nu, \lambda_\varepsilon; \bar{\rho}_{\delta_m})}.$$

Taking $m \to \infty$ gives $L \ge 1/(\Gamma^\star_\beta + \varepsilon)$, with the usual convention $1/0 = +\infty$. Finally let $\varepsilon \downarrow 0$. □

Lemma A.2 (Shared stopped-β argument)**.** *Consider either exact fixed-β Top–Two sampling rule on its counterfactual continuation, and let b be its true best arm. Suppose (i) the number F_n of overridden or guard rounds obeys $F_n \le C(1 + n^\alpha)$ pathwise for some $\alpha < 1$; (ii) after a random leader entrance T_b with $\mathbb{E}T_b < \infty$, every regular round samples b conditionally with probability β; and (iii) $\mathbb{E}\tau_\delta < \infty$ and $\mathbb{E}\tau_\delta \to \infty$ as $\delta \downarrow 0$. Then*

$$\frac{\mathbb{E}N_{\tau_\delta, b}}{\mathbb{E}\tau_\delta} \longrightarrow \beta. \tag{35}$$

Proof. Let $\pi_{t,b} = \mathbb{P}(I_t = b \mid \mathcal{F}_{t-1})$ and $M_{n,b} = N_{n,b} - \sum_{t \le n} \pi_{t,b}$. Predictability and bounded increments make $M_{n,b}$ a martingale. On regular rounds $t > T_b$, $\pi_{t,b} = \beta$; on all others the difference is at most one, so

$$\left| \sum_{t \le n} \pi_{t,b} - \beta n \right| \le T_b + F_n + C_0.$$

Optional isometry after localization at the integrable stopping time and Jensen's inequality give $\mathbb{E}|M_{\tau,b}| \le \sqrt{\mathbb{E}\tau}$ and $\mathbb{E}F_\tau \le C[1 + (\mathbb{E}\tau)^\alpha]$. Thus

$$|\mathbb{E}N_{\tau,b} - \beta\mathbb{E}\tau| \le \mathbb{E}T_b + C_0 + C[1 + (\mathbb{E}\tau)^\alpha] + \sqrt{\mathbb{E}\tau} = o(\mathbb{E}\tau),$$

which proves the assertion. Here τ denotes τ_δ. □

A.1.1 Common finite-confidence inversion

The following model-independent inversion is used in the Model-E and Model-U proofs below, after their respective evidence-entrance lemmas have been established. Both algorithms first test at $n_{\min} = K$.

Throughout,

$$L_\delta := \log(1/\delta).$$

A generic stopping inversion lemma.

Lemma A.3 (Finite-mean evidence entrance implies finite-confidence control)**.** *Let $\Gamma > 0$ and suppose a stopping rule first tests its boundary at integer times $n \ge n_{\min}$, where $n_{\min} \ge 1$ is a deterministic integer, with*

$$b(n, \delta) = L_\delta + d_n, \qquad d_n = o(n).$$

Assume that for every fixed $\gamma_0 < \Gamma$ there is a random time T_{γ_0} with

$$\mathbb{E}[T_{\gamma_0}] < \infty$$

such that, for every integer $n \ge T_{\gamma_0}$, the empirical leader is correct and the minimum correct pairwise evidence is at least $\gamma_0 n$.

For $\delta, \eta \in (0, 1)$ and

$$0 < \gamma < \gamma_0 < \Gamma$$

define

$$n_0(\gamma, \gamma_0) := \min\{m \ge 1 : d_n \le (\gamma_0 - \gamma)n \quad \forall n \ge m\} < \infty.$$

Round T_{γ_0} up to an integer if necessary. Let

$$u_{\gamma_0}(\eta) := \min\{m \in \mathbb{N}_0 : \mathbb{P}(T_{\gamma_0} \le m) \ge 1 - \eta\}$$

be an upper $(1 - \eta)$ quantile. Then

$$\mathbb{P}\left(\tau_\delta \le \max\left\{n_{\min}, u_{\gamma_0}(\eta), n_0(\gamma, \gamma_0), \left\lceil \frac{L_\delta}{\gamma} \right\rceil\right\}\right) \ge 1 - \eta. \tag{36}$$

Moreover,

$$\mathbb{E}[\tau_\delta] \le \mathbb{E}[T_{\gamma_0}] + n_{\min} + n_0(\gamma, \gamma_0) + \left\lceil \frac{L_\delta}{\gamma} \right\rceil. \tag{37}$$

Proof. On $\{T_{\gamma_0} \le u_{\gamma_0}(\eta)\}$, every

$$n \ge \max\left\{n_{\min}, u_{\gamma_0}(\eta), n_0(\gamma, \gamma_0), \left\lceil \frac{L_\delta}{\gamma} \right\rceil\right\}$$

has the correct leader and evidence at least $\gamma_0 n$. By definition of n_0,

$$b(n, \delta) = L_\delta + d_n \le \gamma n + (\gamma_0 - \gamma)n = \gamma_0 n.$$

Hence the stopping condition must already have been met. This proves (36). Pathwise,

$$\tau_\delta \le \max\left\{n_{\min}, T_{\gamma_0}, n_0(\gamma,\gamma_0), \left\lceil \frac{L_\delta}{\gamma} \right\rceil \right\},$$

and the sum bound for the maximum yields (37). □

Corollary A.4 (Near-characteristic finite-confidence form)**.** *Under Lemma A.3, let $T^\star := 1/\Gamma$. For every $\varepsilon > 0$, choose*

$$\gamma = \frac{1}{T^\star + \varepsilon}, \qquad \gamma_0 \in (\gamma, \Gamma).$$

Then

$$\mathbb{E}[\tau_\delta] \le (T^\star + \varepsilon) L_\delta + C_\varepsilon, \tag{38}$$

with

$$C_\varepsilon = \mathbb{E}[T_{\gamma_0}] + n_{\min} + n_0(\gamma,\gamma_0) + 1 < \infty.$$

For every $\eta \in (0,1)$,

$$\mathbb{P}(\tau_\delta \le (T^\star + \varepsilon) L_\delta + C_{\varepsilon,\eta}) \ge 1 - \eta, \tag{39}$$

where

$$C_{\varepsilon,\eta} = \max\{n_{\min}, u_{\gamma_0}(\eta), n_0(\gamma,\gamma_0)\} + 1 < \infty.$$

The constants may depend on the instance, ε and η, but not on δ.

Proof. Apply Lemma A.3 with $\gamma = (T^\star + \varepsilon)^{-1}$ and use

$$\left\lceil \frac{L_\delta}{\gamma} \right\rceil \le (T^\star + \varepsilon) L_\delta + 1.$$

□

Remark A.5 (Optional explicit probability localization)**.** If T_{γ_0} has a finite pth moment, Markov gives the admissible choice

$$u_{\gamma_0}(\eta) \le \left\lceil \left(\frac{\mathbb{E}[T_{\gamma_0}^p]}{\eta} \right)^{1/p} \right\rceil.$$

Model U has moments of every fixed order for its evidence entrance time. The Model E argument below only needs finite expectation. Stronger moment bounds are available but are not needed here.

A.2 Standing assumptions and notation

The one-parameter canonical exponential family, threshold observation $Y_t = \mathbf{1}\{X_t \ge q_t\}$, and one-bit divergence $d_q(u,v)$ are defined in Section 2. Its support is independent of the parameter. Write $I = [\theta_-, \theta_+]$ and let $\theta = (\theta_k)_{k=1}^K$; then $i^\star = \arg\max_k \theta_k$ because $b'' > 0$.

Assumption A.6 (Unique best arm)**.** The best arm is unique, with $\Delta_\star = \min_{j \ne i^\star}(\theta_{i^\star} - \theta_j) > 0$. Suboptimal parameters may tie.

Assumption A.7 (Regular compact one-bit model)**.** The parameter and threshold sets I, Q are compact; $(q,u) \mapsto p_q(u)$ is jointly continuous and $u \mapsto p_q(u)$ is continuously differentiable for every q. For constants $\eta_0 \in (0, 1/2)$ and $G < \infty$, uniformly on $Q \times I$, $\eta_0 \le p_q(u) \le 1 - \eta_0$, and $u \mapsto \log p_q(u)$ and $u \mapsto \log(1 - p_q(u))$ are G-Lipschitz. Finally $(q,u,v) \mapsto d_q(u,v)$ is continuous on $Q \times I^2$.

The identification and exploration proofs use the positive best-arm gap, not pairwise separation of every suboptimal arm.

A.3 Likelihood, GLR statistics, and stopping

For a candidate parameter vector $u = (u_1, \dots, u_K) \in I^K$, define

$$L_n(u) := \sum_{t=1}^{n} \left[Y_t \log p_{q_t}(u_{I_t}) + (1 - Y_t) \log(1 - p_{q_t}(u_{I_t})) \right]. \tag{40}$$

It decomposes arm-wise:

$$L_n(u) = \sum_{k=1}^{K} \ell_{n,k}(u_k),$$

where

$$\ell_{n,k}(a) := \sum_{t \le n : I_t = k} \left[Y_t \log p_{q_t}(a) + (1 - Y_t) \log(1 - p_{q_t}(a)) \right]. \tag{41}$$

Let

$$\widehat{\theta}(n) \in \arg\max_{u \in I^K} L_n(u), \tag{42}$$

with a fixed deterministic tie-breaking rule. By separability, $\widehat{\theta}_k(n) \in \arg\max_{a \in I} \ell_{n,k}(a)$.

For $i \neq j$, define the pairwise generalized likelihood-ratio statistic

$$\begin{aligned} W_n(i,j) := \ell_{n,i}(\widehat{\theta}_i(n)) + \ell_{n,j}(\widehat{\theta}_j(n)) \\ - \sup_{\substack{u_i, u_j \in I \\ u_i \le u_j}} \left[\ell_{n,i}(u_i) + \ell_{n,j}(u_j) \right]. \end{aligned} \tag{43}$$

For arm i, define the alternative set

$$\mathrm{Alt}_i := \left\{ u \in I^K : u_i \le \max_{j \neq i} u_j \right\}, \tag{44}$$

and

$$Z_i(n) := L_n(\widehat{\theta}(n)) - \sup_{u \in \mathrm{Alt}_i} L_n(u). \tag{45}$$

Lemma A.8 (Pairwise decomposition of the GLR)**.** *For every $i \in [K]$,*

$$Z_i(n) = \min_{j \neq i} W_n(i,j).$$

Proof. Because $\mathrm{Alt}_i = \bigcup_{j \neq i} \{u \in I^K : u_i \le u_j\}$, we have

$$\sup_{u \in \mathrm{Alt}_i} L_n(u) = \max_{j \neq i} \sup_{u_i \le u_j} L_n(u).$$

For fixed (i,j), all coordinates $k \notin \{i,j\}$ are unconstrained and are maximized at their coordinatewise MLEs. These terms cancel from the likelihood-ratio gap, leaving exactly $W_n(i,j)$. Taking a maximum of constrained likelihoods therefore becomes a minimum of likelihood-ratio gaps. □

Let

$$\widehat{k}(n) := \arg\max_{k \in [K]} \widehat{\theta}_k(n),$$

with the same fixed deterministic tie-breaking rule as above. We use the stopping rule

$$\tau_\delta := \inf \left\{ n \ge K : Z_{\widehat{k}(n)}(n) \ge c(n,\delta) \right\}, \tag{46}$$

with

$$c(n,\delta) := 1 + \log \frac{\pi^2}{6\delta} + K \log n + K \log(2 + GRn), \qquad R := \theta_+ - \theta_-. \tag{47}$$

The recommendation is $\widehat{k}(\tau_\delta)$.

Proof of Theorem 3. For one observation, the log-likelihood contribution as a function of the active coordinate is G-Lipschitz. Hence $|L_n(u) - L_n(v)| \le Gn\|u - v\|_\infty$. Choose a $(Gn)^{-1}$-net $\mathcal{G}_n \subset I^K$ in the sup norm. Since I has diameter R, it may be chosen so that $|\mathcal{G}_n| \le (2 + GRn)^K$. For every $u \in I^K$ there is $v \in \mathcal{G}_n$ such that $L_n(u) - L_n(\theta) \le 1 + L_n(v) - L_n(\theta)$.

For a fixed v, the process $\exp\{L_n(v) - L_n(\theta)\}$ is a nonnegative likelihood-ratio martingale under P_θ. Adaptivity causes no difficulty because (I_t, q_t) is chosen before Y_t is observed. Thus

$$\mathbb{P}_\theta(L_n(v) - L_n(\theta) \ge c(n,\delta) - 1) \le e^{1-c(n,\delta)}.$$

A union bound over $\mathcal{G}_n$ gives

$$\begin{aligned}\mathbb{P}_\theta\left(\sup_{u\in I^K}\{L_n(u) - L_n(\theta)\} \ge c(n,\delta)\right) &\le (2 + GRn)^K e^{1-c(n,\delta)} \\ &= \frac{6\delta}{\pi^2 n^K} \le \frac{6\delta}{\pi^2 n^2}.\end{aligned}$$

Summing over n gives a total probability at most δ.

If the algorithm stops and recommends a wrong arm $a \ne i^\star$, then $\theta \in \mathrm{Alt}_a$. Consequently

$$\sup_{u\in\mathrm{Alt}_a} L_n(u) \ge L_n(\theta).$$

At stopping,

$$Z_a(n) = L_n(\widehat{\theta}(n)) - \sup_{u\in\mathrm{Alt}_a} L_n(u) \ge c(n,\delta),$$

so $L_n(\widehat{\theta}(n)) - L_n(\theta) \ge c(n,\delta)$. This is contained in the time-uniform likelihood event already controlled above. □

A.4 Shared adaptive beta-design

The central modeling point is that all pairwise GLRs involving the best arm use the same physical best-arm observations. Therefore the best-arm threshold measure must be shared across all challengers.

Definition A.9 (Shared beta-design space)**.** For $b \in [K]$, let $\mathcal{D}_\beta^b$ be the set of tuples

$$\rho = (\rho_1, \dots, \rho_K)$$

of finite nonnegative Borel measures on Q satisfying

$$\sum_{k=1}^K \rho_k(Q) = 1, \qquad \rho_b(Q) = \beta. \tag{48}$$

Equivalently, ρ is a probability measure on $[K] \times Q$ whose arm marginal places mass β on arm b.

For $v, u \in I^K$ and $\rho \in \mathcal{D}_\beta^b$, define

$$\mathcal{I}(v, u; \rho) := \sum_{k=1}^K \int_Q d_q(v_k, u_k)\rho_k(dq). \tag{49}$$

For $b \ne j$, define the pair value under a fixed shared design:

$$V_{b,j}(v; \rho_b, \rho_j) := \inf_{\substack{u_b,u_j\in I\\ u_b\le u_j}} \left[\int_Q d_q(v_b, u_b)\rho_b(dq) + \int_Q d_q(v_j, u_j)\rho_j(dq)\right]. \tag{50}$$

For a vector v with a unique best arm $b(v)$, define

$$F_v(\rho) := \inf_{u\in\mathrm{Alt}_{b(v)}} \mathcal{I}(v, u; \rho), \qquad \rho \in \mathcal{D}_\beta^{b(v)}. \tag{51}$$

Definition A.10 (Shared beta-characteristic value)**.** For v with unique best arm $b(v)$,

$$\Gamma_\beta(v) := \max_{\rho\in\mathcal{D}_\beta^{b(v)}} F_v(\rho), \qquad T_\beta^\star(v) := \Gamma_\beta(v)^{-1}. \tag{52}$$

Lemma A.11 (Pairwise decomposition of the shared objective)**.** *For every v with unique best arm b and every $\rho \in \mathcal{D}_\beta^b$,*

$$F_v(\rho) = \min_{j \neq b} V_{b,j}(v; \rho_b, \rho_j).$$

Consequently,

$$\Gamma_\beta(v) = \max_{\rho \in \mathcal{D}_\beta^b} \min_{j \neq b} V_{b,j}(v; \rho_b, \rho_j). \tag{53}$$

Proof. Use $\mathrm{Alt}_b = \bigcup_{j \neq b} \{u : u_b \leq u_j\}$. For a fixed challenger j, every coordinate $k \notin \{b, j\}$ is unconstrained and is minimized by setting $u_k = v_k$, which contributes zero divergence. The remaining two-coordinate problem is exactly $V_{b,j}(v; \rho_b, \rho_j)$. □

Lemma A.12 (Compactness, continuity, and concavity)**.** *Fix $b \in [K]$.*

(i) $\mathcal{D}_\beta^b$ is convex and weakly compact.

(ii) On any region of I^K where the best arm is b, the map $(v, \rho) \mapsto F_v(\rho)$ is jointly continuous.

(iii) For fixed v, $\rho \mapsto F_v(\rho)$ is concave.

(iv) The maximum in (52) *is attained.*

Proof. The action space $[K] \times Q$ is compact. The space of probability measures on it is weakly compact, and the linear mass constraint $\rho_b(Q) = \beta$ is closed. This proves compactness and convexity.

The function $(v, u, \rho) \mapsto \mathcal{I}(v, u; \rho)$ is jointly continuous because d_q is continuous and bounded on the compact domain. The alternative set Alt_b is compact, so Berge's minimum theorem gives continuity of $F_v(\rho)$.

For fixed u, $\rho \mapsto \mathcal{I}(v, u; \rho)$ is linear. The infimum of linear functions is concave, so F_v is concave. Continuity on the compact feasible set then gives existence of a maximizer. □

Lemma A.13 (Positivity of the shared characteristic value)**.** *Under Assumptions A.6 and A.7,*

$$\Gamma_\beta(\theta) > 0.$$

Proof. Fix any $q \in Q$ and use the feasible design

$$\rho_{i^\star} = \beta \delta_q, \qquad \rho_j = \frac{1 - \beta}{K - 1} \delta_q \quad (j \neq i^\star).$$

For every challenger j, the pair objective is continuous and nonnegative on the compact constraint set $\{(u_i, u_j) : u_i \leq u_j\}$. It cannot vanish: simultaneous vanishing would force $u_i = \theta_{i^\star}$ and $u_j = \theta_j$, which violates $u_i \leq u_j$ because $\theta_{i^\star} > \theta_j$. Thus every pair value is positive and so is their minimum. □

Definition A.14 (Optimal shared-design set)**.** At the true instance, define

$$\mathcal{O}_\beta(\theta) := \arg\max_{\rho \in \mathcal{D}_\beta^{i^\star}} F_\theta(\rho). \tag{54}$$

Lemma A.15 (The raw optimal set is nonempty, compact, and convex)**.** *$\mathcal{O}_\beta(\theta)$ is nonempty, weakly compact, and convex.*

Proof. Nonemptiness and compactness follow from Lemma A.12. If $\rho, \nu \in \mathcal{O}_\beta(\theta)$ and $t \in [0, 1]$, concavity gives

$$F_\theta(t\rho + (1 - t)\nu) \geq tF_\theta(\rho) + (1 - t)F_\theta(\nu) = \Gamma_\beta(\theta).$$

The left-hand side cannot exceed $\Gamma_\beta(\theta)$, so equality holds and the convex combination is also optimal. □

Remark A.16 (Why raw uniqueness is not assumed)**.** Raw uniqueness of the shared optimal arm–threshold measure is false in general. For example, in a Bernoulli canonical family with thresholds $q \in (0, 1)$, $\mathbf{1}\{X \geq q\} = X$ for every such q. Then $p_q(\theta)$ and $d_q(\theta, u)$ are independent of q, so different threshold measures produce exactly the same information. The proof below therefore selects a unique *canonical* optimal design instead of assuming the raw optimal set is a singleton.

A.5 One-bit geometry and strict monotonicity

Lemma A.17 (Strict increase of the threshold success probability)**.** *Under the canonical exponential-family model and Assumption A.7,*

$$p_q'(u) > 0 \qquad \forall (q,u) \in Q \times I.$$

Hence $u \mapsto p_q(u)$ is strictly increasing for every $q \in Q$.

Proof. Differentiating the threshold probability under the regular exponential-family model gives

$$p_q'(u) = \mathbb{E}_u\left[(X - b'(u))\mathbf{1}\{X \geq q\}\right] \tag{55}$$

$$= \mathrm{Cov}_u\left(X, \mathbf{1}\{X \geq q\}\right). \tag{56}$$

Let $p = p_q(u)$ and define

$$m_+ := \mathbb{E}_u[X \mid X \geq q], \qquad m_- := \mathbb{E}_u[X \mid X < q].$$

Then $p_q'(u) = p(1-p)(m_+ - m_-)$. Assumption A.7 gives $p \in [\eta_0, 1-\eta_0]$, so both conditioning events have positive probability. Moreover, $m_+ \geq q > m_-$. Hence $p_q'(u) > 0$. □

Lemma A.18 (Shape of the threshold KL)**.** *For fixed $a \in I$ and $q \in Q$,*

$$\frac{\partial}{\partial x} d_q(a,x) = p_q'(x) \frac{p_q(x) - p_q(a)}{p_q(x)(1 - p_q(x))}. \tag{57}$$

Consequently,

$$\partial_x d_q(a,x) \begin{cases} < 0, & x < a, \\ = 0, & x = a, \\ > 0, & x > a. \end{cases}$$

In particular, $d_q(a,x) = 0$ if and only if $x = a$.

Proof. For Bernoulli KL, $\frac{\partial}{\partial r}\,\mathrm{kl}(p,r) = \frac{r-p}{r(1-r)}$. Substitute $p = p_q(a)$ and $r = p_q(x)$ and apply the chain rule. The sign follows from Lemma A.17. □

A.5.1 Scalar reduction

Let ρ_i, ρ_j be finite nonnegative measures on Q and define

$$G_i(x) := \int_Q d_q(\theta_i, x)\rho_i(dq), \qquad G_j(x) := \int_Q d_q(\theta_j, x)\rho_j(dq).$$

Lemma A.19 (Scalar reduction)**.** *If $\theta_i > \theta_j$, then*

$$\inf_{\substack{u_i, u_j \in I \\ u_i \leq u_j}} \left[G_i(u_i) + G_j(u_j)\right] = \min_{x \in I} \left[G_i(x) + G_j(x)\right]. \tag{58}$$

This remains valid if either measure is the zero measure.

Proof. By Lemma A.18, G_i is nonincreasing on $(-\infty, \theta_i] \cap I$ and nondecreasing on $[\theta_i, \infty) \cap I$. Likewise, G_j is nonincreasing up to θ_j and nondecreasing afterwards. These monotonicities need not be strict when one measure is zero.

Take any feasible (u_i, u_j) with $u_i \leq u_j$.

If $u_j \leq \theta_i$, choose $x = u_j$. Since $u_i \leq u_j \leq \theta_i$, $G_i(x) = G_i(u_j) \leq G_i(u_i)$, while $G_j(x) = G_j(u_j)$. Hence the tied point is no worse.

If $u_j > \theta_i$ and $u_i \geq \theta_j$, choose $x = u_i$. Since $u_j \geq u_i \geq \theta_j$, $G_j(x) = G_j(u_i) \leq G_j(u_j)$, while $G_i(x) = G_i(u_i)$.

Finally, if $u_i < \theta_j < \theta_i < u_j$, choose $x = \theta_j$. Then

$$G_i(\theta_j) \leq G_i(u_i), \qquad G_j(\theta_j) \leq G_j(u_j).$$

Thus every feasible point can be replaced by a tied point with no larger objective. The reverse inequality is immediate because every tied point (x,x) is feasible. □

A.5.2 Strict monotonicity

For $\lambda_i, \lambda_j \in \mathcal{P}(Q)$, define

$$A_i(x) := \int_Q d_q(\theta_i, x)\lambda_i(dq), \qquad A_j(x) := \int_Q d_q(\theta_j, x)\lambda_j(dq).$$

For $a, w \geq 0$, define

$$H_{i,j}(a, w; \lambda_i, \lambda_j) := \min_{x \in I} \left[aA_i(x) + wA_j(x)\right]. \tag{59}$$

Lemma A.20 (Interior minimizer for positive masses)**.** *Suppose $\theta_i > \theta_j$ and $a, w > 0$. Every minimizer of* (59) *lies in*

$$(\theta_j, \theta_i).$$

Proof. At $x = \theta_j$,

$$A_j'(\theta_j) = 0, \qquad A_i'(\theta_j) < 0,$$

because every integrand in $A_i'(\theta_j)$ is strictly negative by Lemma A.18. Differentiation under the probability measure is justified by the uniform boundedness of the derivative of the one-bit log-likelihood on the compact model. Hence the right derivative of $aA_i + wA_j$ at θ_j is strictly negative, so θ_j is not a minimizer.

Similarly,

$$A_i'(\theta_i) = 0, \qquad A_j'(\theta_i) > 0,$$

so θ_i is not a minimizer.

Points below θ_j can be moved right and points above θ_i can be moved left to decrease both terms. Therefore every minimizer belongs to the open interval. □

Proposition A.21 (Strict monotonicity for given pairwise threshold laws)**.** *Fix $a > 0$, $\lambda_i, \lambda_j \in \mathcal{P}(Q)$, and $\theta_i > \theta_j$. Then*

$$w \longmapsto H_{i,j}(a, w; \lambda_i, \lambda_j)$$

is continuous and strictly increasing on $[0, \infty)$.

Proof. Continuity follows from continuity of the objective on compact I.

Let $0 < w < w'$ and let x' minimize the fixed-design objective at w'. Lemma A.20 gives $x' \neq \theta_j$, hence $A_j(x') > 0$. Therefore

$$\begin{aligned} H_{i,j}(a, w') &= aA_i(x') + w'A_j(x') \\ &= aA_i(x') + wA_j(x') + (w' - w)A_j(x') \\ &\geq H_{i,j}(a, w) + (w' - w)A_j(x') \\ &> H_{i,j}(a, w). \end{aligned}$$

At $w = 0$, $H_{i,j}(a, 0) = 0$. For $w' > 0$, the two nonnegative terms cannot vanish simultaneously because $\theta_i \neq \theta_j$, so the minimum is strictly positive. □

For later mathematical reference, define the auxiliary two-arm transportation cost

$$C_{i,j}(a, w; \theta) := \max_{\lambda_i, \lambda_j \in \mathcal{P}(Q)} H_{i,j}(a, w; \lambda_i, \lambda_j). \tag{60}$$

Theorem A.22 (Strict monotonicity of the auxiliary two-arm one-bit cost)**.** *Fix $a > 0$ and $\theta_i > \theta_j$. Then*

$$0 \leq w < w' \quad \Longrightarrow \quad C_{i,j}(a, w; \theta) < C_{i,j}(a, w'; \theta).$$

Proof. The design space $\mathcal{P}(Q)^2$ is weakly compact and the objective is continuous, so an optimizer exists at every w.

For $w > 0$, take an optimizer $(\lambda_i^\star, \lambda_j^\star)$ at w. Proposition A.21 gives

$$\begin{aligned} C_{i,j}(a, w') &\geq H_{i,j}(a, w'; \lambda_i^\star, \lambda_j^\star) \\ &> H_{i,j}(a, w; \lambda_i^\star, \lambda_j^\star) \\ &= C_{i,j}(a, w). \end{aligned}$$

The case $w = 0$ follows because the value is zero at $w = 0$ and strictly positive for every positive challenger mass. □

A.6 Equalization of every shared optimum

Take any

$$\rho^\star \in \mathcal{O}_\beta(\theta).$$

Write $w_k^\star := \rho_k^\star(Q)$.

Lemma A.23 (Every optimal challenger mass is positive)**.** *For every shared optimal design $\rho^\star \in \mathcal{O}_\beta(\theta)$,*

$$w_j^\star > 0 \qquad \forall j \neq i^\star.$$

Proof. By Lemma A.13, $\Gamma_\beta(\theta) > 0$. If $w_j^\star = 0$ for some challenger j, choose $u_{i^\star} = u_j = \theta_{i^\star}$. The best-arm term is zero and there is no challenger term, so the pair value against j is zero, contradicting positivity of the optimal minimum. □

Since $w_{i^\star}^\star = \beta > 0$ and Lemma A.23 gives positive mass to every challenger, define

$$\lambda_k^\star := \rho_k^\star / w_k^\star \in \mathcal{P}(Q), \qquad A_k^\star(x) := \int_Q d_q(\theta_k, x)\lambda_k^\star(dq).$$

For $j \neq i^\star$, define

$$H_j^\star(a, w) := \min_{x \in I} \left[aA_{i^\star}^\star(x) + wA_j^\star(x) \right]. \tag{61}$$

Theorem A.24 (Equal-cost equilibrium)**.** *For every shared optimal design $\rho^\star \in \mathcal{O}_\beta(\theta)$,*

$$H_j^\star(\beta, w_j^\star) = \Gamma_\beta(\theta) \qquad \forall j \neq i^\star. \tag{62}$$

Proof. Suppose a challenger j_0 is slack: $H_{j_0}^\star(\beta, w_{j_0}^\star) > \Gamma_\beta(\theta)$. Let

$$\mathcal{B} := \left\{ j \neq i^\star : H_j^\star(\beta, w_j^\star) = \Gamma_\beta(\theta) \right\}$$

be the nonempty set of binding challengers.

Keep all normalized threshold distributions $\lambda_k^\star$ fixed. For small $\varepsilon > 0$, reduce the mass of j_0 by $|\mathcal{B}|\varepsilon$ and increase every binding challenger mass by ε. The best-arm measure is unchanged, the total mass remains one, and all masses remain nonnegative for sufficiently small ε.

By Proposition A.21, every binding pair value increases strictly. By continuity, the slack pair j_0 remains strictly above the old minimum for sufficiently small ε. All untouched slack pair values are unchanged. The minimum pair value therefore becomes strictly larger than $\Gamma_\beta(\theta)$, contradicting optimality. Thus every challenger is binding. □

A.7 Canonical selection of a shared optimum

Raw optimal designs need not be unique. We now select one canonical exact optimizer by a strictly convex regularizer.

Let the joint action space be

$$\mathcal{A} := [K] \times Q.$$

Identify a tuple $\rho = (\rho_1, \ldots, \rho_K)$ with the probability measure $\bar{\rho}$ on $\mathcal{A}$ satisfying

$$\bar{\rho}(\{k\} \times B) = \rho_k(B).$$

Fix $s > 0$ and define the bounded continuous kernel

$$\kappa((k, q), (\ell, r)) := \mathbf{1}\{k = \ell\} \exp\left(-\frac{(q - r)^2}{s^2} \right). \tag{63}$$

The Gaussian kernel below is strictly positive definite on finite signed measures; the direct sum over the finite arm index preserves this property. Lemma A.25 records the argument explicitly.

Define

$$\mathcal{R}(\rho) := \iint_{\mathcal{A} \times \mathcal{A}} \kappa(a, a')\bar{\rho}(da)\bar{\rho}(da'). \tag{64}$$

Lemma A.25 (Strict convexity of the kernel regularizer)**.** *$\mathcal{R}$ is weakly continuous and strictly convex on the probability measures on $\mathcal{A}$.*

Proof. Bounded continuity of κ gives weak continuity. For distinct probability measures $\bar{\rho}, \bar{\nu}$ on $\mathcal{A}$, let $\sigma = \bar{\rho} - \bar{\nu}$ and let σ_k be its finite signed component on $\{k\} \times Q$. For $g_s(q, r) = \exp(-(q-r)^2/s^2)$, extend each σ_k by zero outside Q. Bochner's formula gives

$$\iint g_s(q, r) d\sigma_k(q) d\sigma_k(r) = c_s \int_{\mathbb{R}} |\widehat{\sigma}_k(\omega)|^2 e^{-s^2\omega^2/4} d\omega, \tag{65}$$

with $c_s > 0$. The right-hand side vanishes only when the Fourier–Stieltjes transform is identically zero, hence only when $\sigma_k = 0$. Since the arm kernel is an orthogonal direct sum,

$$\iint_{\mathcal{A}^2} \kappa(a, a') d\sigma(a) d\sigma(a') = \sum_{k=1}^{K} \iint g_s(q, r) d\sigma_k(q) d\sigma_k(r) > 0$$

whenever $\bar{\rho} \neq \bar{\nu}$. Direct expansion then yields, for $t \in (0, 1)$,

$$\begin{aligned} \mathcal{R}(t\rho + (1-t)\nu) = t\mathcal{R}(\rho) + (1-t)\mathcal{R}(\nu) \\ - t(1-t) \iint \kappa(a, a') d\sigma(a) d\sigma(a'), \end{aligned}$$

which proves strict convexity. □

Definition A.26 (Canonical exact shared optimum)**.** Define

$$\rho^\dagger := \operatorname*{arg\,min}_{\rho \in \mathcal{O}_\beta(\theta)} \mathcal{R}(\rho). \tag{66}$$

Let

$$w_k^\dagger := \rho_k^\dagger(Q), \qquad \lambda_k^\dagger := \frac{\rho_k^\dagger}{w_k^\dagger},$$

where all challenger masses are positive by Lemma A.23, and $w_{i^\star}^\dagger = \beta$. Define

$$A_k^\dagger(x) := \int_Q d_q(\theta_k, x) \lambda_k^\dagger(dq) \tag{67}$$

and

$$H_j^\dagger(a, w) := \min_{x \in I} \left[a A_{i^\star}^\dagger(x) + w A_j^\dagger(x) \right]. \tag{68}$$

Theorem A.27 (Existence and uniqueness of the canonical optimizer)**.** *The minimizer in* (66) *exists and is unique. Moreover,*

$$F_\theta(\rho^\dagger) = \Gamma_\beta(\theta)$$

and

$$H_j^\dagger(\beta, w_j^\dagger) = \Gamma_\beta(\theta) \qquad \forall j \neq i^\star. \tag{69}$$

Proof. The optimal set $\mathcal{O}_\beta(\theta)$ is nonempty, compact, and convex by Lemma A.15. The regularizer is continuous, so a minimizer exists. Strict convexity of $\mathcal{R}$ on the convex set makes the minimizer unique. Because $\rho^\dagger$ belongs to the exact optimal set, $F_\theta(\rho^\dagger) = \Gamma_\beta(\theta)$. The equalization identity follows from Theorem A.24. □

A.8 The regularized shared adaptive oracle and Top-Two rule

The algorithm does not know θ, so it cannot directly use $\rho^\dagger$. We use a vanishing regularization that makes the empirical oracle unique at every round while converging to the canonical exact optimum.

For $\varepsilon \in [0, R]$, define the deterministic information-separation function

$$D(\varepsilon) := \min_{\substack{q \in Q,\ u, v \in I \\ |u - v| \geq \varepsilon}} d_q(u, v), \qquad R := \theta_+ - \theta_-. \tag{70}$$

Then $D(0) = 0$ because the feasible set contains $u = v$, and later we prove $D(\varepsilon) > 0$ for every $\varepsilon > 0$ as well as $D(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$. Set

$$a_n := n^{-1/8}, \tag{71}$$

and define the deterministic radius

$$r_n := \sup\left(\{\varepsilon \in [0, R] : D(\varepsilon) \le 2a_n\} \cup \{0\}\right). \tag{72}$$

Finally set

$$\lambda_n^{\mathrm{reg}} := \sqrt{r_n + n^{-1}}. \tag{73}$$

For a vector v, let $b(v)$ be a deterministic empirical-best tie-breaking rule. When $b(v)$ is fixed, define

$$F_v^{b(v)}(\rho) := \inf_{u \in \mathrm{Alt}_{b(v)}} \mathcal{I}(v, u; \rho), \qquad \rho \in \mathcal{D}_\beta^{b(v)}.$$

Definition A.28 (Vanishingly regularized shared oracle)**.** At round n, for $v = \widehat{\theta}(n)$, the oracle returns

$$\rho_n := \arg\max_{\rho \in \mathcal{D}_\beta^{b(v)}} \left[F_v^{b(v)}(\rho) - \lambda_n^{\mathrm{reg}} \mathcal{R}(\rho)\right]. \tag{74}$$

Write

$$m_{n,k} := \rho_{n,k}(Q).$$

If $m_{n,k} > 0$, define

$$\lambda_{n,k} := \frac{\rho_{n,k}}{m_{n,k}} \in \mathcal{P}(Q).$$

If $m_{n,k} = 0$, use any fixed fallback probability measure $\lambda_0 \in \mathcal{P}(Q)$.

Lemma A.29 (The empirical regularized oracle is unique)**.** *For every n, the maximizer in* (74) *exists and is unique.*

Proof. The feasible design set is compact and convex. The empirical information objective is concave in ρ. By Lemma A.25, $-\lambda_n^{\mathrm{reg}}\mathcal{R}$ is strictly concave because $\lambda_n^{\mathrm{reg}} > 0$. Therefore the sum is strictly concave, continuous, and has a unique maximizer. □

Definition A.30 (Forced-arm regularized-shared β-EB-TC/TCI)**.** Fix an exploration exponent

$$\gamma \in (1/2, 1).$$

Initialize by sampling each arm once using any threshold drawn from a fixed probability measure on Q with full support. At round $n + 1$:

1. Compute $\widehat{\theta}(n)$ and set
$$\widehat{b}_n := \arg\max_{k \in [K]} \widehat{\theta}_k(n)$$
with deterministic tie-breaking.

2. Compute the shared regularized design ρ_n from (74). Write $m_{n,k} := \rho_{n,k}(Q)$. Fix once and for all a probability measure λ° on Q with full support, and define
$$\lambda_{n,k} := \begin{cases} \rho_{n,k}/m_{n,k}, & m_{n,k} > 0, \\ \lambda^\circ, & m_{n,k} = 0. \end{cases}$$
The second line is only a safeguard for early rounds; by Corollary A.43, eventually every $m_{n,k}$ is bounded away from zero.

3. Define the forced-arm set
$$\mathcal{U}_n^{\mathrm{F}} := \{k \in [K] : N_{n,k} < (n+1)^\gamma\}.$$
If $\mathcal{U}_n^{\mathrm{F}} \neq \varnothing$, choose
$$I_{n+1} \in \arg\min_{k \in \mathcal{U}_n^{\mathrm{F}}} N_{n,k}$$
with deterministic tie-breaking. This is called a *forced arm round.*

4. If $\mathcal{U}_n^{\mathrm{F}} = \varnothing$, choose the challenger

$$C_{n+1}^{\mathrm{TC}} \in \arg\min_{j \neq \widehat{b}_n} W_n(\widehat{b}_n, j), \tag{75}$$

$$C_{n+1}^{\mathrm{TCI}} \in \arg\min_{j \neq \widehat{b}_n} \left[W_n(\widehat{b}_n, j) + \log N_{n,j} \right], \tag{76}$$

and draw the Top-Two arm

$$I_{n+1} = \begin{cases} \widehat{b}_n, & \text{with probability } \beta, \\ C_{n+1}, & \text{with probability } 1-\beta. \end{cases}$$

5. In *both* forced and regular rounds, conditional on $I_{n+1} = k$, draw

$$q_{n+1} \sim \lambda_{n,k},$$

then observe $Y_{n+1} = \mathbf{1}\{X_{n+1} \geq q_{n+1}\}$.

Remark A.31 (Why the force acts only on the arm)**.** The two-arm one-bit source uses forced observations at a fixed identifying threshold. In the present canonical exponential-family model on compact Q, Lemma A.32 proves the stronger fact that *every* admissible threshold is uniformly informative away from the truth. Hence it is enough to force only the arm count. Keeping $q_{n+1} \sim \lambda_{n,I_{n+1}}$ even on forced arm rounds preserves the threshold-tracking argument exactly and avoids inserting a fixed-threshold atom into the asymptotic query law. The forced arm choices have density zero and therefore do not alter the limiting shared characteristic value.

A.9 Sublinear forced-arm exploration

The exploration argument now uses only the unique-best assumption. The force is on the arm coordinate; threshold selection remains entirely controlled by the shared oracle.

A.9.1 Uniform informativeness

Lemma A.32 (Uniform informativeness and small-radius behavior)**.** *The function $D : [0, R] \to [0, \infty)$ satisfies*

$$D(0) = 0, \qquad D(\varepsilon) > 0 \quad (\varepsilon > 0),$$

is nondecreasing, and

$$D(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0). \tag{77}$$

Proof. The identity $D(0) = 0$ follows by choosing $u = v$. For $\varepsilon > 0$, the constraint set $\{(q, u, v) : q \in Q,\ u, v \in I,\ |u - v| \geq \varepsilon\}$ is compact. The function $d_q(u, v)$ is continuous. By Lemma A.17, $u \neq v$ implies $p_q(u) \neq p_q(v)$ for every q, hence $d_q(u, v) > 0$. A continuous positive function on a compact set has a positive minimum. Monotonicity follows because increasing ε shrinks the feasible set.

To prove (77), fix an interior point $u_0 \in (\theta_-, \theta_+)$ and any $q_0 \in Q$. For all sufficiently small $\varepsilon > 0$, $u_0 + \varepsilon \in I$, hence

$$0 \leq D(\varepsilon) \leq d_{q_0}(u_0, u_0 + \varepsilon) \longrightarrow 0$$

by continuity of the one-bit divergence. □

Corollary A.33 (The deterministic radii vanish)**.** *The sequence r_n in* (72) *satisfies*

$$r_n \to 0, \qquad \lambda_n^{\mathrm{reg}} \to 0.$$

Proof. Fix $\varepsilon > 0$. Since $D(\varepsilon) > 0$ and $a_n \to 0$, eventually $2a_n < D(\varepsilon)$. By monotonicity of D, no radius at least ε belongs to the set defining r_n, hence eventually $r_n < \varepsilon$. Lemma A.32 also gives $D(\eta) \to 0$ as $\eta \downarrow 0$, so for every fixed n the set in (72) contains positive radii; thus r_n is a genuine (possibly very small) positive radius rather than an eventual exact-equality requirement. Finally $\lambda_n^{\mathrm{reg}} = \sqrt{r_n + n^{-1}} \to 0$. □

A.9.2 Uniform adaptive-threshold likelihood concentration

For arm k, define

$$M_{n,k}(u) := [\ell_{n,k}(\theta_k) - \ell_{n,k}(u)] - \sum_{t \le n : I_t = k} d_{q_t}(\theta_k, u). \tag{78}$$

Lemma A.34 (Uniform martingale envelope)**.** *There exists a nonnegative random variable $\mathcal{W}$ such that*

$$\mathbb{E}_\theta[\mathcal{W}^r] < \infty \qquad \forall r > 0,$$

and, almost surely, for all n, k, u,

$$|M_{n,k}(u)| \le \mathcal{W}\sqrt{(N_{n,k}+1)\log(e+N_{n,k})}. \tag{79}$$

Proof. Index observations of a fixed arm by its local pull count. Let $M_{k,m}(u)$ denote (78) after exactly m pulls of arm k. For fixed u, $M_{k,m}(u)$ is a martingale in m: the threshold used at the mth local pull is predictable, and subtracting $d_q(\theta_k, u)$ centers the corresponding log-likelihood ratio. Assumption A.7 implies deterministic constants $B_0, L_0 < \infty$ such that the centered increments are bounded by B_0 and $u \mapsto M_{k,m}(u)$ is $L_0 m$-Lipschitz.

For every $m \ge 1$, let $\mathcal{G}_m$ be a grid of I of mesh m^{-2}. Since I is compact, $|\mathcal{G}_m| \le C_I(1+m^2)$. Azuma–Hoeffding gives, for any $x \ge 1$,

$$\mathbb{P}\left(\max_{u \in \mathcal{G}_m} |M_{k,m}(u)| > x\sqrt{m\log(e+m)}\right) \le 2C_I(1+m^2)(e+m)^{-x^2/(2B_0^2)}.$$

If $\pi_m(u)$ is a nearest grid point, then $|M_{k,m}(u) - M_{k,m}(\pi_m(u))| \le L_0 m\, m^{-2} \le L_0$. A dyadic peeling over m, with the polynomial grid cardinality absorbed by the $\log(e+m)$ factor, and a union bound over the finite arm set therefore yield constants $C, c, x_0 > 0$ such that the all-time normalized supremum $\mathcal{W}$ obeys

$$\mathbb{P}(\mathcal{W} > x) \le Ce^{-cx^2}, \qquad x \ge x_0.$$

This Gaussian tail implies $\mathbb{E}[\mathcal{W}^r] < \infty$ for every fixed $r > 0$ and proves (79). □

A.9.3 Deterministic exploration supplied by the force

Let

$$F_n := \sum_{t=1}^{n} \mathbf{1}\{\text{round } t \text{ is a forced arm round}\}.$$

Lemma A.35 (Forced arm counts)**.** *For the rule in Definition A.30, there is a deterministic constant $C_{K,\gamma} < \infty$ such that, for all sufficiently large n,*

$$\min_{k \in [K]} N_{n,k} \ge n^\gamma - C_{K,\gamma}. \tag{80}$$

Moreover,

$$F_n \le K(n+1)^\gamma + C_{K,\gamma}, \qquad \frac{F_n}{n} \to 0. \tag{81}$$

Proof. Because $\gamma < 1$, the boundary $n \mapsto n^\gamma$ has increments tending to zero. Choose n_0 so large that

$$(n+K)^\gamma - n^\gamma < 1 \qquad \forall n \ge n_0.$$

Whenever the minimum count lies below the moving boundary, the rule selects an arm with minimum count. During any block of at most K forced selections every arm tied at the current minimum is raised at least once, whereas the boundary grows by less than one. A direct induction over such blocks gives (80) with a deterministic constant absorbing the finite prefix.

A forced pull of arm k can occur only while its count is below the boundary. Consequently the number of forced selections assigned to arm k by time n is at most $(n+1)^\gamma + O(1)$. Summing over k proves (81). □

Remark A.36 (No pairwise suboptimal gap enters exploration)**.** Lemma A.35 is deterministic and contains no quantity of the form $\min_{i \ne j} |\theta_i - \theta_j|$. Equal suboptimal arms are therefore harmless at the exploration stage. The only positive gap used later is $\Delta_\star = \min_{j \ne i^\star}(\theta_{i^\star} - \theta_j) > 0$ after the empirical leader has stabilized.

A.10 MLE consistency and a deterministic shrinking radius

Lemma A.37 (Arm-wise MLE consistency)**.** *If $N_{n,k} \to \infty$, then*

$$\widehat{\theta}_k(n) \to \theta_k \qquad \textit{almost surely}.$$

For every fixed $\varepsilon > 0$, the sample-count threshold after which $|\widehat{\theta}_k(n) - \theta_k| < \varepsilon$ has finite expectation.

Proof. Fix $\varepsilon > 0$. For every u with $|u - \theta_k| \geq \varepsilon$, Lemma A.32 gives $d_{q_t}(\theta_k, u) \geq D(\varepsilon)$ on every pull of arm k. Hence, uniformly over such u,

$$\begin{aligned}\ell_{n,k}(u) - \ell_{n,k}(\theta_k) &= -\sum_{t \leq n : I_t = k} d_{q_t}(\theta_k, u) - M_{n,k}(u) \\ &\leq -D(\varepsilon) N_{n,k} + \mathcal{W}\sqrt{(N_{n,k}+1)\log(e + N_{n,k})}.\end{aligned}$$

For sufficiently large arm sample count the right-hand side is strictly negative. Therefore no MLE can remain outside the ε-ball. Finite moments of the entry time follow from the finite moments of $\mathcal{W}$. □

Theorem A.38 (Shrinking deterministic MLE radius)**.** *Under Lemma A.35, there exists a random N_{rad} with moments of every fixed order such that*

$$n \geq N_{\mathrm{rad}} \quad \Longrightarrow \quad \|\widehat{\theta}(n) - \theta\|_\infty \leq r_n, \tag{82}$$

where r_n is defined in (72)*.*

Proof. Suppose for some arm k that $|\widehat{\theta}_k(n) - \theta_k| > r_n$. By definition of r_n and monotonicity of D, $D\Big(|\widehat{\theta}_k(n) - \theta_k|\Big) > 2a_n$. Since the MLE has likelihood at least that of the truth,

$$0 \geq \ell_{n,k}(\theta_k) - \ell_{n,k}(\widehat{\theta}_k(n)).$$

But the martingale decomposition gives

$$\begin{aligned}&\ell_{n,k}(\theta_k) - \ell_{n,k}(\widehat{\theta}_k(n)) \\ &\qquad \geq 2a_n N_{n,k} - \mathcal{W}\sqrt{(N_{n,k}+1)\log(e + N_{n,k})}.\end{aligned}$$

By Lemma A.35, $N_{n,k} \geq n^\gamma - C_{K,\gamma}$ eventually. Uniformly for such counts,

$$\frac{\mathcal{W}\sqrt{(N_{n,k}+1)\log(e + N_{n,k})}}{a_n N_{n,k}} \leq C\mathcal{W}\, n^{1/8 - \gamma/2}\sqrt{\log(e+n)} \longrightarrow 0, \qquad (\gamma > 1/2).$$

Thus the lower bound becomes strictly positive for all arms and all sufficiently large n, contradicting MLE optimality. The required random time is a polynomial-up-to-logs function of the finite-moment envelopes, so it has moments of every order. □

Corollary A.39 (Eventual correctness of the EB leader)**.** *There exists a random N_{lead} with moments of every fixed order such that*

$$n \geq N_{\mathrm{lead}} \quad \Longrightarrow \quad \widehat{b}_n = i^\star.$$

Proof. Since $r_n \to 0$, eventually $r_n < \frac{\Delta_\star}{4}$. Apply Theorem A.38. Then every empirical parameter is within $\Delta_\star/4$ of its truth, so the unique empirical-best arm is $i^\star$. □

Lemma A.40 (Positive best–challenger transportation separation)**.** *Let*

$$D_\star := D(\Delta_\star/2) > 0.$$

There exists a random $L_\star$ with finite expectation such that, for every $j \neq i^\star$, whenever $N_{n,i^\star} \geq L \geq L_\star$ and $N_{n,j} \geq L$,

$$W_n(i^\star, j) \geq \frac{D_\star}{2} L.$$

Proof. For any feasible pair alternative $u_{i^\star} \le u_j$, at least one coordinate differs from its truth by at least $(\theta_{i^\star} - \theta_j)/2 \ge \Delta_\star/2$. On that arm, say k, every threshold contributes at least $D_\star$ in expected one-bit KL. With $m = N_{n,k} \ge L$, Lemma A.34 gives the local-count bound

$$\ell_{n,k}(\widehat{\theta}_k) - \ell_{n,k}(u_k) \ge D_\star m - C\mathcal{W}\sqrt{(m+1)\log(e+m)}.$$

Choose the random $L_\star$ so that the final term is at most $D_\star m/2$ *for every* $m \ge L_\star$. The other arm contributes nonnegative likelihood loss because its MLE is unconstrained. Hence $W_n(i^\star, j) \ge D_\star m/2 \ge D_\star L/2$. The finite-moment assertion follows from those of $\mathcal{W}$ and the uniform local-count envelope. □

A.11 Convergence of the regularized shared oracle

We first quantify the dependence of the shared information objective on the first parameter argument.

Lemma A.41 (Uniform Lipschitz dependence of the one-bit divergence)**.** *There exists a deterministic $L_d < \infty$ such that*

$$|d_q(v,u) - d_q(v',u)| \le L_d|v - v'|$$

for all $q \in Q$ and $u, v, v' \in I$. Consequently, as long as v and v' have the same best arm b,

$$\sup_{\rho \in \mathcal{D}_\beta^b} |F_v(\rho) - F_{v'}(\rho)| \le L_d\|v - v'\|_\infty. \tag{83}$$

Proof. For $a, r \in [\eta, 1-\eta]$,

$$\frac{\partial}{\partial a}\,\mathrm{kl}(a,r) = \log\frac{a(1-r)}{r(1-a)},$$

whose absolute value is at most $L_{\mathrm{kl}} := 2\log\frac{1-\eta_0}{\eta_0}$. Also,

$$|p_q(v) - p_q(v')| = |e^{\log p_q(v)} - e^{\log p_q(v')}| \le |\log p_q(v) - \log p_q(v')| \le G|v - v'|,$$

because the derivative of e^x is at most one on $(-\infty, 0]$. Thus one may take $L_d := L_{\mathrm{kl}}G$.

For every u and feasible ρ, $|\mathcal{I}(v,u;\rho) - \mathcal{I}(v',u;\rho)| \le L_d\|v - v'\|_\infty$ because the total design mass is one. Taking infima over the same alternative set preserves the bound. □

Theorem A.42 (Convergence of the regularized oracle to the canonical exact optimum)**.** *For the algorithm of Definition A.30,*

$$\rho_n \Longrightarrow \rho^\dagger \qquad \textit{almost surely}. \tag{84}$$

For every weak neighborhood $\mathcal{U}$ of $\rho^\dagger$, the random time after which $\rho_n \in \mathcal{U}$ forever has finite moments of every order.

Proof. After the finite-moment leader-stabilization time, $b(\widehat{\theta}(n)) = i^\star$, so the empirical and true optimization problems use the same compact feasible set $\mathcal{D}_\beta^{i^\star}$ and the same alternative set.

Define

$$\delta_n := \sup_{\rho \in \mathcal{D}_\beta^{i^\star}} |F_{\widehat{\theta}(n)}(\rho) - F_\theta(\rho)|.$$

By Lemma A.41 and Theorem A.38, $\delta_n \le L_d r_n$ eventually. Since $\lambda_n^{\mathrm{reg}} = \sqrt{r_n + n^{-1}}$, we have

$$\lambda_n^{\mathrm{reg}} \to 0, \qquad \frac{\delta_n}{\lambda_n^{\mathrm{reg}}} \le L_d\sqrt{r_n} \longrightarrow 0. \tag{85}$$

Take any subsequence along which ρ_n converges weakly to a cluster point $\bar{\rho}$; compactness guarantees such subsequences. For any exact true optimizer $\rho^\star \in \mathcal{O}_\beta(\theta)$, regularized optimality gives

$$F_{\widehat{\theta}(n)}(\rho_n) - \lambda_n^{\mathrm{reg}}\mathcal{R}(\rho_n) \ge F_{\widehat{\theta}(n)}(\rho^\star) - \lambda_n^{\mathrm{reg}}\mathcal{R}(\rho^\star).$$

Letting n tend along the subsequence and using $\delta_n \to 0$, $\lambda_n^{\mathrm{reg}} \to 0$, and continuity gives

$$F_\theta(\bar{\rho}) \ge F_\theta(\rho^\star) = \Gamma_\beta(\theta).$$

Hence $\bar{\rho} \in \mathcal{O}_\beta(\theta)$.

Now compare specifically with the canonical optimizer $\rho^\dagger$. Because

$$F_{\widehat{\theta}(n)}(\rho_n) \le \Gamma_\beta(\theta) + \delta_n$$

and

$$F_{\widehat{\theta}(n)}(\rho^\dagger) \ge \Gamma_\beta(\theta) - \delta_n,$$

regularized optimality implies

$$\mathcal{R}(\rho_n) \le \mathcal{R}(\rho^\dagger) + 2\frac{\delta_n}{\lambda_n^{\mathrm{reg}}}.$$

Therefore

$$\limsup_n \mathcal{R}(\rho_n) \le \mathcal{R}(\rho^\dagger).$$

Every cluster point belongs to $\mathcal{O}_\beta(\theta)$, and $\rho^\dagger$ is the unique minimizer of $\mathcal{R}$ over that set. Continuity of $\mathcal{R}$ therefore forces every cluster point to equal $\rho^\dagger$. Thus the entire sequence converges to $\rho^\dagger$.

It remains to justify the finite-moment entrance statement. Fix a weak neighborhood $\mathcal{U}$ of $\rho^\dagger$. Choose a weakly open set $\mathcal{V}$ with $\rho^\dagger \in \mathcal{V} \subseteq \mathcal{U}$, and set $C := \mathcal{D}_\beta^{i^\star} \setminus \mathcal{V}$. Since $\mathcal{D}_\beta^{i^\star}$ is weakly compact and $\mathcal{V}$ is open, C is compact. Choose $\eta > 0$ and split

$$C_{\mathrm{sub}} := \{\rho \in C : F_\theta(\rho) \le \Gamma_\beta(\theta) - \eta\}, \qquad C_{\mathrm{near}} := C \setminus C_{\mathrm{sub}}.$$

For sufficiently small η, compactness and uniqueness of the minimizer of $\mathcal{R}$ on $\mathcal{O}_\beta(\theta)$ imply that there exists $\Delta_R > 0$ with

$$\inf_{\rho \in C_{\mathrm{near}}} \mathcal{R}(\rho) \ge \mathcal{R}(\rho^\dagger) + \Delta_R.$$

Indeed, otherwise a sequence in C_{near} with vanishing regularizer gap would have a cluster point in $C \cap \mathcal{O}_\beta(\theta)$ that also minimizes $\mathcal{R}$, contradicting uniqueness.

Let $R_{\max} := \sup_{\rho \in \mathcal{D}_\beta^{i^\star}} \mathcal{R}(\rho) < \infty$. If $\delta_n \le \eta/4$ and $2\lambda_n^{\mathrm{reg}} R_{\max} \le \eta/4$, then every $\rho \in C_{\mathrm{sub}}$ has regularized empirical objective strictly below that of $\rho^\dagger$. On C_{near}, regularized optimality of a point in C (hence outside $\mathcal{V}$) would imply $\lambda_n^{\mathrm{reg}} \Delta_R \le 2\delta_n$, which is impossible once $2\delta_n/\lambda_n^{\mathrm{reg}} < \Delta_R$. Thus $\rho_n \in \mathcal{V} \subseteq \mathcal{U}$ whenever the leader is correct and these deterministic inequalities hold. By Theorem A.38, $\delta_n \le L_d r_n$ after a random time with moments of every fixed order. The remaining inequalities hold after deterministic times because $r_n \to 0$, $\lambda_n^{\mathrm{reg}} \to 0$, and $\delta_n/\lambda_n^{\mathrm{reg}} \to 0$. Taking the maximum with the finite-moment leader-stabilization time proves the claim. □

Corollary A.43 (Convergence of arm masses and threshold marginals)**.** *For every arm k,*

$$m_{n,k} \to w_k^\dagger.$$

Because $w_k^\dagger > 0$ for every arm,

$$\lambda_{n,k} \Longrightarrow \lambda_k^\dagger.$$

For every fixed weak accuracy level, the corresponding convergence time has finite moments of every order.

Proof. Arm masses are continuous linear functionals of the joint measure, so they converge under weak convergence on the disjoint union $[K] \times Q$. Lemma A.23 gives strictly positive limiting masses. Division by the positive convergent masses gives convergence of the normalized threshold marginals. □

A.12 Threshold occupation and empirical rate realization

Define the empirical threshold occupation measure

$$R_{n,k} := \frac{1}{n} \sum_{t=1}^{n} \mathbf{1}\{I_t = k\} \delta_{q_t}. \tag{86}$$

Then

$$R_{n,k}(Q) = \frac{N_{n,k}}{n}.$$

Recall that $\rho_k^\dagger$ is the kth component of the canonical optimal joint design, $w_k^\dagger = \rho_k^\dagger(Q) > 0$, and $\lambda_k^\dagger = \rho_k^\dagger / w_k^\dagger$ is its conditional threshold law. We track the *threshold-averaged KL function*

$$A_k^\dagger(x) := \int_Q d_q(\theta_k, x) \lambda_k^\dagger(dq),$$

not an additional sampling policy. This restates (67) at its point of use.

Lemma A.44 (Threshold-information tracking)**.** *For every arm k,*

$$\sup_{x \in I} \left| \frac{1}{N_{n,k}} \sum_{t \le n : I_t = k} d_{q_t}(\theta_k, x) - A_k^\dagger(x) \right| \longrightarrow 0 \qquad \textit{almost surely}. \tag{87}$$

For every fixed accuracy level, the corresponding convergence time has finite expectation.

Proof. Index the pulls of arm k by the local time $r = 1, 2, \ldots$, and write $Q_{k,r}$ for the threshold used at that pull. Let $\mathcal{H}_{k,r-1}$ be the information available immediately before $Q_{k,r}$ is generated and let $\Lambda_{k,r}$ be its conditional law. Except for the finite fallback prefix, $\Lambda_{k,r}$ equals the current oracle marginal $\lambda_{t-1,k}$ at the corresponding calendar time t.

By Corollary A.43, after a random calendar time with moments of every fixed order, $\lambda_{t-1,k} \Rightarrow \lambda_k^\dagger$. The family $\mathcal{F}_k := \{q \mapsto d_q(\theta_k, x) : x \in I\}$ is a compact subset of $C(Q)$ in the sup norm: joint continuity on the compact set $Q \times I$ implies uniform boundedness and equicontinuity, and the map $x \mapsto d_\cdot(\theta_k, x)$ is sup-norm continuous. Consequently weak convergence of the threshold laws is uniform over this family, namely

$$\sup_{x \in I} \left| \int d_q(\theta_k, x) \lambda_{t-1,k}(dq) - A_k^\dagger(x) \right| \longrightarrow 0. \tag{88}$$

Thus the Cesàro average of the conditional expectations along local pull times converges uniformly to $A_k^\dagger$; the finite fallback prefix contributes $o(1)$.

It remains to control threshold-sampling noise. Define, at local time m,

$$D_{k,m}(x) := \sum_{r=1}^{m} \left[d_{Q_{k,r}}(\theta_k, x) - \mathbb{E}\{ d_{Q_{k,r}}(\theta_k, x) \mid \mathcal{H}_{k,r-1} \} \right].$$

For fixed x, this is a martingale. The increments are uniformly bounded because d is continuous on a compact domain. Moreover $x \mapsto d_q(\theta_k, x)$ is uniformly Lipschitz: the candidate log-likelihood is G-Lipschitz by Assumption A.7, and taking its expectation under the true Bernoulli law preserves that bound. Hence $D_{k,m}$ is Cm-Lipschitz in x for a deterministic C.

Fix $\xi > 0$ and take a deterministic grid of I with mesh at most $\xi/(4C)$. If $\sup_x |D_{k,m}(x)| > \xi m$, then at some grid point $|D_{k,m}| > \xi m/2$. Azuma–Hoeffding and a union bound over this fixed finite grid therefore give

$$\mathbb{P}\left(\sup_{x \in I} |D_{k,m}(x)| > \xi m \right) \le C_\xi e^{-c_\xi m}.$$

A further union bound over all $m \ge m_0$ yields

$$\mathbb{P}\left(\exists m \ge m_0 : \sup_x |D_{k,m}(x)| > \xi m \right) \le C'_\xi e^{-c'_\xi m_0}. \tag{89}$$

Therefore the martingale average converges uniformly almost surely, with an exponentially tailed last-exit local sample size.

Finally Lemma A.35 gives $N_{n,k} \ge n^\gamma - C_{K,\gamma}$ eventually. Equation (89) then yields a calendar last-exit tail of order e^{-cn^γ}. The conditional-expectation part in (88) has a finite-moment entrance time, and the number of local pulls from its finite prefix is at most that entrance time; dividing this prefix by $N_{n,k} \gtrsim n^\gamma$ shows that its fixed-accuracy calendar entrance has finite expectation as well. Combining the two parts proves both assertions. □

Theorem A.45 (Empirical GLR rate realization)**.** *For every challenger $j \ne i^\star$,*

$$\frac{1}{n} W_n(i^\star, j) - H_j^\dagger\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n} \right) \longrightarrow 0 \qquad \textit{almost surely}, \tag{90}$$

uniformly over the finite challenger set. For every $\zeta > 0$, there exists a random $N_{\mathrm{rate}}(\zeta)$ with finite expectation such that, for all $n \ge N_{\mathrm{rate}}(\zeta)$ and all $j \ne i^\star$,

$$\left| \frac{1}{n} W_n(i^\star, j) - H_j^\dagger\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n} \right) \right| \le \zeta. \tag{91}$$

Proof. For a pair $(i^\star, j)$ and candidate (u_i, u_j),

$$\begin{aligned}&\ell_{n,i^\star}(\widehat{\theta}_{i^\star}) - \ell_{n,i^\star}(u_i) + \ell_{n,j}(\widehat{\theta}_j) - \ell_{n,j}(u_j)\\ &= \sum_{k\in\{i^\star,j\}} [\ell_{n,k}(\theta_k) - \ell_{n,k}(u_k)] + \sum_{k\in\{i^\star,j\}} \left[\ell_{n,k}(\widehat{\theta}_k(n)) - \ell_{n,k}(\theta_k)\right].\end{aligned}$$

The first sum equals

$$\sum_{k\in\{i^\star,j\}} \sum_{t:I_t=k} d_{q_t}(\theta_k, u_k) + o(n)$$

uniformly over (u_i, u_j) by Lemma A.34. The second sum is $o(n)$ because

$$0 \le \ell_{n,k}(\widehat{\theta}_k(n)) - \ell_{n,k}(\theta_k) \le GN_{n,k}|\widehat{\theta}_k - \theta_k| = o(n).$$

Thus, uniformly over the constrained alternative set,

$$\begin{aligned}\frac{1}{n}W_n(i^\star, j) = \inf_{u_i\le u_j} \Bigg[&\frac{N_{n,i^\star}}{n}\frac{1}{N_{n,i^\star}} \sum_{t:I_t=i^\star} d_{q_t}(\theta_{i^\star}, u_i)\\ &+ \frac{N_{n,j}}{n}\frac{1}{N_{n,j}} \sum_{t:I_t=j} d_{q_t}(\theta_j, u_j)\Bigg] + o(1).\end{aligned}$$

Lemma A.44 replaces the two empirical threshold averages by $A^\dagger_{i^\star}(u_i)$ and $A^\dagger_j(u_j)$ uniformly. Lemma A.19 then reduces the constrained infimum to a common scalar x, producing exactly

$$H^\dagger_j\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n}\right).$$

The fixed-accuracy finite-mean statement follows from the finite-mean convergence times used in the preceding steps. □

A.13 Oversampled challengers are eventually rejected

Define

$$\psi_{n,k} := \mathbb{P}_\theta(I_n = k \mid \mathcal{F}_{n-1}), \qquad \Psi_{n,k} := \sum_{t=1}^{n} \psi_{t,k}.$$

Lemma A.46 (Sampling-count concentration)**.** *There exists a nonnegative random variable $\mathcal{W}_N$ with moments of every order such that for every n and k,*

$$|N_{n,k} - \Psi_{n,k}| \le \mathcal{W}_N\sqrt{(n+1)\log(e+n)}.$$

Proof. For fixed k,

$$N_{n,k} - \Psi_{n,k} = \sum_{t=1}^{n} [\mathbf{1}\{I_t = k\} - \psi_{t,k}]$$

is a martingale with increments bounded by one. Azuma–Hoeffding, a union bound over the finite arm set, and the same peeling argument as in Lemma A.34 produce the stated all-time envelope and finite moments. □

Lemma A.47 (Best-arm proportion converges to beta)**.** *For every $\varepsilon > 0$, there exists a random $N_\beta(\varepsilon)$ with finite expectation such that*

$$n \ge N_\beta(\varepsilon) \quad\Longrightarrow\quad \left|\frac{N_{n,i^\star}}{n} - \beta\right| \le \varepsilon.$$

The same conclusion holds with $N_{n,i^\star}$ replaced by $\Psi_{n,i^\star}$.

Proof. After the finite-mean time in Corollary A.39, the leader on every regular Top-Two round is $i^\star$. Let F_n be the number of forced arm rounds and let $F_{n,i^\star}$ be those forced rounds assigned to the best arm. Up to the finite prefix, $\Psi_{n,i^\star} = \beta(n - F_n) + F_{n,i^\star} + O(1)$. By Lemma A.35, $0 \le F_{n,i^\star} \le F_n = O(n^\gamma) = o(n)$, so $\Psi_{n,i^\star}/n \to \beta$. The deterministic $O(n^\gamma)$ perturbation has a deterministic fixed-accuracy entrance time, and Lemma A.46 transfers the conclusion to realized counts with finite-mean fixed-accuracy times. □

Theorem A.48 (Oversampled-challenger rejection for TC)**.** *Assume $K \geq 3$; for $K = 2$ the single challenger has asymptotic mass $1 - \beta$ and the oversampling statement is trivial. There exists $\varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0]$ there is a random $N_{\mathrm{rej}}(\varepsilon)$ with finite expectation satisfying: for every $n \geq N_{\mathrm{rej}}(\varepsilon)$ and every $j \neq i^\star$,*

$$\frac{\Psi_{n,j}}{n} \geq w_j^\dagger + \varepsilon \quad \Longrightarrow \quad C_{n+1}^{\mathrm{TC}} \neq j \quad \textit{whenever round } n+1 \textit{ is regular and } \widehat{b}_n = i^\star. \tag{92}$$

Proof. Fix small $\varepsilon > 0$. By Proposition A.21 and the canonical equalization identity (69),

$$H_j^\dagger\left(\beta, w_j^\dagger + \frac{\varepsilon}{2}\right) > \Gamma_\beta(\theta) \qquad \forall j \neq i^\star.$$

Since there are finitely many challengers,

$$g_\varepsilon := \min_{j \neq i^\star}\left[H_j^\dagger\left(\beta, w_j^\dagger + \frac{\varepsilon}{2}\right) - \Gamma_\beta(\theta)\right] > 0. \tag{93}$$

By continuity of the finitely many functions $H_j^\dagger$, choose $\delta > 0$, small compared with $\varepsilon/(K-1)$, such that

$$|a - \beta| \leq \delta, \quad w \geq w_j^\dagger + \frac{\varepsilon}{2} \Longrightarrow H_j^\dagger(a, w) \geq \Gamma_\beta(\theta) + \frac{3g_\varepsilon}{4}, \tag{94}$$

$$|a - \beta| \leq \delta, \quad w \leq w_\ell^\dagger + \delta \Longrightarrow H_\ell^\dagger(a, w) \leq \Gamma_\beta(\theta) + \frac{g_\varepsilon}{4}. \tag{95}$$

Suppose $\frac{\Psi_{n,j}}{n} \geq w_j^\dagger + \varepsilon$. By Lemma A.46, after a finite-mean random time,

$$\frac{N_{n,j}}{n} \geq w_j^\dagger + \frac{3\varepsilon}{4} \geq w_j^\dagger + \frac{\varepsilon}{2}.$$

By Lemma A.47,

$$\left|\frac{N_{n,i^\star}}{n} - \beta\right| \leq \delta.$$

There must exist another challenger $\ell \neq i^\star, j$ such that

$$\frac{N_{n,\ell}}{n} \leq w_\ell^\dagger + \delta. \tag{96}$$

Indeed, if every $\ell \neq i^\star, j$ satisfied $N_{n,\ell}/n > w_\ell^\dagger + \delta$, then (A.13) and $\sum_{r \neq i^\star} w_r^\dagger = 1 - \beta$ would give

$$\begin{aligned} 1 - \frac{N_{n,i^\star}}{n} &= \sum_{r \neq i^\star} \frac{N_{n,r}}{n} \\ &> (1 - \beta) + \frac{\varepsilon}{2} + (K-2)\delta, \end{aligned}$$

whereas $N_{n,i^\star}/n \geq \beta - \delta$ from (A.13) implies $1 - \frac{N_{n,i^\star}}{n} \leq 1 - \beta + \delta$. This is impossible as soon as, for example, $\delta < \frac{\varepsilon}{2(K-1)}$. Thus a comparator $\ell \neq j$ exists.

Equations (94)–(96) give

$$\begin{aligned} H_j^\dagger\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n}\right) &\geq \Gamma_\beta(\theta) + \frac{3g_\varepsilon}{4}, \\ H_\ell^\dagger\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,\ell}}{n}\right) &\leq \Gamma_\beta(\theta) + \frac{g_\varepsilon}{4}. \end{aligned}$$

Use Theorem A.45 with error at most $g_\varepsilon/8$. Then eventually

$$\frac{1}{n}\left[W_n(i^\star, j) - W_n(i^\star, \ell)\right] \geq \frac{g_\varepsilon}{4} > 0.$$

Therefore j cannot minimize the TC score. □

Theorem A.49 (Oversampled-challenger rejection for TCI)**.** *Under the same assumptions and range of ε as Theorem A.48,*

$$\frac{\Psi_{n,j}}{n} \geq w_j^\dagger + \varepsilon \quad \Longrightarrow \quad C_{n+1}^{\mathrm{TCI}} \neq j$$

eventually on regular Top-Two rounds whenever the leader is $i^\star$, with a finite-mean convergence time.

Proof. The TC proof gives a comparator ℓ and a constant $c_\varepsilon > 0$ such that $W_n(i^\star, j) - W_n(i^\star, \ell) \geq c_\varepsilon n$ eventually. For the TCI scores,

$$\begin{aligned}&[W_n(i^\star, j) + \log N_{n,j}] - [W_n(i^\star, \ell) + \log N_{n,\ell}] \\ &\quad \geq c_\varepsilon n - \log n,\end{aligned}$$

because initialization ensures $1 \leq N_{n,k} \leq n$. The right-hand side is positive for sufficiently large n, so j cannot minimize the TCI score. □

A.14 Convergence of the arm allocation

Theorem A.50 (Finite-mean convergence to the canonical shared allocation)**.** *For every $\varepsilon > 0$, there exists a random T_β^ε with finite expectation such that*

$$n \geq T_\beta^\varepsilon \quad \Longrightarrow \quad \max_{k \in [K]} \left| \frac{N_{n,k}}{n} - w_k^\dagger \right| \leq \varepsilon. \tag{97}$$

Consequently,

$$\frac{N_{n,k}}{n} \to w_k^\dagger \qquad \forall k.$$

Proof. The best-arm coordinate follows from Lemma A.47. If $K = 2$, then $\Psi_{n,j}/n = 1 - \Psi_{n,i^\star}/n$ for the unique challenger j, so Lemma A.47 directly gives $\Psi_{n,j}/n \to 1 - \beta = w_j^\dagger$ with a finite-mean fixed-accuracy entrance time; Lemma A.46 then transfers the conclusion to $N_{n,j}/n$. Hence assume $K \geq 3$ below.

We prove the challenger coordinates through a last-crossing argument that explicitly keeps the forced rounds. Fix a target accuracy $\varepsilon > 0$. Choose

$$0 < \eta < \min\left\{\varepsilon_0, \frac{\varepsilon}{8K}\right\},$$

where ε_0 is the range in Theorems A.48–A.49. Let T_0 be a finite-mean random time after which (i) the leader on every regular round is $i^\star$, (ii) the oversampled-challenger rejection property holds simultaneously for all challengers at tolerance η, and (iii) the fixed-accuracy best-arm predictable proportion bound $\left|\frac{\Psi_{n,i^\star}}{n} - \beta\right| \leq \eta$ holds. Such a T_0 exists by Corollary A.39, Theorems A.48–A.49, and Lemma A.47.

Fix a challenger $j \neq i^\star$. Consider any $n \geq T_0$. If there exists a last time $s \in [T_0, n]$ such that $\frac{\Psi_{s,j}}{s} \leq w_j^\dagger + \eta$, then for every $t = s+1, \ldots, n$ for which the ratio remains above $w_j^\dagger + \eta$, Theorems A.48–A.49 forbid j from being selected on a regular round. At the single crossing round immediately after s, the predictable mass of j can increase by at most one before the oversampling condition becomes active. Thereafter, as long as the ratio stays above $w_j^\dagger + \eta$, j can receive predictable mass only on forced rounds. Therefore

$$\Psi_{n,j} \leq \Psi_{s,j} + 1 + F_n - F_s \leq s(w_j^\dagger + \eta) + 1 + F_n - F_s.$$

Consequently

$$\frac{\Psi_{n,j}}{n} \leq w_j^\dagger + \eta + \frac{1 + F_n}{n}. \tag{98}$$

If no such s exists after T_0, then $\Psi_{t,j}/t > w_j^\dagger + \eta$ throughout the interval $[T_0, n]$. Consequently the rejection property prevents any regular-round service to j after T_0; only forced rounds can increase its predictable count. Hence

$$\Psi_{n,j} \leq \Psi_{T_0,j} + F_n - F_{T_0} \leq T_0 + F_n.$$

The random prefix T_0 must be accounted for explicitly: merely requiring $n \geq T_0$ does not make $\Psi_{T_0,j}/n$ small. Choose a *deterministic* integer $n_{\mathrm{F}}(\eta)$ such that $F_n/n \leq \eta$ and $(1 + F_n)/n \leq \eta$ for every $n \geq n_{\mathrm{F}}(\eta)$; it exists by Lemma A.35. Define the common random entrance

$$T_{\mathrm{up}}(\eta) := \max\left\{n_{\mathrm{F}}(\eta),\, T_0,\, \lceil T_0/\eta \rceil\right\}.$$

For all $n \geq T_{\mathrm{up}}(\eta)$, the last-crossing case obeys (98) and the no-crossing case satisfies $\Psi_{n,j}/n \leq T_0/n + F_n/n \leq 2\eta$. Since $w_j^\dagger \geq 0$, both cases therefore give, simultaneously for all challengers,

$$\frac{\Psi_{n,j}}{n} \leq w_j^\dagger + 2\eta, \qquad n \geq T_{\mathrm{up}}(\eta). \tag{99}$$

Crucially, the entrance time still has finite expectation for each *fixed* $\eta > 0$: indeed

$$\mathbb{E}T_{\mathrm{up}}(\eta) \leq n_{\mathrm{F}}(\eta) + (1 + \eta^{-1})\mathbb{E}T_0 + 1 < \infty.$$

The factor η^{-1} is harmless here because this theorem asserts finite-mean entrance at a fixed accuracy; no uniform-in-η entrance bound is claimed.

For the lower bound, use $\sum_{k=1}^K \frac{\Psi_{n,k}}{n} = 1$. Combining the best-arm upper bound $\Psi_{n,i^\star}/n \leq \beta + \eta$ with (99) for all challengers except j gives

$$\begin{aligned}\frac{\Psi_{n,j}}{n} &\geq 1 - (\beta + \eta) - \sum_{\ell \neq i^\star, j} (w_\ell^\dagger + 2\eta) \\ &= w_j^\dagger - (2K - 3)\eta.\end{aligned}$$

Our choice of η makes both the upper and lower predictable-proportion errors at most $\varepsilon/2$ for all arms after a finite-mean time.

Finally Lemma A.46 implies $\max_k \left|\frac{N_{n,k} - \Psi_{n,k}}{n}\right| \to 0$ with a finite-mean fixed-accuracy entrance time (indeed the envelope has moments of every order). After enlarging the preceding random time once more, we obtain

$$\max_{k \in [K]} \left| \frac{N_{n,k}}{n} - w_k^\dagger \right| \leq \varepsilon,$$

which proves (97). □

A.15 Achievability at the shared β-characteristic rate

Lemma A.51 (Linear growth of the correct stopping statistic)**.** *For every $\zeta > 0$, there exists a random $T_Z(\zeta)$ with finite expectation such that*

$$n \geq T_Z(\zeta) \quad \Longrightarrow \quad Z_{i^\star}(n) \geq n\left[\Gamma_\beta(\theta) - \zeta\right].$$

Proof. Theorem A.50 gives $\frac{N_{n,k}}{n} \to w_k^\dagger$. Theorem A.45 gives

$$\frac{1}{n} W_n(i^\star, j) - H_j^\dagger\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n}\right) \to 0$$

uniformly over challengers. Continuity and canonical equalization yield

$$\frac{1}{n} W_n(i^\star, j) \to H_j^\dagger(\beta, w_j^\dagger) = \Gamma_\beta(\theta)$$

uniformly over the finite challenger set. Lemma A.8 then gives $\frac{1}{n} Z_{i^\star}(n) \to \Gamma_\beta(\theta)$. All fixed-accuracy entry times used here have finite expectation. □

Theorem A.52 (Achievability at the shared adaptive β-characteristic rate)**.** *Under Assumptions A.6 and A.7, the forced-arm regularized-shared β-EB-TC and β-EB-TCI algorithms with the stopping rule* (46) *are δ-correct, stop almost surely, and satisfy*

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}_\theta[\tau_\delta]}{L_\delta} \leq T_\beta^\star(\theta) = \frac{1}{\Gamma_\beta(\theta)}. \tag{100}$$

Proof. Finite-time δ-correctness is Theorem 3. Put

$$\Gamma := \Gamma_\beta(\theta).$$

Fix

$$0 < g < g_0 < \Gamma.$$

By Lemma A.51 and the finite-mean entrance results, there exists a random time T_{g_0} with

$$\mathbb{E}_\theta[T_{g_0}] < \infty$$

such that, for every $n \geq T_{g_0}$, the empirical leader is $i^\star$ and

$$Z_{i^\star}(n) \geq g_0 n.$$

The stopping boundary has the form

$$c(n, \delta) = L_\delta + d_n, \qquad d_n = o(n),$$

and testing begins at $n_{\min} = K$. Lemma A.3 therefore gives

$$\mathbb{E}_\theta[\tau_\delta] \leq \mathbb{E}_\theta[T_{g_0}] + K + n_0(g, g_0) + \left\lceil \frac{L_\delta}{g} \right\rceil.$$

After division by L_δ and letting $\delta \downarrow 0$,

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}_\theta[\tau_\delta]}{L_\delta} \leq \frac{1}{g}.$$

Finally let $g \uparrow \Gamma$ to obtain

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}_\theta[\tau_\delta]}{L_\delta} \leq \frac{1}{\Gamma} = T^\star_\beta(\theta).$$

The same eventual linear evidence growth implies $\tau_\delta < \infty$ almost surely for every fixed δ. □

Remark A.53 (From achievability to optimality)**.** The theorem above gives the matching upper bound. The K-arm change-of-measure lower bound in Appendix A.1, combined with the stopped best-arm fraction result in Section A.17, gives the corresponding fixed-β lower bound and hence asymptotic fixed-β optimality.

A.16 Continuum design and numerical discretization

The theoretical query set is the compact continuum Q, and feasible designs are finite measures on $[K] \times Q$. A numerical implementation may replace Q by a threshold grid Q_M and the scalar alternative set by a finite grid I_L. Under joint continuity of $d_q(u, v)$ on the relevant compact sets, both discretization errors vanish as the grid meshes go to zero.

The approximation need not be one-sided: restricting the learner to Q_M may decrease the max–min value, whereas restricting Nature to I_L may increase it. Numerical validation should therefore report grid refinement rather than interpret one finite-grid value as a certified bound.

Equal suboptimal arms are allowed. The forced-arm safeguard contributes only $O(n^\gamma) = o(n)$ rounds, and on those rounds the algorithm retains the oracle's conditional threshold law. No uniqueness of the raw shared optimum is assumed: the kernel regularizer selects a unique canonical design, and the vanishingly regularized empirical oracle converges to it.

A pairwise relaxation (theoretical only). For a best arm $i^\star$ and challenger j, define

$$C_{i^\star,j}(\beta, w; \theta) := \max_{\lambda_\star, \lambda_j \in \mathcal{P}(Q)} \min_{x \in I} \left\{ \beta \int_Q d_q(\theta_{i^\star}, x) \lambda_\star(dq) + w \int_Q d_q(\theta_j, x) \lambda_j(dq) \right\},$$

and set

$$\Gamma^{\mathrm{pair}}_\beta(\theta) := \max_{\substack{w_j \geq 0 \\ \sum_{j \neq i^\star} w_j = 1 - \beta}} \min_{j \neq i^\star} C_{i^\star,j}(\beta, w_j; \theta).$$

Here each pair is allowed a different best-arm threshold distribution; this relaxes only the common-query constraint, while retaining the common best-arm mass β and total challenger budget $1 - \beta$. For any $q_0 \in Q$, let $\Gamma^{\mathrm{E}}_{\beta,q_0}$ be (11) restricted to designs $\rho_k = w_k \delta_{q_0}$ for every arm, with $w_{i^\star} = \beta$. The best *common fixed-threshold* information rate is $\Gamma^{\mathrm{fixed}}_\beta := \max_{q_0 \in Q} \Gamma^{\mathrm{E}}_{\beta,q_0}$. This is a theoretical feasible subclass, not a separate experiment.

Proposition A.54 (Fixed, feasible shared, and pairwise-relaxed designs)**.**

$$\Gamma^{\mathrm{fixed}}_\beta \leq \Gamma_\beta(\theta) \leq \Gamma^{\mathrm{pair}}_\beta(\theta).$$

Proof. The left inequality follows by restricting the feasible shared design in (11) to $\rho_k = w_k \delta_{q_0}$ for a common q_0 and then maximizing over q_0. For the right inequality, for any feasible joint design ρ, let $w_j = \rho_j(Q)$ and normalize its positive arm masses; choose arbitrary conditional query laws when a mass vanishes. For each challenger, this produces a candidate in $C_{i^\star,j}(\beta, w_j; \theta)$ with exactly the feasible design's pair information. Thus the latter cannot exceed the pairwise maximum. Take the minimum over challengers and then the maximum over the *one* shared design ρ; the resulting masses obey the budget above. Neither equality nor strictness is asserted. □

A.17 Matching lower bound for the adaptive-threshold algorithm

The preceding proof gives the matching limsup. For the liminf, Corollary A.39 supplies a permanent correct-leader entrance with finite mean; Lemma A.35 supplies $F_n = O(n^\gamma)$ with $\gamma < 1$. The expected stopping time is finite by the upper bound and tends to infinity by change of measure against any finite-KL strict wrong-best alternative. Hence the *shared* Lemma A.2 yields

$$\mathbb{E}_\theta N_{\tau_\delta, i^\star} / \mathbb{E}_\theta \tau_\delta \to \beta. \tag{101}$$

The corresponding strict/closed alternative values agree: any closed pair-order alternative can be perturbed to strict order with arbitrarily small change in the continuous d_q on compact $Q \times \Theta^2$. Any ties among the finitely many remaining challengers can be removed by arbitrarily small coordinate changes, lowering tied competitors at an upper endpoint if necessary. The shared pair decomposition (Lemma A.11) then gives

$$\inf_{u \in \mathrm{Alt}_{i^\star}} \mathcal{I}(\theta, u; \rho) = \min_{j \neq i^\star} V_{i^\star,j}(\theta; \rho_{i^\star}, \rho_j).$$

Maximizing over one shared $\rho \in \mathcal{D}_\beta^{i^\star}$ identifies the generic fixed-β value as $\Gamma_\beta(\theta)$. Combine (101) with Corollary A.1 to obtain

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\theta \tau_\delta}{\log(1/\delta)} \geq 1/\Gamma_\beta(\theta),$$

matching the established limsup.

B MODEL E: FINITE-CONFIDENCE RESULTS

The model-independent stopping-time inversion is Lemma A.3 in the common-tools subsection of the preceding Model-E asymptotic part. This part applies it to Model E and then proves the separate identifying-threshold companion correction.

B.1 Main Model-E rule

Fix $\gamma_0 < \Gamma_\beta(\theta)$. Lemma A.51 and Corollary A.39 give the integrable entrance $T_{\gamma_0}^{\mathrm{E}} := \max\{T_Z(\Gamma_\beta(\theta) - \gamma_0), N_{\mathrm{lead}}\}$, after which the empirical leader is $i^\star$ and $Z_{i^\star}(n) \geq \gamma_0 n$. Since $b_{\mathrm{E}}(n, \delta) = L_\delta + O(\log n)$, Corollary A.4 directly yields, for any $\varepsilon, \eta > 0$,

$$\mathbb{E}_\theta \tau_\delta^{\mathrm{E}} \leq (\Gamma_\beta(\theta)^{-1} + \varepsilon) L_\delta + C_\varepsilon,$$
$$\mathbb{P}_\theta\{\tau_\delta^{\mathrm{E}} \leq (\Gamma_\beta(\theta)^{-1} + \varepsilon) L_\delta + C_{\varepsilon,\eta}\} \geq 1 - \eta.$$

Here the constants depend on the Model-E entrance and the boundary, not on δ.

B.2 Explicit Model-E companion refinement

This section records the sharper bounds displayed in the main paper. The Model E result concerns a separate fixed-β global-design rule, not the Top-Two rule of the main asymptotic theorem; this distinction is essential.

Lemma B.1 (Uniform tail-quantile integration)**.** *Suppose a nonnegative random variable T and an increasing, locally absolutely continuous $q : [0, \infty) \to [0, \infty)$ satisfy $\mathbb{P}(T > q(x)) \leq e^{-x}$ for all $x \geq 0$. Then*

$$\mathbb{E}T \leq q(0) + \int_0^\infty e^{-x} q'(x)\, dx,$$

provided the integral is finite. In particular, if $q(x) = b + \sum_{r=1}^R a_r (h + x)^{p_r}$ for $h \geq 1$, $a_r \geq 0$, and fixed $p_r > 0$, then $\mathbb{E}T \leq b + C_{\boldsymbol{p}} \sum_r a_r h^{p_r}$.

Proof. The layer-cake identity and the monotone change of variables $y = q(x) - q(0)$ give $\mathbb{E}(T - q(0))_+ \leq \int_0^\infty e^{-x} q'(x)\, dx$; flat pieces follow by approximation. Finally $\int_0^\infty e^{-x}(h + x)^{p-1}\, dx = O_p(h^{p-1})$ for $h \geq 1$. □

B.3 Model E: explicit $2/3$ correction for a fixed-β companion rule

For this subsection write $\Theta := I = [\theta_-, \theta_+]$. Fix an identifying threshold $q_0 \in Q$ and assume

$$m_0 := \inf_{\theta \in \Theta} \partial_\theta p_{q_0}(\theta) > 0.$$

Initialize by pulling every arm once at q_0. Fix $\gamma \in (1/2, 1)$ and a fixed scheduling coefficient $a_0 > 0$. Before every action at calendar time $t+1$, if any arm satisfies $M^F_{k,0}(t) < a_0(t+1)^\gamma$, force the least-scheduled such arm at q_0 (deterministic tie-breaking). Otherwise run the global design rule below. Here $M^F_{k,0}(t)$ counts only forced identification pulls of arm k, not regular oracle pulls that happen to select q_0. This deterministic sublinear target schedule has, for all sufficiently large t,

$$\min_k M^F_{k,0}(t) = \Omega_{K,a_0}(t^\gamma), \qquad F(t) = \sum_k M^F_{k,0}(t) = O_{K,a_0}(t^\gamma).$$

The counting bounds follow from the following deterministic argument. Put $b_t = \lceil a_0(t+1)^\gamma \rceil$ and $D_t = \sum_k [b_t - M^F_{k,0}(t)]_+$. A forced round occurs whenever $D_t > 0$, and the next target can create at most $K(b_{t+1} - b_t)$ additional deficit. Consequently,

$$D_{t+1} \le D_t - \mathbf{1}\{D_t > 0\} + K(b_{t+1} - b_t).$$

Because $b_t = O(t^\gamma) = o(t)$, D_t cannot remain positive forever: otherwise summing the display from any fixed time would make its right-hand side eventually negative. Moreover, for all sufficiently large t, successive integer increases of b_t are separated by more than K rounds, since $a_0[(t+K+2)^\gamma - (t+1)^\gamma] \to 0$. After the first zero deficit beyond this deterministic transient, each new unit target increase creates at most K units of deficit, which the scheduler clears before the next target increase. Hence $D_t \le K$ eventually and $M^F_{k,0}(t) \ge b_t - K$. Each forced pull made at time $t+1$ leaves the updated arm count at most b_t; monotonicity of b_t therefore also gives $M^F_{k,0}(t) \le b_t + C_0$ for a fixed initialization constant C_0. Summing across arms proves the two count bounds. The practical code uses a separate finite coefficient and a discretized oracle.

After observing the first t rounds, and before drawing the action at time $t+1$, compute the MLE $\widehat{\theta}(t)$ and the deterministically tie-broken empirical best arm

$$\widehat{b}_t \in \arg\max_k \widehat{\theta}_k(t).$$

Let

$$\mathcal{D}^b_\beta := \{\rho \in \mathcal{P}([K] \times Q) : \rho_b(Q) = \beta\},$$

and define the fixed-β global value

$$C_\beta(v, b) := \sup_{\rho \in \mathcal{D}^b_\beta} \inf_{u \in \mathrm{Alt}_b} \mathcal{I}(v, u; \rho).$$

Choose a measurable ε_t-optimal design $\rho_t \in \mathcal{D}^{\widehat{b}_t}_\beta$ with $\varepsilon_t = t^{-1}$ and draw the next arm–threshold action from ρ_t using fresh internal randomness. Equivalently, on a regular calendar round s the predictable action distribution ρ_s is computed from $\widehat{\theta}(s-1)$ (not from the yet-unobserved round-s data). Such a selector exists by compactness and joint continuity of the design objective; equivalently, one may select deterministically from a countable dense family of designs using the first element meeting the ε_t-optimality criterion. At the true instance,

$$C_\beta(\theta, i^\star) = \Gamma^{\mathrm{E}}_\beta(\theta).$$

The scheduled rounds are sublinear, hence they do not change this first-order fixed-β information constant.

Use the same pairwise GLRs and boundary as the main Model E stopping rule:

$$\tau^{\mathrm{E,ft}}_\delta := \inf\{t \ge K : Z_{\widehat{b}_t}(t) \ge c_{\mathrm{E}}(t, \delta)\}, \qquad \widehat{i}_{\tau^{\mathrm{E,ft}}_\delta} := \widehat{b}_{\tau^{\mathrm{E,ft}}_\delta}.$$

Finite-time validity follows from the likelihood-covering theorem because it holds under arbitrary predictable arm–threshold choices. Finally define

$$T^\star_{\mathrm{E}} := \frac{1}{\Gamma^{\mathrm{E}}_\beta(\theta)}.$$

For the proofs below fix a numerical constant $C_K \ge 12$ depending only on the compact model and K, and set

$$H_{K,t}(\eta) := 1 + (2K+4)\log(e+t) + \log(C_K/\eta).$$

All probability allocations below can be covered by increasing this one constant C_K once; it never depends on t, δ or η.

Lemma B.2 (Time-uniform MLE localization for the companion rule)**.** *Under the compact one-bit regularity assumption and $m_0 = \inf_{\theta\in\Theta} \partial_\theta p_{q_0}(\theta) > 0$, there are deterministic constants $C_M, C_I < \infty$, independent of η, such that for each $\eta \in (0, 1/3)$, with probability at least $1-\eta$, simultaneously for all integers $t \geq 1$,*

$$\|\widehat{\theta}(t) - \theta\|_\infty \leq r_t(\eta) := C_M(t \vee 1)^{-\gamma/2}\sqrt{H_{K,t}(\eta)}.$$

If the true best arm is unique, one may choose the deterministic *localization time*

$$t_{\mathrm{id}}(\eta) = \left\lceil C_I[1 + \log(C_I/\eta)]^{1/\gamma} \right\rceil$$

so that the empirical leader is correct for every $t \geq t_{\mathrm{id}}(\eta)$ on the same event. The fixed constants also absorb the finite initialization/scheduling prefix.

Proof. For fixed $u \in \Theta^K$, let $\Lambda_t(u) = L_t(u) - L_t(\theta)$ and $\Delta_k = u_k - \theta_k$. At a forced q_0 pull, the conditional Bernoulli affinity obeys

$$\mathbb{E}_\theta[e^{(\Lambda_t(u)-\Lambda_{t-1}(u))/2} \mid \mathcal{F}_{t-1}] = A(p_{q_0}(\theta_k), p_{q_0}(u_k)) \leq e^{-c_H\Delta_k^2},$$

where $c_H > 0$ is deterministic: the derivative lower bound implies $|p_{q_0}(u_k) - p_{q_0}(\theta_k)| \geq m_0|\Delta_k|$, and the standard Bernoulli affinity inequality gives the final bound. At other pulls the affinity is at most one. Thus

$$\mathcal{E}_t(u) = \exp\left\{\tfrac{1}{2}\Lambda_t(u) + c_H\sum_k M_{k,0}^F(t)\Delta_k^2\right\}$$

is a nonnegative supermartingale starting from one. The forced schedule gives a deterministic t_0 and $c_F > 0$ with $\min_k M_{k,0}^F(t) \geq c_F t^\gamma$ for all $t \geq t_0$.

At a fixed $t \geq t_0$, choose a deterministic $(Gt)^{-1}$-net $\mathcal{N}_t \subset \Theta^K$ with $|\mathcal{N}_t| \leq (Ct)^K$, where the finite constant incorporates the compact parameter diameter. The complete log likelihood is Gt-Lipschitz, so a nearest net point v to the MLE satisfies $\Lambda_t(v) \geq -1$. If $\|\widehat{\theta}(t) - \theta\|_\infty > r$, then $\|v - \theta\|_\infty \geq r - (Gt)^{-1}$ and hence

$$\mathcal{E}_t(v) \geq \exp\{-\tfrac{1}{2} + c_H c_F t^\gamma[r - (Gt)^{-1}]_+^2\}.$$

Markov's inequality for each *fixed* net point, followed by a union bound over $\mathcal{N}_t$, shows that the probability of this MLE event is at most

$$(Ct)^K \exp\{\tfrac{1}{2} - c_H c_F t^\gamma[r - (Gt)^{-1}]_+^2\}.$$

The stated $r_t(\eta)$ makes this at most $6\eta/[\pi^2(t+1)^2]$ after increasing C_M and C_K once. Summing over integer times and absorbing the finite times $t < t_0$ into the radius establishes the simultaneous claim. In particular, no data-dependent net candidate is inserted into the martingale before the fixed-net union bound.

Let $\Delta_\star = \min_{j\neq i^\star}(\theta_{i^\star} - \theta_j) > 0$. It suffices to have $r_t(\eta) < \Delta_\star/4$ for all subsequent t. The function $t^{-\gamma/2}\sqrt{H_{K,t}(\eta)}$ is eventually decreasing, and for a fixed, sufficiently large C_I, $t_{\mathrm{id}}(\eta) = \lceil C_I[1 + \log(C_I/\eta)]^{1/\gamma}\rceil$ satisfies this inequality for all $t \geq t_{\mathrm{id}}(\eta)$, uniformly in η. Indeed t_{id}^γ is a sufficiently large multiple of $1 + \log(C_I/\eta)$, which also dominates the additional $\log t_{\mathrm{id}}$ term after the fixed multiplier is enlarged. This proves permanent correct leadership on the same event. □

Lemma B.3 (Fixed-β oracle transfer)**.** *Suppose v and θ have the same best arm $i^\star$. If $\rho \in \mathcal{D}_\beta^{i^\star}$ is ε-optimal for $C_\beta(v, i^\star)$, then*

$$\inf_{u\in \mathrm{Alt}_{i^\star}} \mathcal{I}(\theta, u; \rho) \geq \Gamma_\beta^{\mathrm{E}}(\theta) - 2L_d\|v - \theta\|_\infty - \varepsilon.$$

Proof. Assumption A.7 makes $p_q(\cdot)$ uniformly Lipschitz, since its log is uniformly Lipschitz and all its values lie in a compact subinterval of $(0, 1)$. Bernoulli KL is uniformly Lipschitz in its first probability argument on this compact square. The needed first-parameter Lipschitzness of d_q thus follows from the stated assumptions and gives

$$|\mathcal{I}(v, u; \rho) - \mathcal{I}(\theta, u; \rho)| \leq L_d\|v - \theta\|_\infty$$

uniformly over u and $\rho \in \mathcal{D}_\beta^{i^\star}$. Taking the infimum over alternatives and then the supremum over the same fixed feasible set shows

$$|C_\beta(v, i^\star) - C_\beta(\theta, i^\star)| \leq L_d\|v - \theta\|_\infty.$$

Combine this with ε-optimality and $C_\beta(\theta, i^\star) = \Gamma_\beta^{\mathrm{E}}(\theta)$. □

Theorem B.4 (Evidence lower envelope for the companion rule)**.** *There is a fixed $C_{\theta,K} < \infty$ such that, for every $\eta \in (0, 1/3)$, with probability at least $1 - 3\eta$, simultaneously for all integers $t \geq t_{\mathrm{id}}(\eta) + 1$,*

$$Z_{i^\star}(t) \geq t\Gamma^{\mathrm{E}}_{\beta}(\theta) - C_{\theta,K}\big[t^{\gamma} + t^{1-\gamma/2}\sqrt{H_{K,t}(\eta)} + \sqrt{tH_{K,t}(\eta)} + t_{\mathrm{id}}(\eta) + \log(e+t)\big].$$

The constant and the function $H_{K,t}$ are independent of δ and have no hidden dependence on η.

Proof. Use the infinite counterfactual companion trajectory, continued past any stopping decision, and set $\mathcal{F}_{s-1}$ to be the history *before* choosing round s. On its regular rounds the predictable design ρ_s is determined from $\widehat{\theta}(s-1)$. Let $\mathcal{R}_t$ comprise only regular rounds with $s - 1 \geq t_{\mathrm{id}}(\eta)$. On the localization event of Lemma B.2, the leader is then $i^\star$ and Lemma B.3 yields for each such s

$$\inf_{u \in \mathrm{Alt}_{i^\star}} \mathcal{I}(\theta, u; \rho_s) \geq \Gamma^{\mathrm{E}}_{\beta}(\theta) - 2L_d r_{s-1}(\eta) - (s-1)^{-1}.$$

For any fixed alternative u, write the round-s log LR as

$$D_s(u) = \log \frac{P^{q_s}_{\theta_{I_s}}(Y_s)}{P^{q_s}_{u_{I_s}}(Y_s)}, \qquad \bar{D}_s(u) = \mathbb{E}_\theta[D_s(u) \mid \mathcal{F}_{s-1}].$$

The fresh random action is included in this conditional expectation. On a regular round $\bar{D}_s(u) = \mathcal{I}(\theta, u; \rho_s)$; on *every* round excluded from $\mathcal{R}_t$ (including initial regular rounds with an incorrect leader and all forced rounds), it is a nonnegative KL divergence. Therefore

$$\begin{aligned}\inf_{u \in \mathrm{Alt}_{i^\star}} \sum_{s \leq t} \bar{D}_s(u) &\geq \inf_{u \in \mathrm{Alt}_{i^\star}} \sum_{s \in \mathcal{R}_t} \mathcal{I}(\theta, u; \rho_s) \\ &\geq \sum_{s \in \mathcal{R}_t} \inf_{u \in \mathrm{Alt}_{i^\star}} \mathcal{I}(\theta, u; \rho_s) \\ &\geq t\Gamma^{\mathrm{E}}_{\beta}(\theta) - C\big[t^{\gamma} + t_{\mathrm{id}}(\eta) + t^{1-\gamma/2}\sqrt{H_{K,t}(\eta)} + \log(e+t)\big].\end{aligned}$$

Here $|\mathcal{R}_t| \geq t - Ct^{\gamma} - t_{\mathrm{id}}(\eta) - 1$; $\sum_{s \in \mathcal{R}_t}(s-1)^{-1} \leq 1 + \log t$; and $\sum_{s \leq t} r_{s-1}(\eta)$, omitting any initial zero index, is bounded by $Ct^{1-\gamma/2}\sqrt{H_{K,t}(\eta)}$.

For the *same* fixed parameter alternative u, put $M_t(u) = \sum_{s \leq t}(D_s(u) - \bar{D}_s(u))$. Compactness and the interior Bernoulli-probability bounds make its increments uniformly bounded. Moreover, $u \mapsto M_t(u)$ is deterministically Ct-Lipschitz in the sup norm. For each *deterministic* calendar time t use a deterministic t^{-2}-net of the entire K-dimensional parameter box, containing at most $C(t+1)^{2K}$ points. Azuma–Hoeffding for each *fixed* net point, a failure allocation $6\eta/[\pi^2(t+1)^2]$, and the nearest-net interpolation error $O(t \cdot t^{-2})$ give the single, simultaneous event

$$\sup_{u \in \Theta^K} |M_t(u)| \leq C\sqrt{tH_{K,t}(\eta)} + C/t \quad \text{for every } t \geq 1$$

with probability at least $1 - \eta$, after increasing the fixed constants in $H_{K,t}$. The net may include parameters outside the alternative set, since we cover Θ^K first.

Finally, for $t \geq t_{\mathrm{id}}(\eta) + 1$,

$$Z_{i^\star}(t) \geq L_t(\theta) - \sup_{u \in \mathrm{Alt}_{i^\star}} L_t(u) = \inf_{u \in \mathrm{Alt}_{i^\star}} \sum_{s \leq t} D_s(u).$$

Combining the predictable-information and martingale events by a union bound proves the theorem (indeed their stated budgets total $2\eta \leq 3\eta$); the counterfactual prefix agrees with the actually observed prefix up to stopping. □

Balancing

$$t^{\gamma} \quad \text{and} \quad t^{1-\gamma/2}$$

gives

$$\gamma^\star = 2/3.$$

Define

$$A^{\mathrm{E}}_{\delta,\eta} := K\log(e + L_\delta) + \log(1/\eta) + 1.$$

Lemma B.5 (Uniform trial-time inversion for the E companion)**.** *Take $\gamma = 2/3$, $L = \log(1/\delta) \geq \log 2$ and $x = 1+\log(1/\eta)$. Put*

$$h_L = 1 + K\log(e+L), \quad A = h_L + x, \quad P = L^{2/3}\sqrt{A} + \sqrt{LA} + A + (1+x)^{3/2} + 1.$$

There is one fixed $D < \infty$, independent of L, x and hence of δ, η, such that

$$t = \lceil T_{\mathrm{E}}^{\star} L + DP \rceil$$

is beyond $t_{\mathrm{id}}(\eta)$, obeys $t \geq K$, and satisfies

$$t\Gamma_{\beta}^{\mathrm{E}}(\theta) - C_{\theta,K}\big[t^{2/3}\sqrt{H_{K,t}(\eta)} + \sqrt{tH_{K,t}(\eta)} + t_{\mathrm{id}}(\eta) + \log(e+t)\big] \geq c_{\mathrm{E}}(t,\delta).$$

Thus the same D works even when $x \gg L$; no implicit constants in this assertion depend on δ or η.

Proof. The deterministic leader time of Lemma B.2 obeys $t_{\mathrm{id}}(\eta) \leq C_I'(1+x)^{3/2}$. Crucially,

$$A^{3/2} \leq C\{L^{2/3}\sqrt{A} + (1+x)^{3/2}\} \leq CP.$$

Indeed, with $a = h_L$, $b = x$,

$$(a+b)^{3/2} \leq \sqrt{2}(a^{3/2} + b^{3/2}).$$

$h_L^{3/2} \leq C_K L^{2/3}\sqrt{A}$ for all $L \geq \log 2$ because $\sup_{L \geq \log 2} h_L/L^{2/3} < \infty$. The $x^{3/2}$ term belongs to P.

For any $D \geq 1$, $t \leq C(1+D)(L+P)$, and elementary polynomial growth bounds give $\log(e+t) \leq C\{\log(e+L) + \log(e+x) + \log(e+D)\}$. Since $\log(e+x) \leq Cx$, we obtain $H_{K,t}(\eta) \leq C[A + \log(e+D)]$. Moreover the preceding inequality implies $P^{2/3}\sqrt{A} \leq CP$, $\sqrt{PA} \leq P$ and $A \leq P$. Combining these with subadditivity of powers and $\sqrt{\log(e+D)} \leq C(1+D)^{1/12}$ gives, for a *fixed* C independent of D, L, x,

$$t^{2/3}\sqrt{H_{K,t}(\eta)} \leq C(1+D)^{3/4}P,$$
$$\sqrt{tH_{K,t}(\eta)} \leq C(1+D)^{3/4}P,$$
$$t^{2/3} + t_{\mathrm{id}}(\eta) + \log(e+t) \leq C(1+D)^{3/4}P.$$

The theoretical boundary has the deterministic envelope $c_{\mathrm{E}}(t,\delta) \leq L + B_0 + B_1\log(e+t)$ with constants fixed by the model. Because $T_{\mathrm{E}}^{\star}\Gamma_{\beta}^{\mathrm{E}}(\theta) = 1$, our trial time provides surplus at least $D\Gamma_{\beta}^{\mathrm{E}}(\theta)P$. Choose a single D so large that this dominates $C'(1+D)^{3/4}P$ and the boundary logarithm; this is possible since $3/4 < 1$. Also $DP \geq t_{\mathrm{id}}(\eta) + K + 1$ for D sufficiently large, because $P \geq 1$ and K is fixed. This establishes the claimed *uniform* inversion without choosing an implicit constant that later changes with D. □

Proof of Theorem 11. On the simultaneous evidence event of Theorem B.4, Lemma B.5 guarantees that the *correct* leader passes the exact theoretical stopping boundary at the integer trial time. Hence the counterfactual first stopping time is no later than that trial time, including paths that have already stopped. The lemma's constant D is fixed uniformly in δ, η, giving the displayed probability bound.

For the expectation, let $x \geq 0$ and take $\eta_x = e^{-(x+3)}$, so $3\eta_x \leq e^{-x}$. After increasing one constant C, the high-probability statement provides the *same increasing differentiable* majorant for every x,

$$q(x) = T_{\mathrm{E}}^{\star}L + C\big\{L^{2/3}\sqrt{h_L + x} + \sqrt{L(h_L+x)} + (h_L+x) + (1+x)^{3/2} + 1\big\}.$$

In particular $\mathbb{P}_{\theta}(\tau_{\delta}^{\mathrm{E,ft}} > q(x)) \leq e^{-x}$ holds with *one* fixed C for all $x \geq 0$. Apply the layer-cake Lemma B.1:

$$\mathbb{E}\tau_{\delta}^{\mathrm{E,ft}} \leq q(0) + \int_0^{\infty} e^{-x}q'(x)\,dx.$$

The integral is bounded by $C\{L^{2/3}h_L^{-1/2} + \sqrt{L}\,h_L^{-1/2} + 1\}$; together with $q(0)$ this is at most the asserted bound. For fixed K the dominant term is $O_{\theta,K}(L^{2/3}\sqrt{\log(e+L)})$. □

Remark B.6. Because

$$\log(e + L_{\delta}) = \Theta(\log\log(e/\delta)),$$

the dominant expected correction may equivalently be written

$$O_{\theta,K}\Big(L_{\delta}^{2/3}\sqrt{\log\log(e/\delta)}\Big)$$

for fixed K.

C MODEL U: ASYMPTOTIC OPTIMALITY

C.1 Model and observation law

Fix $F_k \in \mathcal{P}([0,B])$ with interior mean μ_k and a unique $i^\star = \arg\max_k \mu_k$ (suboptimal means may tie); write

$$\Delta_\star := \min_{j \neq i^\star}(\mu_{i^\star} - \mu_j) > 0. \tag{102}$$

After sampling I_t, the algorithm retains (Q_t, Y_t), where $Q_t \sim \mathrm{Unif}[0,B]$ is independent of $X_t \sim F_{I_t}$ and $Y_t = \mathbf{1}\{X_t \geq Q_t\}$. Write $S_F(q) = \mathbb{P}_F(X \geq q)$.

Lemma C.1 (Exact marginal Bernoulliization)**.** *For independent $X \sim F$ and $Q \sim \mathrm{Unif}[0,B]$,*

$$Y = \mathbf{1}\{X \geq Q\} \sim \mathrm{Ber}(\mu(F)/B). \tag{103}$$

Proof. Conditioning on $X = x$ gives $\mathbb{P}(Y = 1 \mid X = x) = x/B$; integrate over F. □

Consequently the leader uses

$$\widehat{p}_{n,k} = N_{n,k}^{-1} \sum_{t \leq n : I_t = k} Y_t, \qquad \widehat{\mu}_{n,k} = B\widehat{p}_{n,k}, \tag{104}$$

$$\widehat{b}_n \in \arg\max_k \widehat{p}_{n,k}. \tag{105}$$

The guard ensures every local count diverges, so these local-sample estimates are consistent. In contrast to the marginal Bernoulli law,

$$Y \mid Q = q \sim \mathrm{Ber}(S_F(q)), \tag{106}$$

which retains information about the entire survival profile.

C.2 Information in one Uniform-Observed sample

The law of one observed pair is

$$P_F^U(dq, dy) = \frac{dq}{B} S_F(q)^y (1 - S_F(q))^{1-y}, \qquad y \in \{0,1\}.$$

Proposition C.2 (Uniform-Observed KL)**.** *For $F, G \in \mathcal{P}([0,B])$,*

$$D_U(F,G) := \mathrm{KL}(P_F^U, P_G^U) = \frac{1}{B} \int_0^B \mathrm{kl}(S_F(q), S_G(q))\, dq. \tag{107}$$

Moreover,

$$\mathrm{kl}\left(\frac{\mu(F)}{B}, \frac{\mu(G)}{B}\right) \leq D_U(F,G). \tag{108}$$

Proof. The threshold has the same uniform law under F and G, so the chain rule for KL leaves the conditional Bernoulli KL and integration over Q. Inequality (108) is data processing under $(Q,Y) \mapsto Y$. □

C.3 One-sided transportation: exact scalar projection

For $x \in [0,B]$ define the closed one-sided costs

$$K_U^-(F,x) := \inf_{G:\, \mu(G) \leq x} D_U(F,G), \tag{109}$$

$$K_U^+(F,x) := \inf_{G:\, \mu(G) \geq x} D_U(F,G). \tag{110}$$

Clearly $K_U^-(F,x) = 0$ for $x \geq \mu(F)$ and $K_U^+(F,x) = 0$ for $x \leq \mu(F)$.

For $p \in [0,1]$ and $\lambda \in \mathbb{R}$, define

$$T_\lambda(p) := \arg\min_{r \in [0,1]} \{\mathrm{kl}(p,r) + \lambda r\}. \tag{111}$$

For $p \in (0,1)$ the minimizer is interior and satisfies the first-order condition

$$\frac{r-p}{r(1-r)} + \lambda = 0. \tag{112}$$

Set $T_0(p) = p$. For $\lambda \neq 0$,

$$T_\lambda(p) = \frac{1+\lambda - \sqrt{(1+\lambda)^2 - 4\lambda p}}{2\lambda}. \tag{113}$$

At a removable endpoint convention one may redefine the survival at $q = 0$ to be one; this changes neither means nor D_U.

Lemma C.3 (Basic properties of T_λ)**.** *For each λ, $p \mapsto T_\lambda(p)$ is nondecreasing. For each p, $\lambda \mapsto T_\lambda(p)$ is nonincreasing. If $p \in (0,1)$, then $T_\lambda(p) < p$ for $\lambda > 0$ and $T_\lambda(p) > p$ for $\lambda < 0$.*

Proof. The first statement follows by differentiating the minimizing branch in p. The second follows directly from monotone comparative statics: if r_a, r_b minimize $\mathrm{kl}(p,r) + \lambda_a r$ and $\mathrm{kl}(p,r) + \lambda_b r$ with $\lambda_a < \lambda_b$, adding the two optimality inequalities gives $(\lambda_b - \lambda_a)(r_b - r_a) \leq 0$. The final statement follows from (112). □

Define

$$m_F(\lambda) := \int_0^B T_\lambda(S_F(q))\, dq.$$

For completeness, the scalar minimizer is continuous in λ for *every* $p \in [0,1]$. Indeed, $r \mapsto \mathrm{kl}(p,r)$ is lower semicontinuous on the compact interval $[0,1]$, and $\mathrm{kl}(p,r) + \lambda r$ has a unique minimizer. If $\lambda_m \to \lambda$, any convergent subsequence of minimizers r_m has a limit r; lower semicontinuity and optimality against the fixed minimizer at λ show that r also minimizes the limit objective. Uniqueness forces $r = T_\lambda(p)$, so the entire sequence converges. At the extreme probabilities the minimizers are explicit: $T_\lambda(0) = 0$ for $\lambda \geq -1$ and $T_\lambda(0) = 1 + 1/\lambda$ for $\lambda < -1$; $T_\lambda(1) = 1$ for $\lambda \leq 1$ and $T_\lambda(1) = 1/\lambda$ for $\lambda > 1$. For any fixed $p \in [0,1]$ and any $\rho \in (0,1)$, comparison with the finite-cost candidate $r = \rho$ and nonnegativity of Bernoulli KL give

$$T_\lambda(p) \leq \rho + \frac{\mathrm{kl}(p,\rho)}{\lambda} \quad (\lambda > 0), \qquad T_\lambda(p) \geq \rho - \frac{\mathrm{kl}(p,\rho)}{|\lambda|} \quad (\lambda < 0).$$

Sending λ to the corresponding infinity and then $\rho \downarrow 0$ or $\rho \uparrow 1$ proves the limits $T_\lambda(p) \to 0$ and $T_\lambda(p) \to 1$ even at $p = 0, 1$. Because $0 \leq T_\lambda(p) \leq 1$, dominated convergence now gives

$$\lim_{\lambda \to -\infty} m_F(\lambda) = B, \qquad m_F(0) = \mu(F), \qquad \lim_{\lambda \to +\infty} m_F(\lambda) = 0,$$

and continuity of m_F at every finite λ (including when S_F takes the values 0 or 1 on sets of positive measure). For arbitrary $[0,B]$-supported laws, m_F need not be strictly decreasing; hence the multiplier need not be unique. Continuity and the limits below nevertheless guarantee existence for every interior target, and all solutions yield the same primal variational value.

Theorem C.4 (Exact scalar representation)**.** *Let $F \in \mathcal{P}([0,B])$ with $0 < \mu(F) < B$.*

If $0 < x < \mu(F)$, choose any finite $\lambda_x \geq 0$ satisfying $m_F(\lambda_x) = x$. Then

$$K_U^-(F,x) = \frac{1}{B} \int_0^B \mathrm{kl}(S_F(q), T_{\lambda_x}(S_F(q)))\, dq. \tag{114}$$

If $\mu(F) < x < B$, choose any finite $\lambda_x \leq 0$ satisfying $m_F(\lambda_x) = x$. Then

$$K_U^+(F,x) = \frac{1}{B} \int_0^B \mathrm{kl}(S_F(q), T_{\lambda_x}(S_F(q)))\, dq. \tag{115}$$

At $x = \mu(F)$ the relevant one-sided value is 0. At the active endpoints, $K_U^-(F,0) = K_U^+(F,B) = +\infty$ under $0 < \mu(F) < B$; equivalently the displayed formulas are interpreted as the extended-value limits $\lambda \to +\infty$ and $\lambda \to -\infty$. If several finite multipliers solve an interior mean equation, they give the same variational value.

Proof. Write $p(q) = S_F(q)$ and $r_\lambda(q) = T_\lambda(p(q))$. Since S_F is left-continuous and nonincreasing and T_λ is continuous and nondecreasing in p, the composition $r_\lambda = T_\lambda \circ S_F$ is left-continuous and nonincreasing. Hence it is the survival function $q \mapsto G_\lambda([q,B])$ of a probability law G_λ on $[0,B]$ (with the usual endpoint convention).

For every $r \in [0,1]$, pointwise optimality gives $\mathrm{kl}(p,r) + \lambda r \geq \mathrm{kl}(p, r_\lambda) + \lambda r_\lambda$. Integrating and using $\int r_\lambda = x$ gives

$$\frac{1}{B}\int \mathrm{kl}(p,r) \geq \frac{1}{B}\int \mathrm{kl}(p, r_\lambda) + \frac{\lambda}{B}\Big(x - \int r\Big).$$

If $0 < x < \mu(F)$ then $\lambda \geq 0$ and every feasible r satisfies $\int r \leq x$, so the last term is nonnegative. Thus r_λ is globally optimal. The K_U^+ case is identical with $\lambda \leq 0$ and $\int r \geq x$. Continuity of m_F, together with $m_F(0) = \mu(F)$ and its limits 0 and B at $+\infty$ and $-\infty$, gives a finite multiplier for every interior target by the intermediate-value theorem. The value is independent of the chosen multiplier because every such profile is a primal optimizer.

For $x = 0$, the only law with mean at most zero is δ_0. Since $\mu(F) > 0$, the set $\{q : S_F(q) > 0\}$ has positive Lebesgue measure, and hence $D_U(F, \delta_0) = +\infty$. The argument for $K_U^+(F, B)$ is symmetric. Finally the value at $x = \mu(F)$ is zero by taking $G = F$. □

C.4 Pair reduction and fixed-β characteristic

For masses $a, b \geq 0$ define

$$V_{ij}^U(a,b) := \inf_{\mu(G_i) \leq \mu(G_j)} \{aD_U(F_i, G_i) + bD_U(F_j, G_j)\}. \tag{116}$$

Lemma C.5 (Scalar separator)**.** *For every pair (i,j) and $a, b \geq 0$,*

$$V_{ij}^U(a,b) = \inf_{x \in [0,B]} \{aK_U^-(F_i, x) + bK_U^+(F_j, x)\}, \tag{117}$$

where a zero coefficient multiplying an extended value is interpreted as zero.

Proof. First suppose $a, b > 0$. For any feasible pair (G_i, G_j) with $\mu(G_i) \leq \mu(G_j)$, choose $x \in [\mu(G_i), \mu(G_j)]$. By definition of the one-sided costs, $aD_U(F_i, G_i) + bD_U(F_j, G_j) \geq aK_U^-(F_i, x) + bK_U^+(F_j, x)$. Taking the infimum over feasible pairs gives the lower bound in (117). Conversely, fix x for which the displayed scalar objective is finite and let $\varepsilon > 0$. Choose $G_i^\varepsilon, G_j^\varepsilon$ satisfying $\mu(G_i^\varepsilon) \leq x \leq \mu(G_j^\varepsilon)$ and

$$D_U(F_i, G_i^\varepsilon) \leq K_U^-(F_i, x) + \varepsilon, \qquad D_U(F_j, G_j^\varepsilon) \leq K_U^+(F_j, x) + \varepsilon.$$

The pair is feasible, so

$$V_{ij}^U(a,b) \leq aK_U^-(F_i, x) + bK_U^+(F_j, x) + (a+b)\varepsilon.$$

Letting $\varepsilon \downarrow 0$ and then taking the infimum over x proves the result for positive masses.

If $b = 0$, choose $G_i = F_i$ and any G_j with mean at least $\mu(F_i)$ to obtain $V_{ij}^U(a, 0) = 0$; the scalar side is also zero by taking $x = \mu(F_i)$. The case $a = 0$ is symmetric, and $a = b = 0$ is immediate. This proves all boundary cases. □

Fix $\beta \in (0,1)$. For $j \neq i^\star$ and $w \geq 0$, let

$$h_j(w) := I_{j,U}^\beta(w) := V_{i^\star j}^U(\beta, w) = \inf_x \{\beta K_U^-(F_{i^\star}, x) + wK_U^+(F_j, x)\}. \tag{118}$$

Define

$$\Gamma_\beta^U(\mathbf{F}) := \max_{\substack{w_j \geq 0 \\ \sum_{j \neq i^\star} w_j = 1-\beta}} \min_{j \neq i^\star} h_j(w_j), \qquad T_\beta^U := 1/\Gamma_\beta^U. \tag{119}$$

Lemma C.6 (Robust pair geometry)**.** *For every challenger j, h_j is finite, continuous, nondecreasing and concave on $[0, \infty)$, with $h_j(0) = 0$. Under the unique-best assumption, $h_j(w) > 0$ for every $w > 0$.*

Proof. By (118), h_j is the pointwise infimum over x of affine functions of w with nonnegative slope. Hence it is concave and nondecreasing. It is finite for every $w > 0$: choose $x \in (\mu_j, \mu_{i^\star})$, for which both one-sided costs are finite by Theorem C.4. A finite concave function is continuous on the interior of its domain. At the origin, $h_j(0) = 0$ by Lemma C.5, and choosing $x = \mu_{i^\star}$ gives

$$0 \leq h_j(w) \leq wK_U^+(F_j, \mu_{i^\star}) \longrightarrow 0, \qquad w \downarrow 0,$$

where the constant is finite because $\mu_j < \mu_{i^\star} < B$. Thus continuity also holds at zero.

Finally data processing under $(Q, Y) \mapsto Y$ gives $D_U(F, G) \geq \mathrm{kl}\Big(\frac{\mu(F)}{B}, \frac{\mu(G)}{B}\Big)$. Therefore $h_j(w)$ is bounded below by the two-arm Bernoulli transportation cost with positive masses β and w. Since $\mu_j/B < \mu_{i^\star}/B$, that cost is strictly positive for every $w > 0$. □

Remark C.7 (No strict-monotonicity assumption)**.** For arbitrary $[0, B]$-supported laws, h_j may have flat pieces. We therefore do not assume a unique optimal allocation and do not base the proof on equalization. The allocation analysis below uses critical information levels, which only require Lemma C.6.

C.5 Finite-data current-status profile GLR

For each arm k, let

$$(Q_{k,r}, Y_{k,r})_{r\geq 1}$$

denote the threshold–bit observations indexed by the local pull count of arm k. At global time n, the data from arm k are the first $N_{n,k}$ terms of this sequence. Because every threshold has a continuous uniform law and there are only countably many local thresholds, the event

$$\Omega_Q^\circ := \{Q_{k,r} \in (0, B) \text{ for all } k, r\} \tag{120}$$

has probability one. All current-status profile and Stieltjes arguments below are carried out on Ω_Q°; this removes endpoint bookkeeping without changing any probability statement. Under a candidate law $G \in \mathcal{P}([0, B])$, the conditional distribution of the bit is

$$Y_{k,r} \mid Q_{k,r} = q \sim \mathrm{Ber}(S_G(q)),$$

where

$$S_G(q) := \mathbb{P}_{X\sim G}(X \geq q)$$

is the survival function of G. Since the threshold law is always $\mathrm{Unif}[0, B]$ and does not depend on G, its density cancels from all likelihood ratios. We therefore work with the conditional likelihood of the labels given the realized thresholds.

C.5.1 Armwise current-status likelihood

For a local sample size $m \geq 0$, define

$$\bar{\ell}_{k,m}(G) := \sum_{r=1}^{m} \Big[Y_{k,r} \log S_G(Q_{k,r}) + (1 - Y_{k,r}) \log(1 - S_G(Q_{k,r}))\Big], \tag{121}$$

$$\ell_{n,k}(G) := \bar{\ell}_{k,N_{n,k}}(G). \tag{122}$$

Equivalently,

$$L_{n,k}(G) := \exp\{\ell_{n,k}(G)\} = \prod_{r=1}^{N_{n,k}} S_G(Q_{k,r})^{Y_{k,r}} \big(1 - S_G(Q_{k,r})\big)^{1-Y_{k,r}}. \tag{123}$$

We use the usual extended-real conventions for $\log 0$. Define

$$\bar{\mathcal{M}}_{k,m} := \sup_{G\in\mathcal{P}([0,B])} \bar{\ell}_{k,m}(G), \qquad \mathcal{M}_{n,k} := \bar{\mathcal{M}}_{k,N_{n,k}}. \tag{124}$$

Thus $\mathcal{M}_{n,k}$ is the unrestricted maximum log likelihood, i.e., the best current-status fit obtainable for arm k without a mean constraint. Because the likelihood depends on G only through the values of the nonincreasing survival profile S_G at the observed thresholds, sorting the thresholds transforms (122) into a Bernoulli isotonic likelihood problem. Hence an unconstrained maximizer is a current-status nonparametric MLE and may be computed by the pool-adjacent-violators algorithm (PAVA), up to the harmless endpoint conventions needed to represent a bounded-law survival function; see, e.g., the current-status likelihood literature [12].

C.5.2 Pairwise profile GLR

For an ordered pair (i, j), consider the pairwise null hypothesis

$$H_0^{ij} : \qquad \mu(G_i) \leq \mu(G_j). \tag{125}$$

The unrestricted pair log likelihood is $\mathcal{M}_{n,i} + \mathcal{M}_{n,j}$, whereas the best log likelihood under the null is

$$\sup_{\mu(G_i)\leq\mu(G_j)} \big\{\ell_{n,i}(G_i) + \ell_{n,j}(G_j)\big\}.$$

This motivates the empirical pairwise transportation score

$$W_n^U(i, j) := \mathcal{M}_{n,i} + \mathcal{M}_{n,j} - \sup_{\mu(G_i)\leq\mu(G_j)} \big\{\ell_{n,i}(G_i) + \ell_{n,j}(G_j)\big\}. \tag{126}$$

Thus $W_n^U(i,j)$ is a profile generalized likelihood-ratio statistic for testing H_0^{ij} against the unrestricted alternative.

If

$$L_{0,n}^{ij} := \sup_{\mu(G_i)\le\mu(G_j)} L_{n,i}(G_i)L_{n,j}(G_j),$$

then

$$\exp\{W_n^U(i,j)\} = \frac{\sup_{G_i,G_j} L_{n,i}(G_i)L_{n,j}(G_j)}{L_{0,n}^{ij}}. \tag{127}$$

A large value of $W_n^U(i,j)$ therefore means that forcing the data to obey $\mu(G_i) \le \mu(G_j)$ incurs a large likelihood loss.

C.5.3 Empirical one-sided profile losses

For $x \in [0,B]$, define

$$\mathcal{K}_{n,i}^-(x) := \mathcal{M}_{n,i} - \sup_{\mu(G_i)\le x} \ell_{n,i}(G_i), \tag{128}$$

$$\mathcal{K}_{n,j}^+(x) := \mathcal{M}_{n,j} - \sup_{\mu(G_j)\ge x} \ell_{n,j}(G_j). \tag{129}$$

These quantities are nonnegative. The first is the empirical likelihood cost of forcing arm i to have mean at most x; the second is the cost of forcing arm j to have mean at least x.

Proposition C.8 (Exact empirical scalar-separator representation)**.** *For every ordered pair $i \ne j$,*

$$W_n^U(i,j) = \inf_{x\in[0,B]} \left\{\mathcal{K}_{n,i}^-(x) + \mathcal{K}_{n,j}^+(x)\right\}. \tag{130}$$

Proof. The order constraint admits the exact scalar-separator representation

$$\{(G_i,G_j) : \mu(G_i) \le \mu(G_j)\} = \bigcup_{x\in[0,B]} \{(G_i,G_j) : \mu(G_i) \le x \le \mu(G_j)\}.$$

Therefore

$$\sup_{\mu(G_i)\le\mu(G_j)} \{\ell_{n,i}(G_i) + \ell_{n,j}(G_j)\}$$
$$= \sup_{x\in[0,B]} \left\{ \sup_{\mu(G_i)\le x} \ell_{n,i}(G_i) + \sup_{\mu(G_j)\ge x} \ell_{n,j}(G_j) \right\}.$$

At fixed x the two candidate laws are optimized independently, so the joint supremum separates into the sum of the two marginal suprema; no supremum need be attained.

Subtracting both sides from $\mathcal{M}_{n,i} + \mathcal{M}_{n,j}$ and using $C - \sup_x f(x) = \inf_x\{C - f(x)\}$ gives

$$W_n^U(i,j) = \inf_{x\in[0,B]} \left\{\mathcal{K}_{n,i}^-(x) + \mathcal{K}_{n,j}^+(x)\right\},$$

which is (130). □

Universal measurability and definability of the raw-profile policy. Equip $\mathcal{P}([0,B])$ with the weak topology. This is a compact Polish space, and the mean map $G \mapsto \mu(G)$ is continuous. Moreover $(q,G) \mapsto S_G(q) = G([q,B])$ is Borel: it is the parameterized integral of the bounded Borel function $(q,x) \mapsto \mathbf{1}\{x \ge q\}$. With the usual extended-real conventions for $\log 0$, the finite-data score $(\mathsf{d},G) \mapsto \ell_{n,k}(G;\mathsf{d})$ is therefore Borel in the finite transcript d and the candidate law. The unrestricted profile value and the mean-ordered two-arm profile value are suprema of Borel functions over, respectively, a Polish candidate space and a closed Borel constraint set (the latter is closed because the mean map is continuous). By the measurable-projection theorem these value functions are upper semianalytic, hence universally measurable, as functions of the finite transcript. On Ω_Q° these profile values are finite: every log score is at most zero, while the common candidate $(\delta_0+\delta_B)/2$ has survival $1/2$ at every interior threshold and also gives a finite feasible mean-ordered pair. Thus $W_n^U(i,j)$ is a finite universally measurable statistic. Throughout Model U we take the universally completed natural filtration; finite-arm argmin/argmax operations with deterministic tie-breaking and the threshold comparison in the stopping rule preserve universal measurability. Hence the exact arm choices, recommendation, and stopping time used below are formally well defined. This completion changes none of the probability statements under any admissible instance.

C.6 Exact guarded raw-profile EB–TCI rule

For reference, the *exact* policy analyzed below is defined in Section 2 and the Model-U algorithm subsection of the main text. After one initial pull per arm, at each time n it computes the marginal-bit leader $\widehat{b}_n$ and checks the guard $\min_k N_{n,k} < \sqrt{n/K}$. A guard round forces a least-sampled arm; otherwise, it nominates the minimizer of $W_n^U(\widehat{b}_n, j) + \log N_{n,j}$ over $j \neq \widehat{b}_n$ *before* drawing the independent Top-Two coin, which selects the leader with probability β and the nominee with probability $1-\beta$. Only after arm selection does nature independently draw $Q \sim \mathrm{Unif}[0, B]$ and supply $(Q, \mathbf{1}\{X \geq Q\})$. All finite-set choices use deterministic tie-breaking. The exact raw-profile statistic is used for both nomination and stopping; the finite sieve below calibrates its boundary but is not run online.

The guard makes every count diverge while using only $O(\sqrt{Kn}) = o(n)$ interventions. Later stabilization and tracking proofs quantify the guard and the exceptional wrong-leader rounds; no argument silently treats a random last-exit time as a stopping time.

C.7 Certified raw-profile stopping rule

We use the same raw profile GLR for challenging and stopping. To state the certified boundary without a forward reference, define the proof-only sieve schedules

$$\alpha_n := \frac{1}{4}(n+1)^{-1/3}, \qquad \xi_n := \frac{1}{16}(n+1)^{-2/3}, \tag{131}$$

and

$$J_n := \left\lceil \frac{1-\alpha_n/2}{\xi_n} \right\rceil, \qquad M_n := \binom{n+J_n}{J_n}, \qquad a_n := -\log(1-\alpha_n) + \frac{4\xi_n}{\alpha_n}, \qquad c_n := na_n + 2\log M_n. \tag{132}$$

The finite monotone sieve producing this penalty is used only in the calibration proof below; it is not fitted by the online algorithm.

Sublinearity of the certified penalty. From (131), $\alpha_n = \Theta(n^{-1/3})$ and $\xi_n/\alpha_n = \Theta(n^{-1/3})$. Since $-\log(1-u) = O(u)$ as $u \downarrow 0$, $a_n = O(n^{-1/3})$ and hence $na_n = O(n^{2/3})$. Also $J_n = O(n^{2/3})$, and

$$\log M_n \leq J_n \log\left(\frac{e(n+J_n)}{J_n}\right) = O\Big(n^{2/3}\log(n+1)\Big).$$

Therefore

$$c_n = O\Big(n^{2/3}\log(n+1)\Big) = o(n). \tag{133}$$

Define

$$b(n,\delta) := c_n + \log\frac{(K-1)\pi^2 n^2}{6\delta} = \log(1/\delta) + d_n, \qquad d_n := c_n + \log\frac{(K-1)\pi^2 n^2}{6}. \tag{134}$$

By (133), $d_n = o(n)$ for fixed K. The stopping time and recommendation are

$$\tau_\delta := \inf\left\{ n \geq K : \min_{j \neq \widehat{b}_n} W_n^U(\widehat{b}_n, j) \geq b(n,\delta) \right\}, \qquad \widehat{\imath}_{\tau_\delta} := \widehat{b}_{\tau_\delta}. \tag{135}$$

We now prove, still in Part C, that this exact boundary is anytime δ-correct under arbitrary predictable arm sampling that respects the arm-before-threshold protocol. The proof is separate from the sampling analysis: it uses a finite monotone sieve only as a calibration device, while the online algorithm always evaluates the raw profile GLR. After this calibration proof, the remainder of Part C uses only the already established first-order fact $d_n = o(n)$.

C.7.1 Pairwise calibration target

For an ordered pair (i, j), recall the null hypothesis

$$H_0^{ij}: \quad \mu_i \leq \mu_j.$$

The key result needed for the stopping rule is the following fixed-time inequality.

Theorem C.9 (Fixed-time pairwise calibration)**.** *For every ordered pair (i, j) satisfying $\mu_i \le \mu_j$, every $n \ge 1$, and every $x > 0$,*

$$\mathbb{P}_{\mathbf{F}}\left(W_n^U(i,j) \ge c_n + x\right) \le e^{-x}. \tag{136}$$

The bound is valid under arbitrary predictable arm sampling that respects the arm-before-threshold protocol.

For fixed-time analysis, extend any policy beyond its first stopping time by an arbitrary predictable arm rule. This counterfactual continuation does not change the realized transcript or the stopping event up to τ_δ, and makes the all-time GLR events below well defined.

Once Theorem C.9 is established, anytime incorrect-stop control is immediate. Indeed, define

$$\omega_n := \frac{6}{\pi^2 n^2}, \qquad \sum_{n=1}^{\infty} \omega_n = 1.$$

For a pairwise error level $\alpha \in (0, 1)$, choosing

$$x_n := \log \frac{1}{\alpha \omega_n} = \log \frac{\pi^2 n^2}{6\alpha}$$

in (136) gives

$$\mathbb{P}_{\mathbf{F}}\left(W_n^U(i,j) \ge c_n + \log \frac{\pi^2 n^2}{6\alpha}\right) \le \alpha\omega_n. \tag{137}$$

A union bound over n therefore yields

$$\mathbb{P}_{\mathbf{F}}\left(\exists n \ge 1 :\ W_n^U(i,j) \ge c_n + \log \frac{\pi^2 n^2}{6\alpha}\right) \le \alpha, \tag{138}$$

whenever $\mu_i \le \mu_j$.

Setting $\alpha = \delta/(K-1)$ gives exactly the boundary (134).

Theorem C.10 (Anytime incorrect-stop control)**.** *For every instance with a unique best arm $i^\star$, the stopping rule* (135) *satisfies*

$$\mathbb{P}_{\mathbf{F}}\left(\tau_\delta < \infty,\ \widehat{i}_{\tau_\delta} \ne i^\star\right) \le \delta. \tag{139}$$

Proof. Suppose a wrong arm $i \ne i^\star$ is recommended. By the definition of the stopping rule, at the stopping time we must have $W_{\tau_\delta}^U(i, i^\star) \ge b(\tau_\delta, \delta)$. Because $i^\star$ is the true best arm, the null

$$H_0^{i,i^\star} :\quad \mu_i \le \mu_{i^\star}$$

is true. Applying the anytime pairwise bound (138) with $\alpha = \delta/(K-1)$ shows that, for each fixed wrong arm i,

$$\mathbb{P}_{\mathbf{F}}\left(\exists n \ge 1 :\ W_n^U(i,i^\star) \ge b(n,\delta)\right) \le \frac{\delta}{K-1}.$$

Finally, a union bound over the $K-1$ wrong arms gives

$$\mathbb{P}_{\mathbf{F}}\left(\tau_\delta < \infty,\ \widehat{i}_{\tau_\delta} \ne i^\star\right) \le (K-1)\frac{\delta}{K-1} = \delta.$$

□

It remains to prove Theorem C.9.

C.7.2 Finite monotone sieve for calibration

The construction in this subsection is used only in the proof. It is not part of the algorithm. For every arm k, imagine an infinite local sequence

$$(Q_{k,r}, X_{k,r})_{r\geq 1}, \qquad Q_{k,r} \sim \mathrm{Unif}[0, B], \qquad X_{k,r} \sim F_k,$$

independent across arms and local indices. The rth pull of arm k reveals $Q_{k,r}$ and $Y_{k,r} := \mathbf{1}\{X_{k,r} \geq Q_{k,r}\}$. For a candidate distribution G, write

$$\phi_G(q, y) := y \log S_G(q) + (1 - y) \log(1 - S_G(q)).$$

Fix a global horizon n. Use the schedules $\alpha_n, \xi_n, J_n, M_n, a_n, c_n$ already defined in (131)–(132), and put

$$\Delta_n := \frac{1 - \alpha_n/2}{J_n} \leq \xi_n, \qquad \mathcal{Z}_n := \left\{ \frac{\alpha_n}{4} + \ell\Delta_n : \ell = 0, 1, \ldots, J_n \right\}. \tag{140}$$

Then $\mathcal{Z}_n \subset [\alpha_n/4, 1 - \alpha_n/4]$, and every $u \in [\alpha_n/2, 1 - \alpha_n/2]$ has a grid point $z \in \mathcal{Z}_n$ with $z \leq u$ and $0 \leq u - z \leq \xi_n$.

The grid performs two operations: α_n prevents log-likelihood singularities, while ξ_n controls the downward-rounding loss. Use the same coupled infinite local threshold sequence introduced above. At fixed n, the proof may inspect its first n entries, including thresholds not yet revealed by the algorithm; the algorithm itself never sees those unused future entries. Because any arm has at most n local pulls by global time n, this single sequence contains every threshold at which the likelihood can be evaluated up to that horizon.

For a fixed arm k, sort the first n proof-only thresholds as

$$Q_{k,(1)} < \cdots < Q_{k,(n)}.$$

Ties and endpoints occur with probability zero and may be resolved arbitrarily. Let $\mathfrak{G}_{n,k}$ be the class of canonical probability laws whose survival values on these points form a nonincreasing grid-valued sequence

$$z_1 \geq z_2 \geq \cdots \geq z_n, \qquad z_r \in \mathcal{Z}_n.$$

Every such sequence is realizable by a law on $[0, B]$. For example, one may assign mass $1 - z_1$ at 0, mass $z_r - z_{r+1}$ at $Q_{k,(r)}$ for $1 \leq r < n$, and mass z_n at $Q_{k,(n)}$. Hence

$$|\mathfrak{G}_{n,k}| \leq M_n. \tag{141}$$

The calibration now has only two ingredients left to prove: every raw profile can be replaced by a sieve profile at cost at most na_n, and a union bound over at most M_n^2 two-arm sieve candidates contributes the term $2 \log M_n$. These are exactly the two terms in $c_n = na_n + 2 \log M_n$ from (132).

C.7.3 Approximation of the unrestricted profile

Lemma C.11 (Finite-sieve approximation)**.** *For every arm k, every integer $0 \leq m \leq n$, and every candidate law G on $[0, B]$, there exists $H \in \mathfrak{G}_{n,k}$, depending on G and on the proof-only threshold values but not on the labels, such that*

$$\bar{\ell}_{k,m}(H) \geq \bar{\ell}_{k,m}(G) - ma_n. \tag{142}$$

Consequently,

$$\bar{\mathcal{M}}_{k,m} \leq \max_{H \in \mathfrak{G}_{n,k}} \bar{\ell}_{k,m}(H) + ma_n. \tag{143}$$

Proof. For $m = 0$ both scores are empty sums equal to zero; choose any sieve law. Thus consider $1 \leq m \leq n$. Write $r(q) := S_G(q)$. First move the survival profile slightly away from the singular values 0 and 1 by defining

$$u(q) := (1 - \alpha_n) r(q) + \frac{\alpha_n}{2}. \tag{144}$$

Then $\frac{\alpha_n}{2} \leq u(q) \leq 1 - \frac{\alpha_n}{2}$. Moreover,

$$u(q) \geq (1 - \alpha_n) r(q), \qquad 1 - u(q) \geq (1 - \alpha_n)(1 - r(q)).$$

Hence, for every $y \in \{0, 1\}$,

$$\begin{aligned} &y \log u(q) + (1 - y) \log(1 - u(q)) \\ &\quad \geq y \log r(q) + (1 - y) \log(1 - r(q)) + \log(1 - \alpha_n). \end{aligned} \tag{145}$$

Now evaluate u on the first n threshold points and round each value downward to the nearest point of $\mathcal{Z}_n$. Downward rounding preserves monotonicity, so the rounded values define a law $H \in \mathfrak{G}_{n,k}$. If $z \le u$ denotes the rounded value, then

$$0 \le u - z \le \xi_n, \qquad z \ge \frac{\alpha_n}{4}.$$

For $y = 1$,

$$\begin{aligned} \log z &= \log u - \log \frac{u}{z} \\ &\ge \log u - \frac{u - z}{z} \\ &\ge \log u - \frac{4\xi_n}{\alpha_n}. \end{aligned} \tag{146}$$

For $y = 0$, downward rounding can only improve the log likelihood because $1 - z \ge 1 - u$, so

$$\log(1 - z) \ge \log(1 - u). \tag{147}$$

Combining (145), (146), and (147) gives $\phi_H(q, y) \ge \phi_G(q, y) - a_n$ at every one of the first n local threshold points. Summing over the first $m \le n$ revealed observations yields (142). Taking the supremum over G gives (143). □

C.7.4 Proof of the pairwise calibration theorem

Proof of Theorem C.9. Fix an ordered pair (i, j) satisfying the null $\mu_i \le \mu_j$. Let

$$L_{0,n}^{ij} := \sup_{\mu(G_i) \le \mu(G_j)} L_{n,i}(G_i) L_{n,j}(G_j)$$

denote the constrained profile likelihood. Since the true pair (F_i, F_j) is feasible under the null,

$$L_{0,n}^{ij} \ge L_{n,i}(F_i) L_{n,j}(F_j). \tag{148}$$

By the definition of $W_n^U(i, j)$,

$$W_n^U(i, j) = \mathcal{M}_{n,i} + \mathcal{M}_{n,j} - \log L_{0,n}^{ij}. \tag{149}$$

Applying Lemma C.11 with $m = N_{n,i}$ and $m = N_{n,j}$ and using $N_{n,i} + N_{n,j} \le n$ gives

$$\begin{aligned} W_n^U(i, j) \le\ & n a_n \\ & + \max_{H_i \in \mathfrak{G}_{n,i},\, H_j \in \mathfrak{G}_{n,j}} \log \frac{L_{n,i}(H_i) L_{n,j}(H_j)}{L_{n,i}(F_i) L_{n,j}(F_j)}. \end{aligned} \tag{150}$$

Let $\mathcal{Q}_\infty$ denote the sigma-field generated by all proof-only threshold sequences. For a fixed terminal horizon n, put $\mathcal{H}_t^{\mathcal{Q}} := \sigma(\mathcal{Q}_\infty, \mathcal{F}_t)$, where $\mathcal{F}_t$ contains the revealed labels, realized actions and Top–Two coins through time t; the next coin is revealed in the pre-label sigma-field $\mathcal{H}_{t+1,-}^{\mathcal{Q}}$. Only the proof uses this enlarged filtration. Since the actual policy selects the next arm without inspecting an unrevealed threshold or its label, the next reward remains independent of $\mathcal{H}_{t+1,-}^{\mathcal{Q}}$ conditionally on the selected arm. Conditional on $\mathcal{Q}_\infty$, the sorted-threshold vectors, quantized grid profiles, and their canonical law realizations are measurable and fixed before any reward label is revealed; only the proof, not the actual policy, sees unused future thresholds. For a fixed pair (H_i, H_j), define

$$Z_t(H_i, H_j) := \frac{L_{t,i}(H_i) L_{t,j}(H_j)}{L_{t,i}(F_i) L_{t,j}(F_j)}.$$

Conditionally on $\mathcal{Q}_\infty$, sieve candidates are fixed functions of thresholds and do not depend on any reward labels. Work with $\mathcal{H}_t^{\mathcal{Q}}$ and the pre-label field $\mathcal{H}_{t+1,-}^{\mathcal{Q}}$ defined above. The next arm may depend on past labels and observed thresholds but not on the new label; its next local threshold is fixed in the enlarged filtration. Thus, conditional on the chosen arm, its next bit still has parameter $p = S_{F_k}(q)$. For a sieve candidate let $h = S_{H_k}(q)$, and define the likelihood factor on the true label support only. Its conditional expectation equals

$$\sum_{y:\, p^y(1-p)^{1-y} > 0} p^y (1-p)^{1-y} \frac{h^y (1-h)^{1-y}}{p^y (1-p)^{1-y}} = \sum_{y:\, p^y(1-p)^{1-y} > 0} h^y (1-h)^{1-y} \le 1.$$

When $0 < p < 1$ the last sum is one; at $p = 0$ or $p = 1$ it may be strictly smaller. This calculation applies equally when an arm outside the tested pair is drawn (then the product is unchanged). Therefore $(Z_t(H_i, H_j))_{t\geq 0}$ is a nonnegative supermartingale conditionally on $\mathcal{Q}_\infty$. In particular,

$$\mathbb{E}_{\mathbf{F}}\left[Z_n(H_i, H_j) \mid \mathcal{Q}_\infty\right] \leq 1. \tag{151}$$

By Markov's inequality,

$$\mathbb{P}_{\mathbf{F}}\left(Z_n(H_i, H_j) \geq e^y \mid \mathcal{Q}_\infty\right) \leq e^{-y}.$$

There are at most M_n^2 candidate pairs. Therefore a union bound gives

$$\begin{aligned}\mathbb{P}_{\mathbf{F}}\left(\max_{H_i,H_j} \log Z_n(H_i, H_j) \geq 2\log M_n + x \middle| \mathcal{Q}_\infty\right) \\ \leq e^{-x}.\end{aligned} \tag{152}$$

Integrating over $\mathcal{Q}_\infty$ and combining (150), (152), and the definition $c_n = na_n + 2\log M_n$ proves

$$\mathbb{P}_{\mathbf{F}}\left(W_n^U(i, j) \geq c_n + x\right) \leq e^{-x}.$$

□

C.8 Quantitative stabilization of the leader and the current-status profile

C.8.1 Guard and stable empirical-best leader

Let G_n be the number of guard interventions among the first n pulls.

Lemma C.12 (Deterministic guard bounds)**.** *There exists a numerical constant $C_K < \infty$ depending only on K such that, for every sample path and every n,*

$$\min_{k\in[K]} N_{n,k} \geq \left\lfloor \sqrt{\frac{n}{K}} \right\rfloor - C_K, \qquad G_n \leq C_K(1 + \sqrt{Kn}). \tag{153}$$

Proof. The rule can activate only when the least count lies below the moving boundary $\sqrt{n/K}$, and a guard pull increases a least count by one. Between two successive integer levels of the boundary, each arm needs at most one forced increment to keep pace with that level. Summing over the $O(\sqrt{n/K})$ levels reached by time n gives at most $K\sqrt{n/K} + O(K) = O(\sqrt{Kn})$ guard pulls and the stated lower bound on every count. Enlarging the constant handles the finitely many initial times. □

Write $p_k := \mu_k/B$ and

$$\Delta_p := p_{i^\star} - \max_{j\neq i^\star} p_j > 0.$$

Lemma C.13 (Stable empirical-best leader)**.** *There exists a random time T_{lead} with finite moments of every fixed order such that*

$$\widehat{b}_n = i^\star \qquad \forall n \geq T_{\mathrm{lead}}.$$

More precisely, there exist constants $C, c > 0$ such that

$$\mathbb{P}(T_{\mathrm{lead}} > n) \leq Ce^{-c\sqrt{n}}.$$

Proof. For every arm, index the observations in local pull order: $Y_{k,1}, Y_{k,2}, \ldots$ are iid $\mathrm{Ber}(p_k)$ by exact Bernoulliization. If $\widehat{b}_n \neq i^\star$, then for some $j \neq i^\star$ either

$$\widehat{p}_{n,i^\star} \leq p_{i^\star} - \frac{\Delta_p}{2} \quad \text{or} \quad \widehat{p}_{n,j} \geq p_j + \frac{\Delta_p}{2}.$$

We do not condition on the adaptive count $N_{n,k}$. Instead, for any deterministic $m_0 \geq 1$, Hoeffding's inequality at each local sample size and a union bound give

$$\begin{aligned}\mathbb{P}\left(\exists m \geq m_0 : \left|\frac{1}{m}\sum_{r=1}^{m} Y_{k,r} - p_k\right| \geq \frac{\Delta_p}{2}\right) \\ \leq 2\sum_{m\geq m_0} \exp\left(-\frac{m\Delta_p^2}{2}\right) \leq C_0 e^{-c_0 m_0}.\end{aligned}$$

By Lemma C.12, for all sufficiently large n every local count satisfies $N_{n,k} \geq c_g\sqrt{n}$ for a deterministic $c_g > 0$. Therefore

$$\mathbb{P}(\widehat{b}_n \neq i^\star) \leq C_1 e^{-c_1\sqrt{n}}.$$

Consequently

$$\mathbb{P}(T_{\mathrm{lead}} > n) \leq \sum_{m \geq n} \mathbb{P}(\widehat{b}_m \neq i^\star) \leq C e^{-c\sqrt{n}}.$$

Since $\sum_n n^r e^{-c\sqrt{n}} < \infty$ for every fixed r, all fixed moments are finite. □

C.8.2 Predictable Top-Two counts and the best-arm fraction

Put $s := 1 - \beta$. The analysis must use predictable quantities only; in particular, the last-exit time T_{lead} is not a stopping time and must not be inserted inside martingale increments. For each challenger $j \neq i^\star$, define

$$J_{t+1,j} := \mathbf{1}\{t+1 \text{ is non-guard},\ \widehat{b}_t = i^\star,\ C_{t+1} = j\}, \qquad \Psi_{n,j} := s \sum_{t<n} J_{t+1,j}. \tag{154}$$

During the K initialization rounds set $J_{t+1,j} := 0$; a leader or challenger nominee need not be defined there. On later rounds, $\widehat{b}_t$, the guard decision, and C_{t+1} are $\mathcal{F}_t$-measurable, so $J_{t+1,j}$ is predictable. When $\pi_{t+1,\star}$ is used on an initialization round, it denotes the actual predictable initialization selection probability. Also put

$$L_n := \sum_{t<n} \mathbf{1}\{\widehat{b}_t \neq i^\star\}, \qquad L_\infty := \sup_n L_n.$$

By Lemma C.13, $L_\infty \leq T_{\mathrm{lead}} + K$ up to the deterministic initialization convention, and hence L_∞ has finite moments of every fixed order.

Lemma C.14 (Predictable-to-actual tracking)**.** *For every $\varepsilon > 0$ there exists a random time $T_{\mathrm{tr}}(\varepsilon)$ with finite moments of every fixed order such that, for all $n \geq T_{\mathrm{tr}}(\varepsilon)$,*

$$\max_{j \neq i^\star} \left| \frac{N_{n,j}}{n} - \frac{\Psi_{n,j}}{n} \right| \leq \varepsilon, \qquad \left| \frac{N_{n,i^\star}}{n} - \beta \right| \leq \varepsilon. \tag{155}$$

Consequently $N_{n,i^\star}/n \to \beta$ almost surely.

Proof. For $j \neq i^\star$ define the genuine martingale $M_{n,j} := \sum_{t<n} J_{t+1,j}\big(\mathbf{1}\{I_{t+1} = j\} - s\big)$. Indeed $J_{t+1,j}$ is $\mathcal{F}_t$-measurable and, on the event $J_{t+1,j} = 1$, the Top-Two randomization pulls the challenger with conditional probability s. All pulls of arm j not represented by this martingale occur during initialization, guard rounds, or rounds with an incorrect empirical leader. Thus for a deterministic constant K_0,

$$|N_{n,j} - \Psi_{n,j}| \leq |M_{n,j}| + K_0 + G_n + L_\infty. \tag{156}$$

The increments of $M_{n,j}$ are bounded by one. Azuma–Hoeffding followed by a union bound over $m \geq n$ gives, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\Big(\exists m \geq n :\ |M_{m,j}| > \frac{\varepsilon m}{4}\Big) \leq C_\varepsilon e^{-c_\varepsilon n}.$$

Together with the deterministic bound $G_n = O(\sqrt{n})$ and the stretched-exponential tail of L_∞, this yields a finite-moment entrance time after which the first inequality in (155) holds simultaneously for all challengers.

For the best arm, let

$$\pi_{t+1,\star} := \mathbb{P}(I_{t+1} = i^\star \mid \mathcal{F}_t), \qquad A_{n,\star} := \sum_{t<n} \pi_{t+1,\star},$$

and define $M_{n,\star} := N_{n,i^\star} - A_{n,\star}$. Then $M_{n,\star}$ is a bounded-increment martingale. On every non-guard round with $\widehat{b}_t = i^\star$, one has $\pi_{t+1,\star} = \beta$; only initialization, guard rounds, and incorrect-leader rounds contribute to the discrepancy from β. Hence

$$|A_{n,\star} - \beta n| \leq K_1 + G_n + L_\infty \tag{157}$$

for a deterministic constant K_1. Applying the same uniform Azuma bound to $M_{n,\star}$ and using (157) proves the second inequality and the almost-sure convergence. □

C.8.3 One profile approximation theorem: why the empirical W_n^U converges to population transportation

The empirical profile statistic and the population transportation are linked by a quantitative bound using only the bounded variation of the two logarithmic score functions after common law-level interiorization. Write $\varphi_r(q,y) := y\log r(q) + (1-y)\log(1-r(q))$. The only population identity needed below is

$$\mathbb{E}_{F_k}\varphi_{S_G}(Q,Y) = \mathbb{E}_{F_k}\varphi_{S_{F_k}}(Q,Y) - D_U(F_k,G). \tag{158}$$

A CDF-to-score transfer fact. Let $\widehat{\nu}$ and ν be finite subprobability measures on $(0,B)$, write $\widehat{F}(x) = \widehat{\nu}((0,x])$ and $F(x) = \nu((0,x])$, and let f be bounded, left-continuous, and of bounded variation. Then

$$\left|\int f\, d(\widehat{\nu}-\nu)\right| \le \big(|f(B-)| + \mathrm{TV}(f)\big)\|\widehat{F}-F\|_\infty. \tag{159}$$

Indeed, with $\Delta = \widehat{\nu}-\nu$ and $D(x) = \Delta((0,x])$, the finite-variation Stieltjes representation gives pathwise

$$\int_{(0,B)} f(q)\,\Delta(dq) = f(B-)D(B-) - \int_{(0,B)} D(x)\,df(x).$$

A right jump contributes $df(\{a\}) = f(a+) - f(a)$, so the inclusive endpoint in $D(a) = \Delta((0,a])$ matches the convention $S_G(q) = G([q,B])$ even when a candidate law has an atom at an observed threshold. Taking absolute values gives (159).

Lemma C.15 (Time-uniform current-status profile comparison)**.** *For every $\eta \in (0,1)$ there is an event of probability at least $1-\eta$ on which, simultaneously for all calendar times $n \ge K$ and ordered pairs $i \ne j$,*

$$\left|W_n^U(i,j) - nV_{ij}^U\left(\frac{N_{n,i}}{n}, \frac{N_{n,j}}{n}\right)\right| \le C\left[1 + \log(en)\sqrt{nH_n(\eta)}\right], \qquad H_n(\eta) = \log\frac{CK(en)^8}{\eta}. \tag{160}$$

No lower bound on the realized local counts or on the true survival probabilities is required.

Proof of Lemma C.15. The proof has three steps: marked-CDF concentration, a bounded-variation transfer to log scores, and a common smoothing argument that removes the singularities at survival probabilities 0 and 1.

Step 1: all-prefix marked-CDF concentration. For one arm and $y \in \{0,1\}$, let

$$\widehat{F}_{m,y}(x) := \frac{1}{m}\sum_{r=1}^{m} \mathbf{1}\{Q_r \le x, Y_r = y\}, \qquad F_y(x) := \mathbb{P}(Q \le x, Y = y),$$

for $m \ge 1$. To apply the ordinary DKW inequality, view the marked observation as the scalar random variable that equals Q when $Y = y$ and $B+1$ otherwise. Its CDF on $[0,B]$ is exactly F_y, and its empirical CDF is exactly $\widehat{F}_{m,y}$. Thus, for every deterministic prefix m, DKW applies without requiring the two marks to be independent. Allocating a summable failure budget proportional to $\eta/[K(m+1)^2]$ over the two marks, all arms, and all deterministic prefixes gives one event of probability at least $1-\eta$ on which, for every calendar horizon n and every $1 \le m \le n$,

$$\max_{y\in\{0,1\}} \|\widehat{F}_{m,y} - F_y\|_\infty \le \sqrt{\frac{H_n(\eta)}{2m}}. \tag{161}$$

This event is built on complete iid local streams before any realized adaptive count $N_{n,k}$ is inserted.

Step 2: transfer the CDF error to every smoothed log score. Let r be a nonincreasing candidate survival profile and, for a common $u \in (0,1)$, set

$$r^{(u)}(q) := (1-u)r(q) + u/2.$$

This is the survival profile on interior thresholds of the realizable law

$$G^{(u)} = (1-u)G + \frac{u}{2}(\delta_0 + \delta_B).$$

Its mean satisfies

$$\mu(G^{(u)}) = (1-u)\mu(G) + \frac{uB}{2}.$$

Hence applying the same u to both arms multiplies their mean difference by $1-u$ and therefore preserves the mean order. The common smoothing thus controls logarithmic singularities without changing the constrained profile problem. Moreover $r^{(u)} \in [u/2, 1-u/2]$, so the two score functions $\log r^{(u)}$ and $\log(1-r^{(u)})$ are bounded, monotone, left-continuous BV functions and each satisfies

$$|f(B-)| + \mathrm{TV}(f) \le 2\log(2/u).$$

The transfer bound (159) and (161) therefore give, uniformly over every such candidate and every deterministic local prefix m,

$$\left| \sum_{r=1}^{m} \varphi_{r^{(u)}}(Q_r, Y_r) - m\mathbb{E}\varphi_{r^{(u)}}(Q, Y) \right| \le C\log(2/u)\sqrt{mH_n(\eta)}. \tag{162}$$

The case $m = 0$ is the empty sum and needs no concentration argument.

Step 3: compare raw empirical and population profile suprema. Choose $u = u_n = (n+1)^{-1}$. For every observation and every raw candidate, common smoothing loses at most $-\log(1-u_n)$ in log score, because $r^{(u_n)} \ge (1-u_n)r$ and $1 - r^{(u_n)} \ge (1-u_n)(1-r)$. Consequently, for either the unrestricted pair class or the mean-ordered pair class, its raw empirical supremum and its common-smoothed empirical supremum differ by at most $n[-\log(1-u_n)] \le 1$; the same is true for the corresponding population expected-score suprema. Since $\log(2/u_n) = O(\log(en))$, applying (162) to the two arms and using $m_i + m_j \le n$ gives a uniform empirical-to-population supremum error

$$C\big[1 + \log(en)\sqrt{nH_n(\eta)}\big].$$

Finally, by (158), the unrestricted population supremum is the true expected score, whereas the mean-ordered population supremum is smaller by exactly

$$nV_{ij}^{U}(m_i/n, m_j/n).$$

Subtracting the two profile suprema yields (160) for every deterministic pair (m_i, m_j). We may now substitute $(m_i, m_j) = (N_{n,i}, N_{n,j})$ on the all-prefix event, without conditioning on these adaptive counts. □

Why the smoothing is proof-only. A fixed restriction $\kappa \le S_G \le 1-\kappa$ would change the admissible current-status experiment. The common mixture used above is instead only a comparison device: it keeps the original raw profile class and preserves the mean order while making the two logarithmic score functions bounded.

Theorem C.16 (Uniform profile approximation with finite-moment entrance)**.** *For any ordered pair (i, j) and fixed $\varepsilon > 0$, there is a random time $T_{ij,\varepsilon}$ with finite moments of every fixed order such that, for all $n \ge T_{ij,\varepsilon}$,*

$$\left| \frac{W_n^U(i,j)}{n} - V_{ij}^U\left(\frac{N_{n,i}}{n}, \frac{N_{n,j}}{n}\right) \right| \le \varepsilon. \tag{163}$$

Proof. For each deterministic n invoke the fixed-horizon event in the preceding proof with $\eta_n = e^{-n^{1/3}}$. Its normalized error is $O(n^{-1/3}\log(en))$, uniformly in all local counts. For n large enough this is below fixed ε. The probability of a violation at some calendar time $m \ge n$ is then bounded by $\sum_{m \ge n} e^{-m^{1/3}}$, which has a stretched-exponential tail up to a polynomial factor. The last violating time therefore has moments of every fixed order. The result requires no passage from a pointwise almost-sure assertion to a moment bound. □

Corollary C.17 (Population-index fidelity)**.** *For every $\varepsilon > 0$ there exists a finite-moment random time $T_{\mathrm{ind}}(\varepsilon)$ such that for all $n \ge T_{\mathrm{ind}}(\varepsilon)$ and all $j \ne i^\star$,*

$$\left| \frac{W_n^U(i^\star, j) + \log N_{n,j}}{n} - h_j\left(\frac{N_{n,j}}{n}\right) \right| \le \varepsilon. \tag{164}$$

Proof. Theorem C.16 gives

$$\frac{W_n^U(i^\star, j)}{n} = V_{i^\star j}^U\left(\frac{N_{n,i^\star}}{n}, \frac{N_{n,j}}{n}\right) + o(1)$$

with finite-moment fixed-accuracy entrance times. Lemma C.14 gives $N_{n,i^\star}/n \to \beta$ with the same type of entrance control. The map $(a, w) \mapsto V_{i^\star j}^U(a, w)$ is the infimum of affine functions of (a, w) and is finite on $(0,\infty)^2$, hence continuous there. It also extends continuously to $w = 0$ uniformly for a in any compact subset of $(0,\infty)$, because $0 \le V_{i^\star j}^U(a, w) \le wK_U^+(F_j, \mu_{i^\star})$ by choosing the scalar separator $x = \mu_{i^\star}$. Since eventually $N_{n,i^\star}/n \in [\beta/2, (1+\beta)/2]$, uniform continuity on the resulting compact strip allows the first argument to be replaced by β, uniformly in $w \in [0, 1]$. Finally $0 \le \log N_{n,j}/n \le \log n/n \to 0$, uniformly in j. □

C.9 Critical-level TCI allocation without strict monotonicity

Recall

$$h_j(w) = V^U_{i^\star j}(\beta, w), \qquad \Gamma^U_\beta = \max_{\substack{w_j \geq 0 \\ \sum_{j\neq i^\star} w_j = s}} \min_{j \neq i^\star} h_j(w_j), \qquad s = 1-\beta.$$

Only continuity and monotonicity of h_j will be used.

For $0 < \gamma < \sup_w h_j(w)$ define the generalized inverse

$$c_j(\gamma) := \inf\{w \geq 0 : h_j(w) \geq \gamma\}. \tag{165}$$

Lemma C.18 (Generalized-inverse geometry)**.** *For every challenger j:*

(i) if $0 < \gamma < \gamma'$ are both below the maximal attainable level, then $c_j(\gamma) < c_j(\gamma')$;

(ii) c_j is left-continuous: if $\gamma_m \uparrow \gamma$, then $c_j(\gamma_m) \uparrow c_j(\gamma)$;

(iii) if $h_j(w) < \gamma$, then $w < c_j(\gamma)$, while if $w \geq c_j(\gamma)$ then $h_j(w) \geq \gamma$.

Proof. Part (iii) is immediate from monotonicity and continuity. For (i), suppose $c_j(\gamma) = c_j(\gamma') =: c$. Since $\gamma > 0$ and $h_j(0) = 0$, we have $c > 0$. By continuity, $h_j(c) \geq \gamma' > \gamma$, hence for all $w < c$ sufficiently close to c, $h_j(w) > \gamma$, contradicting the definition of $c_j(\gamma)$. For (ii), monotonicity gives a limit $\ell \leq c_j(\gamma)$. Because h_j is continuous, each closed upper level set $\{w : h_j(w) \geq \gamma_m\}$ contains its left endpoint, so $h_j(c_j(\gamma_m)) \geq \gamma_m$. If $\ell < c_j(\gamma)$, then $c_j(\gamma_m) \uparrow \ell$ and continuity yields $h_j(\ell) \geq \gamma$, contradicting the definition of $c_j(\gamma)$. □

Lemma C.19 (Strict critical-budget slack)**.** *For every $0 < \gamma < \Gamma^U_\beta$,*

$$\sum_{j \neq i^\star} c_j(\gamma) < s. \tag{166}$$

Proof. Let $w^\star$ be any maximizer in the definition of Γ^U_β. Since $\Gamma^U_\beta > \gamma > 0$, $h_j(w^\star_j) \geq \Gamma^U_\beta > \gamma$ for every j, and necessarily $w^\star_j > 0$. By continuity there exists $u_j < w^\star_j$ with $h_j(u_j) > \gamma$, hence $c_j(\gamma) \leq u_j < w^\star_j$. Summing over j gives (166). □

C.9.1 A deterministic deficit barrier

Lemma C.20 (Critical-level deficit barrier)**.** *For every $0 < \gamma < \Gamma^U_\beta$, there exists a random time $T^{\mathrm{alloc}}_\gamma$ with finite moments of every fixed order such that*

$$\min_{j \neq i^\star} h_j\left(\frac{N_{n,j}}{n}\right) \geq \gamma \qquad \forall n \geq T^{\mathrm{alloc}}_\gamma. \tag{167}$$

Proof of Lemma C.20. Fix $\gamma < \Gamma^U_\beta$. Choose one higher "safe" level $\bar\gamma \in (\gamma, \Gamma^U_\beta)$ and set

$$d_j := c_j(\bar\gamma), \qquad D := \sum_{j \neq i^\star} d_j < s.$$

Thus d_j is a safe challenger fraction and the total safe challenger mass still fits strictly inside the available budget $s = 1-\beta$. By Lemma C.18,

$$\kappa := \min_j\{d_j - c_j(\gamma)\} > 0.$$

Using left-continuity of the c_j, choose $\gamma < \gamma_0 < \gamma_1 < \bar\gamma$ so that

$$\sum_j \big(d_j - c_j(\gamma_0)\big) < \kappa/8, \qquad \rho := \min_j\left[d_j - \sup\{w : h_j(w) \leq \gamma_1\}\right] > 0. \tag{168}$$

Choose a deterministic $\varepsilon > 0$ so small that $K\varepsilon < \kappa/16$, $\varepsilon < \rho/4$, and $3\varepsilon < \gamma_1 - \gamma_0$.

Let T be the maximum of T_{lead} and the finite-moment entrance times in Corollary C.17 and Lemma C.14, enlarged so that for every $n \geq T$,

$$\max_j \left|\frac{N_{n,j} - \Psi_{n,j}}{n}\right| \leq \varepsilon, \qquad \max_j \left|\frac{W^U_n(i^\star, j) + \log N_{n,j}}{n} - h_j(N_{n,j}/n)\right| \leq \varepsilon, \tag{169}$$

and the empirical leader is $i^\star$.

We track only one additional quantity,

$$\mathcal{D}_n := \sum_{j \neq i^\star} [d_j n - \Psi_{n,j}]_+, \tag{170}$$

the total predictable challenger mass still missing relative to the safe targets $d_j n$.

Step 1: below γ_0, TCI serves an active deficit. Suppose a post-T non-guard round satisfies $\min_j h_j(N_{n,j}/n) < \gamma_0$. If j_0 attains this minimum, the TCI choice and (169) imply

$$h_{C_{n+1}}(N_{n,C_{n+1}}/n) \le h_{j_0}(N_{n,j_0}/n) + 2\varepsilon < \gamma_1.$$

By (168), $N_{n,C_{n+1}}/n \le d_{C_{n+1}} - \rho$ and therefore

$$\Psi_{n,C_{n+1}}/n \le d_{C_{n+1}} - \rho + \varepsilon \le d_{C_{n+1}} - 3\rho/4.$$

Hence the selected coordinate has a linear positive deficit and, for all sufficiently large n, remains positive after receiving predictable service s. Since every target $d_j n$ increases by d_j in one calendar step while the nominated challenger receives service s, the aggregate deficit obeys

$$\mathcal{D}_{n+1} - \mathcal{D}_n \le D - s = -a, \qquad a := s - D > 0, \tag{171}$$

on such a non-guard round. On a guard round no Ψ coordinate is served, so simply

$$\mathcal{D}_{n+1} - \mathcal{D}_n \le D < s. \tag{172}$$

Step 2: in the γ_0-good region the normalized deficit is small. If $\min_j h_j(N_{n,j}/n) \ge \gamma_0$, then $N_{n,j}/n \ge c_j(\gamma_0)$ for every j. By (169),

$$\frac{\mathcal{D}_n}{n} \le \sum_j [d_j - c_j(\gamma_0) + \varepsilon]_+ < \frac{3\kappa}{16}.$$

Step 3: a γ-bad time forces a large linear deficit. If $\min_j h_j(N_{n,j}/n) < \gamma$, then for some j, $N_{n,j}/n < c_j(\gamma)$. Again using (169),

$$\frac{\mathcal{D}_n}{n} \ge d_j - \frac{\Psi_{n,j}}{n} \ge d_j - c_j(\gamma) - \varepsilon > \frac{3\kappa}{4}. \tag{173}$$

It remains to show that Steps 1–3 are incompatible at large times. Consider a maximal post-T interval $[u, v]$ on which the minimum information stays below γ_0. Combining (171) and (172), and using $D + a = s$, gives for every $t \in [u, v]$

$$\mathcal{D}_t \le \mathcal{D}_u - a(t-u) + s\{G_t - G_u\}. \tag{174}$$

If the interval starts immediately after a γ_0-good time, then (C.9.1) and the one-step bound give $\mathcal{D}_u \le (3\kappa/16)u + O(1)$. Since $G_t \le C_K(1 + \sqrt{Kt})$, the right-hand side of (174) is at most $(\kappa/4)t$ for every sufficiently large t in the interval. Such a time cannot be γ-bad, because (173) would require $\mathcal{D}_t > (3\kappa/4)t$.

If instead the process is already below γ_0 at time T, then $\mathcal{D}_T \le DT$. If it stayed there for another L rounds, (174) gives

$$0 \le \mathcal{D}_{T+L} \le DT - aL + sC_K\sqrt{K(T+L)} + O(1).$$

For a deterministic C_γ, the right-hand side is negative whenever $L \ge C_\gamma(T+1)$ and T is large, a contradiction. Thus a γ_0-good time is reached by an affine function of T, and after that the preceding excursion argument rules out all sufficiently late γ-bad times. The last γ-bad time is therefore bounded by a deterministic affine function of T plus an instance-dependent constant. Since T has moments of every fixed order, so does $T_\gamma^{\mathrm{alloc}}$. □

Theorem C.21 (Information-level allocation convergence)**.** *Under guarded TCI,*

$$\min_{j \neq i^\star} h_j\left(\frac{N_{n,j}}{n}\right) \longrightarrow \Gamma_\beta^U \tag{175}$$

almost surely. Moreover, for every $\gamma < \Gamma_\beta^U$, the last calendar time at which the left-hand side is below γ has finite moments of every fixed order.

Proof. Lemma C.20 gives

$$\liminf_{n\to\infty} \min_j h_j(N_{n,j}/n) \geq \gamma$$

for every $0 < \gamma < \Gamma^U_\beta$, hence the liminf is at least Γ^U_β.

For the opposite inequality, take any subsequence along which $\min_j h_j(N_{n,j}/n)$ attains its limsup. By compactness of $[0,1]^{K-1}$, extract a further subsequence such that

$$N_{n,j}/n \to w_j \qquad (j \neq i^\star).$$

Lemma C.14 gives $N_{n,i^\star}/n \to \beta$, hence $\sum_{j\neq i^\star} w_j = s$. By continuity of the h_j,

$$\limsup_n \min_j h_j(N_{n,j}/n) = \min_j h_j(w_j) \leq \Gamma^U_\beta$$

by the definition of Γ^U_β. This proves convergence. The finite-moment last-exit statement is exactly Lemma C.20. □

C.10 Evidence rate and asymptotic upper bound

By the certified stopping rule (134),

$$b(n,\delta) = \log(1/\delta) + d_n, \qquad d_n = o(n),$$

where the sublinearity was proved already in (133). Thus only the linear growth rate of the raw profile GLR matters at first order.

Lemma C.22 (Raw profile evidence grows at every rate below Γ^U_β)**.** *For every $0 < \gamma < \Gamma^U_\beta$ there exists a random time T_γ with finite moments of every fixed order such that, for all $n \geq T_\gamma$,*

$$\min_{j\neq i^\star} W^U_n(i^\star, j) \geq n\gamma. \tag{176}$$

Proof. Choose deterministic levels

$$\gamma < \gamma_0 < \gamma_1 < \Gamma^U_\beta, \qquad \rho_0 := \gamma_1 - \gamma_0 > 0.$$

The finite-moment last-exit statement in Theorem C.21, applied at level γ_1, gives a finite-moment time after which

$$\min_{j\neq i^\star} h_j\left(\frac{N_{n,j}}{n}\right) \geq \gamma_1 = \gamma_0 + \rho_0.$$

Apply Corollary C.17 with accuracy $\rho_0/3$, and enlarge the entrance time deterministically so that $\log n/n \leq \rho_0/3$. Then, for every challenger and all later n,

$$\frac{W^U_n(i^\star, j)}{n} \geq h_j\left(\frac{N_{n,j}}{n}\right) - \frac{\rho_0}{3} - \frac{\log N_{n,j}}{n} \geq \gamma_0 + \frac{\rho_0}{3} > \gamma.$$

Taking the minimum over j proves (176); the maximum of the finitely many entrance times still has finite moments of every fixed order. □

Theorem C.23 (Asymptotic upper bound)**.** *For every fixed $\delta \in (0,1)$, the guarded EB–TCI algorithm with direct stopping rule* (135) *stops almost surely, and it satisfies*

$$\limsup_{\delta\downarrow 0} \frac{\mathbb{E}_{\mathbf{F}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{\Gamma^U_\beta(\mathbf{F})}.$$

Proof. Fix numbers $\gamma < \gamma_0 < \Gamma^U_\beta$. By Lemma C.22, there is a finite-moment time $T^W_{\gamma_0}$ such that

$$\min_{j\neq i^\star} W^U_n(i^\star, j) \geq n\gamma_0 \qquad \forall n \geq T^W_{\gamma_0}.$$

Let

$$\widetilde{T}_{\gamma_0} := \max\{T^W_{\gamma_0}, T_{\mathrm{lead}}\}.$$

Then $\widetilde{T}_{\gamma_0}$ has finite expectation, and for every $n \geq \widetilde{T}_{\gamma_0}$ the empirical leader used by the stopping rule is $i^\star$. By (C.10), there exists a deterministic n_0 such that

$$d_n \leq n(\gamma_0 - \gamma) \qquad \forall n \geq n_0.$$

Therefore, whenever $n \geq \max\{\widetilde{T}_{\gamma_0}, n_0\}$ and $n\gamma \geq \log(1/\delta)$,

$$b(n,\delta) = \log(1/\delta) + d_n \leq n\gamma + n(\gamma_0 - \gamma) = n\gamma_0,$$

so the stopping condition is satisfied with the true best arm as leader. Hence

$$\tau_\delta \leq \max\left\{\widetilde{T}_{\gamma_0}, n_0, \left\lceil \frac{\log(1/\delta)}{\gamma} \right\rceil \right\}.$$

Taking expectations, dividing by $\log(1/\delta)$, letting $\delta \downarrow 0$, and finally letting $\gamma \uparrow \Gamma^U_\beta$ proves the claim. □

C.11 Change-of-distribution lower bound

We next prove that no δ-correct realization of the present fixed-β sampling architecture can have a smaller first-order constant.

Lemma C.24 (Strict alternatives with finite observed KL are dense)**.** *Fix a pair (i,j) and weights $a > 0$ and $b \geq 0$. The closed transportation value*

$$V^U_{ij}(a,b) = \inf_{\mu(G_i) \leq \mu(G_j)} \{aD_U(F_i,G_i) + bD_U(F_j,G_j)\}$$

is unchanged if the infimum is approached by pairs with strict order $\mu(G_i) < \mu(G_j)$ and with observed laws dominating the true observed laws, so that the change-of-distribution KL is well-defined and finite whenever the target value is finite.

Proof. The zero-weight case requires a separate finite-KL construction. If $b = 0$, then $V^U_{ij}(a,0) = 0$: leave the first arm unchanged, $G_i = F_i$, and choose $G_j = (1-t)F_j + t\delta_B$ with

$$\max\left\{0, \frac{\mu(F_i) - \mu(F_j)}{B - \mu(F_j)}\right\} < t < 1.$$

This is possible because both true means are interior. It ensures $\mu(G_i) < \mu(G_j)$ and $D_U(F_j,G_j) \leq -\log(1-t) < \infty$, so the zero cost is achieved by an admissible strict, finite-KL pair.

Now suppose $a, b > 0$. Take an ε-optimal feasible pair with finite divergence on both arms. If its means are equal, mix G_i by a weight $\xi > 0$ with δ_0 and G_j by the same weight with δ_B. The means then become strictly ordered. The observed experiment is affine in the arm law: $P^U_{(1-\xi)G+\xi R} = (1-\xi)P^U_G + \xi P^U_R$. Hence, whenever $D_U(F,G) < \infty$, $D_U\big(F,(1-\xi)G+\xi R\big) \leq D_U(F,G) - \log(1-\xi)$, because the alternative mixture measure dominates $(1-\xi)P^U_G$. Thus this strict-order perturbation increases the weighted objective by $o(1)$ as $\xi \downarrow 0$.

Next mix both perturbed laws, with the same weight $\eta > 0$, with the common interior law $H_0 = (\delta_0 + \delta_B)/2$. The common affine transformation of the two means preserves their strict order, while for Lebesgue-a.e. $q \in (0,B)$ the alternative conditional success probabilities belong to $[\eta/2, 1-\eta/2]$. Therefore the true observed laws are absolutely continuous with respect to the resulting alternatives. The same mixture-measure inequality shows that this second step increases each finite observed KL by at most $-\log(1-\eta) = o(1)$. Letting first $\xi, \eta \downarrow 0$ and then the original $\varepsilon \downarrow 0$ proves the claim. □

Lemma C.25 (Expected best-arm fraction at the stopping time)**.** *For the guarded fixed-β U rule,*

$$\mathbb{E}N_{\tau_\delta, i^\star} / \mathbb{E}\tau_\delta \to \beta. \tag{177}$$

Proof. Lemma C.12 gives $G_n = O(\sqrt{n})$; the incorrect-leader prefix from Lemma C.14 has finite mean. The finite-expectation upper bound is established above, while a finite-positive-KL strict alternative and the stopped change-of-measure inequality show $\mathbb{E}\tau_\delta \to \infty$. All hypotheses of the shared stopped-β Lemma A.2 therefore hold, proving (177). □

Proof of Theorem 10. Anytime incorrect-stop control is Theorem C.10; almost-sure stopping and the limsup are Theorem C.23. It remains to identify the fixed-β characteristic in Corollary A.1.

Let w be any arm-allocation vector with $w_{i^\star} = \beta$. For the closed order-reversal set, we claim

$$\inf_{\substack{\mathbf{G}:\,\exists j\neq i^\star \\ \mu(G_{i^\star})\le\mu(G_j)}} \sum_{k=1}^{K} w_k D_U(F_k, G_k) = \min_{j\neq i^\star} V^U_{i^\star j}(\beta, w_j). \tag{178}$$

For the lower bound, every closed alternative contains some $j \neq i^\star$ with $\mu(G_{i^\star}) \le \mu(G_j)$; discarding all nonnegative terms except those of $(i^\star, j)$ gives at least $V^U_{i^\star j}(\beta, w_j)$ and hence at least the right-hand side. For the reverse inequality, choose a challenger j attaining the right-hand side and an ε-optimal feasible pair for $V^U_{i^\star j}(\beta, w_j)$, leaving the other arms unchanged. This already belongs to the closed order-reversal set because $\mu(G_j) \ge \mu(G_{i^\star})$. Letting $\varepsilon \downarrow 0$ proves (178).

The generic lower bound uses strict unique-best alternatives rather than the closed set. The two infima nevertheless agree. First apply Lemma C.24 to make the selected pair strict and to interiorize its observed laws with arbitrarily small KL overhead. If the resulting instance does not yet have a unique best arm, choose an arm $r \neq i^\star$ of maximal alternative mean and replace its alternative law $\widetilde{G}_r$ by

$$G_r^{(\eta)} = (1-\eta)\widetilde{G}_r + \eta\delta_B, \qquad \eta > 0.$$

All means before this final perturbation are strictly below B (the common interiorization in Lemma C.24 ensures this), so for arbitrarily small η one may break any finite tie at the maximum while keeping $i^\star$ non-best. Moreover

$$D_U(F_r, G_r^{(\eta)}) \le D_U(F_r, \widetilde{G}_r) - \log(1-\eta),$$

with the convention that $\widetilde{G}_r = F_r$ for an unchanged arm. Thus the additional weighted KL tends to zero. Because r was chosen at the previous maximal mean, this single positive perturbation makes r strictly larger than every arm that was tied at that maximum. Hence it gives a strict alternative with a unique best arm different from $i^\star$ and arbitrarily small excess cost. Hence the strict and closed inner infima coincide.

Using Lemma C.5, (178) becomes

$$\inf_{\mathbf{G}\in\mathcal{A}(\mathbf{F})} \sum_k w_k D_U(F_k, G_k) = \min_{j\neq i^\star} h_j(w_j).$$

Maximizing over $w_{i^\star} = \beta$ and $\sum_{j\neq i^\star} w_j = 1-\beta$ therefore gives exactly $\Gamma^\star_\beta(\mathbf{F}) = \Gamma^U_\beta(\mathbf{F})$. Lemma C.25 verifies the stopped-fraction hypothesis of Corollary A.1, which yields

$$\liminf_{\delta\downarrow 0} \frac{\mathbb{E}_{\mathbf{F}}[\tau_\delta]}{\log(1/\delta)} \ge \frac{1}{\Gamma^U_\beta(\mathbf{F})}.$$

Combining this with Theorem C.23 proves (26). □

D MODEL U: FINITE-CONFIDENCE RESULTS

Part C already contains the complete certified stopping-rule argument: the finite-sieve calibration, anytime δ-correctness, and the first-order $L_\delta + o(n)$ boundary. This part therefore contains only the sharper finite-confidence stopping-time analysis. It reuses the all-prefix profile, leader, and certified-boundary results from Part C; no second stopping rule or second calibration argument is introduced.

D.1 Subcritical finite-confidence refinement for Model U

The preceding calibrated rule and profile approximation also yield a finite-confidence bound with leading term $T^\star_U L_\delta$ and an explicit sublinear correction. Unlike a global-slope approach, the argument below allows flat pair-information curves and requires no pre-established linear-count entrance. We analyze the *infinite counterfactual continuation of the same exact guarded sampling rule*, ignoring a possible earlier stopping decision. This continuation agrees with the executed experiment up to its stopping time; it is distinct from the arbitrary predictable continuation that suffices for fixed-time null calibration. Thus all guard, all-prefix, and positive-deficit claims below concern the same pre-stop policy. Write $s = 1-\beta$, $\Gamma = \Gamma^U_\beta(\mathbf{F})$, $V_j(a,w) := V^U_{i^\star j}(a,w)$, and $h_j(w) := V_j(\beta, w)$. These are the population rates for the *same* exact raw-profile procedure and certified boundary already analyzed in the Model-U asymptotic part and above.

Lemma D.1 (Subcritical allocation geometry)**.** *Fix any maximizing challenger vector $w^\star$ in* (119)*. For $0 < \varepsilon \le \min\{\Gamma/4, 1\}$ put $\gamma = \Gamma - \varepsilon$ and $c_j = c_j(\gamma)$ as defined in* (165)*. There are fixed finite constants $c_0, U, C_a > 0$,* independent of ε*, for which*

$$c_0 \le c_j < w_j^\star, \quad h_j(c_j) = \gamma, \quad \sum_{j \ne i^\star} c_j \le s, \tag{179}$$

$$h_j(w) \le \gamma - \frac{\varepsilon}{s}(c_j - w) \quad (0 \le w \le c_j), \qquad h_j(w) \ge \gamma - U[c_j - w]_+ \quad (w \ge 0), \tag{180}$$

$$|V_j(a, w) - V_j(\beta, w)| \le C_a|a - \beta| \quad (a, w \in [0, 1]). \tag{181}$$

No strict monotonicity or differentiability of h_j is assumed.

Proof. By Lemma C.6, h_j is continuous, concave, nondecreasing, $h_j(0) = 0$, and positive at any positive mass. The compact allocation simplex has positive max–min value, hence every maximizer has positive coordinates and $h_j(w_j^\star) \ge \Gamma$. Continuity gives $0 < c_j < w_j^\star$ and $h_j(c_j) = \gamma$. Moreover, $\gamma \ge 3\Gamma/4 > \Gamma/2$, so $c_0 := \min_j c_j(\Gamma/2) > 0$ is independent of ε. Summing $c_j \le w_j^\star$ proves the budget in (179). For $w < c_j$, decreasing secant slopes of a concave function yield

$$\frac{\gamma - h_j(w)}{c_j - w} \ge \frac{h_j(w_j^\star) - \gamma}{w_j^\star - c_j} \ge \frac{\varepsilon}{s}.$$

Concavity on $[0, c_j]$ also implies $h_j(w) \ge \gamma w/c_j$; monotonicity gives $h_j(w) \ge \gamma$ above c_j. Hence the global lower chord holds with $U = \Gamma/c_0$. For the last claim, the scalar-separator identity (Lemma C.5) permits restricting its infimum to $x \in [\mu_j, \mu_{i^\star}]$, even at zero weights. On this interval $0 \le K_U^-(F_{i^\star}, x) \le K_U^-(F_{i^\star}, \mu_j) < \infty$: the two-point law $(1 - x/B)\delta_0 + (x/B)\delta_B$ shows finiteness for every interior x. Comparing the same infimum at a and β yields (181), uniformly with $C_a = \max_j K_U^-(F_{i^\star}, \mu_j)$. □

Lemma D.2 (Joint quantitative coin tracking)**.** *Let T_{lead} and G_n be the permanent-leader time and guard count from Lemmas C.13 and C.12; in particular $\mathbb{E}T_{\mathrm{lead}}^4 < \infty$ and $G_n \le C_K(1 + \sqrt{Kn})$ deterministically. Define $\Psi_{n,j}$ by* (154) *and $R_n = s^{-1}\sum_j \Psi_{n,j}$; initialization indicators are zero. For every $\eta \in (0, 1/4)$ there is an event of probability at least $1 - \eta$ on which, for* all *$n \ge K$,*

$$\begin{aligned} \max_{j \ne i^\star} |N_{n,j} - \Psi_{n,j}| + |N_{n,i^\star} - \beta n| &\le C\big[T_{\mathrm{lead}} + G_n + K + \sqrt{nH_n(\eta)}\big], \\ 0 \le n - R_n &\le K + G_n + T_{\mathrm{lead}}, \qquad H_n(\eta) = \log \frac{C_0 K (en)^8}{\eta}. \end{aligned} \tag{182}$$

The random T_{lead} is a last-exit time, not *a stopping time, and is never used inside a martingale conditional expectation.*

Proof. For each challenger put $J_{t+1,j} = \mathbf{1}\{\text{non-guard}, \widehat{b}_t = i^\star, C_{t+1} = j\}$, nominated *before* the independent Top–Two coin. The bounded-increment process $M_{n,j} = \sum_{K \le t < n} J_{t+1,j}(\mathbf{1}\{I_{t+1} = j\} - s)$ is a martingale. Likewise the best-arm coin fluctuations over correct-leader non-guard rounds are a bounded-increment martingale. Azuma followed by a union bound with summable weights $\eta/[CK(n+1)^2]$ gives $\max_j |M_{n,j}| \le C\sqrt{nH_n(\eta)}$ for every n on one event. Every other realized challenger pull is an initialization, guard, or pre-stabilization round, at most $K + G_n + T_{\mathrm{lead}}$ in total. Hence $|N_{n,j} - \Psi_{n,j}| \le |M_{n,j}| + K + G_n + T_{\mathrm{lead}}$. The best-arm predictable count is βR_n on regular correct-leader rounds and differs from βn only by the same exceptional rounds. Finally R_n counts precisely the non-guard correct-leader rounds after initialization, proving the second assertion pathwise. All Azuma events were established at deterministic times, without conditioning on T_{lead}. □

Lemma D.3 (Subcritical deficit envelope)**.** *Intersect the $1 - \eta$ events of Lemmas C.15 and D.2. On the* same two underlying concentration events*, for* every *$0 < \varepsilon \le \min\{\Gamma/4, 1\}$ and simultaneously for all $n \ge T_{\mathrm{lead}} \vee K$,*

$$\min_{j \ne i^\star} W_n^U(i^\star, j) \ge n(\Gamma - \varepsilon) - \frac{C}{\varepsilon}\left[T_{\mathrm{lead}} + K + 1 + \sqrt{n} + \log(en)\sqrt{nH_n(\eta)} + \log(en)\right]. \tag{183}$$

The confidence event does not depend on ε.

Proof of Lemma D.3. Fix ε and write $\gamma = \Gamma - \varepsilon$ and $c_j = c_j(\gamma)$. The proof tracks only one new object: the virtual allocation debt

$$D_{n,j} := c_j R_n - \Psi_{n,j}, \qquad D_n^+ := \sum_{j \ne i^\star} [D_{n,j}]_+, \tag{184}$$

where R_n is the number of regular correct-leader rounds from Lemma D.2. Thus $c_j R_n$ is the challenger service that arm j should have received by those rounds in order to sustain the subcritical level γ, while $\Psi_{n,j}$ is the predictable service it actually received. Since $\sum_j c_j \le s$, $D^+_{T_{\mathrm{lead}}} \le sT_{\mathrm{lead}}$ pathwise. Guard and incorrect-leader rounds leave every $D_{n,j}$ unchanged. On a regular round that nominates challenger a,

$$D_{n+1,j} - D_{n,j} = c_j - s\mathbf{1}\{j = a\}. \tag{185}$$

On the intersection of the two concentration events, enlarge one nondecreasing envelope e_n so that, simultaneously for every challenger,

$$|(nc_j - N_{n,j}) - D_{n,j}| \le e_n, \tag{186}$$

$$\left|W^U_n(i^\star, j) + \log N_{n,j} - nh_j(N_{n,j}/n)\right| \le e_n, \tag{187}$$

with

$$e_n \le C\Big[T_{\mathrm{lead}} + G_n + K + 1 + \log(en)\sqrt{nH_n(\eta)} + \log(en)\Big]. \tag{188}$$

Indeed, (186) follows from $D_{n,j} = c_j R_n - \Psi_{n,j}$ and Lemma D.2; (187) follows from Lemma C.15, the best-mass bound (181), Lemma D.2, and $0 \le \log N_{n,j} \le \log(en)$.

Consider now a post-stabilization regular round. Let j_+ maximize $D_{n,j}$ and let a be the challenger actually nominated by TCI. If $D_{n,j_+} > e_n$, then (186) gives $nc_{j_+} - N_{n,j_+} \ge D_{n,j_+} - e_n > 0$. The upper secant in (180) therefore yields

$$nh_{j_+}(N_{n,j_+}/n) \le n\gamma - \frac{\varepsilon}{s}(D_{n,j_+} - e_n).$$

Because a minimizes the empirical penalized index and both empirical indices are within e_n of their population counterparts,

$$nh_a(N_{n,a}/n) \le nh_{j_+}(N_{n,j_+}/n) + 2e_n.$$

The global lower chord in (180), together with (186), gives the opposite bound

$$nh_a(N_{n,a}/n) \ge n\gamma - U[D_{n,a} + e_n]_+.$$

Combining the last three displays gives

$$U[D_{n,a} + e_n]_+ \ge \frac{\varepsilon}{s}(D_{n,j_+} - e_n) - 2e_n. \tag{189}$$

Choose a fixed C_0 large enough that $C_0/s - 1/s - 2 > U$ and $C_0/s > Us$. If

$$D_{n,j_+} > C_0\varepsilon^{-1}(e_n + 1),$$

then (189) implies $[D_{n,a} + e_n]_+ > e_n + s$, hence $D_{n,a} > s$. The nominated coordinate therefore stays positive after (185) and decreases by $s - c_a$. All other coordinates together can increase their positive parts by at most $\sum_{j\neq a} c_j \le s - c_a$, because $\sum_j c_j \le s$. Thus

$$D^+_{n+1} \le D^+_n \qquad \text{whenever} \qquad \max_j D_{n,j} > C_0\varepsilon^{-1}(e_n + 1). \tag{190}$$

If the displayed threshold is not exceeded, then $D^+_n \le (K-1)C_0\varepsilon^{-1}(e_n+1)$, and one regular update can increase D^+_n by at most s; exceptional rounds leave it unchanged. Since e_n is nondecreasing, a pathwise induction from T_{lead} gives

$$D^+_n \le sT_{\mathrm{lead}} + (K-1)C_0\varepsilon^{-1}(e_n+1) + s. \tag{191}$$

No optional stopping is used here: T_{lead} is only a random pathwise starting index.

Finally, (186) and (191) imply

$$N_{n,j} \ge nc_j - D^+_n - e_n.$$

The lower chord in (180) therefore gives

$$nh_j(N_{n,j}/n) \ge n\gamma - U(D^+_n + e_n).$$

Using (187) and removing the penalty $\log N_{n,j} \le \log(en)$ yields

$$W^U_n(i^\star, j) \ge n\gamma - U(D^+_n + e_n) - e_n - \log(en).$$

Substitute (191), (188), and the deterministic guard bound $G_n = O(\sqrt{n})$. Since $\gamma = \Gamma - \varepsilon$ and $0 < \varepsilon \le 1$, all lower-order terms are absorbed into the right-hand side of (183), uniformly over challengers. □

Proof of Theorem 12. Apply Lemma D.3 at each deterministic horizon n with

$$\varepsilon_n = \min\{\Gamma/4, 1, n^{-1/4}\}.$$

This choice balances the two dominant losses in Lemma D.3: the information loss $n\varepsilon$ and the tracking/profile term of order $\sqrt{n}/\varepsilon$, up to logarithmic factors. Both are therefore of order $n^{3/4}$ when $\varepsilon \asymp n^{-1/4}$. The lemma holds for all admissible ε on the same concentration event, so this deterministic choice requires no extra union bound. Since $n\varepsilon_n \le n^{3/4}$ and $\varepsilon_n^{-1} \le C_\Gamma n^{1/4}$, simultaneously for all $n \ge T_{\mathrm{lead}} \vee K$,

$$\min_j W_n^U(i^\star, j) \ge n\Gamma - C\left[n^{3/4}\{1 + \log(en)\sqrt{H_n(\eta)}\} + n^{1/4}(T_{\mathrm{lead}} + 1)\right].$$

Because $\mathbb{E}T_{\mathrm{lead}}^4 < \infty$, Markov's inequality gives a deterministic

$$t_\eta = C_T \eta^{-1/4} \ge K \qquad \text{with} \qquad \mathbb{P}(T_{\mathrm{lead}} > t_\eta) \le \eta.$$

For $n \ge t_\eta$, enlarge C_T if necessary so that $H_n(\eta) \le C\log(en)$. On the joint event formed by the profile, tracking, and leader bounds, whose probability is at least $1 - 3\eta$, Young's inequality

$$t_\eta n^{1/4} \le n^{3/4}/3 + 2t_\eta^{3/2}/3$$

and the certified boundary estimate

$$c_n + \log\frac{(K-1)\pi^2 n^2}{6} \le C[1 + n^{2/3}\log(en)]$$

yield, for all $n \ge t_\eta$,

$$\min_j W_n^U(i^\star, j) - b(n, e^{-L}) \ge n\Gamma - L - C\left[R(n) + t_\eta^{3/2} + 1\right]. \tag{192}$$

Set

$$x := 1 + L/\Gamma + t_\eta^{3/2}$$

and choose

$$n_* := \left\lceil L/\Gamma + A\{t_\eta^{3/2} + R(x) + 1\}\right\rceil.$$

Since $t_\eta \ge 1$, $n_* \ge t_\eta$. Also $R(z)/z$ is bounded on $[1, \infty)$, hence $n_* \le C_A x$, and therefore $R(n_*) \le C_A^{3/4}(1 + \log C_A)^{3/2} R(x)$. For one fixed sufficiently large A, the positive term $\Gamma A\{t_\eta^{3/2} + R(x) + 1\}$ dominates the remainder in (192), uniformly over L and η, because

$$\frac{A}{A^{3/4}(1 + \log A)^{3/2}} \longrightarrow \infty.$$

Thus the true-leader stopping condition holds by time n_*; if the rule stopped earlier, the same upper bound is automatic. Since $t_\eta^{3/2} = C_T^{3/2}\eta^{-3/8}$ and fixed multiplicative constants can be absorbed into A and into the argument of R, this gives the quantitative high-probability bound announced in Theorem 12.

For the mean bound, integrate the upper-tail quantile. For upper quantiles use the preceding probability bound with $\eta = (1-u)/3$; for the remaining fixed lower portion of the quantile function use one fixed admissible η_0. Near $\eta = 0$,

$$\eta^{-3/8} \quad \text{and} \quad R(1 + \eta^{-3/8}) = O\Big(\eta^{-9/32}[1 + \log(1/\eta)]^{3/2}\Big)$$

are integrable because $9/32 < 1$. Using $R(a+b) \le C[R(a) + R(b)]$ and integrating gives

$$\mathbb{E}\tau_\delta^{\mathrm{U}} \le \frac{L}{\Gamma} + C\left[1 + R\left(1 + \frac{L}{\Gamma}\right)\right] \le \frac{L}{\Gamma} + C'\big[1 + L^{3/4}\log^{3/2}(e + L)\big].$$

The δ-error guarantee used here is the certified stopping theorem proved in Part C for this same raw-profile rule. □

E NUMERICAL AND EXPERIMENTAL DETAILS

E.1 Population characteristic values

For the bounded benchmarks,

$$F_k = \mathrm{Beta}(12\mu_k, 12(1-\mu_k)), \qquad B = 1,$$

is used only to generate rewards. Model U's one-bit population constants are computed from the raw-current-status divergence, while the full-feedback constants use the bounded-support nonparametric $K_{\inf}^{\pm}$ duals. For $\beta = 1/2$, the representative HARD5 and TIE5 constants are

Instance	Γ_β^U	T_U	$\Gamma_\beta^{\mathrm{full}}$	T_{full}	T_U/T_{full}
HARD5 $(.7, .6, .5, .4, .3)$	.01344268	74.3899	.04620421	21.6431	3.437
TIE5 $(.7, .6, .6, .4, .3)$	.01002120	99.7884	.03716686	26.9057	3.709

Since $T = 1/\Gamma$, the final column is also the first-order one-bit/full sample-complexity ratio.

For the Gaussian representative CLUSTER5 job at $\beta = .5$, the numerical population ratio used in Figure 1 is

$$\frac{T^\star_{\mathrm{1bit}}}{T^\star_{\mathrm{full}}} \approx 1.571976.$$

These numerical constants are attached to the corresponding population oracle calculations; constants from one instance or numerical query grid must not be reused for a different simulation configuration.

Model-U refinement check. For HARD5, refining the current-status quadrature from 501 to 2001 points leaves the displayed Γ_β^U unchanged to the shown precision and yields challenger masses approximately

$$(.351974, .079156, .041787, .027083)$$

beside the fixed best-arm mass $\beta = .5$. This check concerns the population characteristic value, not the empirical stopping rule.

E.2 Broad paired benchmark

Instances. The Gaussian instances are

$$\mathrm{SEP3} = (.70, .60, .45), \quad \mathrm{CLUSTER5} = (.70, .64, .63, .62, .61), \quad \mathrm{TIE4} = (.70, .64, .64, .60),$$

with $\sigma = .2$. The bounded instances are

$$\mathrm{EASY3} = (.75, .45, .25), \quad \mathrm{HARD5} = (.70, .60, .50, .40, .30), \quad \mathrm{TIE5} = (.70, .60, .60, .40, .30),$$

with Beta concentration 12 for data generation. Each instance is run at

$$\beta \in \{.3, .5, .7\}, \qquad \delta \in \{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}\},$$

with 200 replication IDs per instance–β job. The five confidence levels within one method trajectory are dependent and are not counted as five independent simulations.

Model E records `full_asym`, `onebit_asym`, and separate exploratory finite-confidence companion traces. Model U records nonparametric `full` and raw-current-status `onebit`; the asymptotic and practical-calibration analyses reuse these same Model-U trajectories rather than rerunning them.

Issued seed and cap schedule.

E instance	β	seed	cap	U instance	β	seed	cap
SEP3	.3	710101	60000	EASY3	.3	720101	40000
SEP3	.5	710102	60000	EASY3	.5	720102	40000
SEP3	.7	710103	60000	EASY3	.7	720103	40000
CLUSTER5	.3	710104	160000	HARD5	.3	720104	80000
CLUSTER5	.5	710105	160000	HARD5	.5	720105	80000
CLUSTER5	.7	710106	160000	HARD5	.7	720106	80000
TIE4	.3	710107	140000	TIE5	.3	720107	120000
TIE4	.5	710108	140000	TIE5	.5	720108	120000
TIE4	.7	710109	140000	TIE5	.7	720109	120000

The two representative jobs used in Figure 1, namely E–CLUSTER5–$\beta = .5$ and U–HARD5–$\beta = .5$, have been re-audited directly from the raw `trials.csv` files. The remaining archived grid-level calibration summaries should be accompanied by the corresponding raw job folders in the final reproducibility package.

Practical boundary. The legacy practical benchmark checks

$$b_{\mathrm{prac}}(n, \delta) = \log\left(\frac{(K-1)\pi^2 \max(n,1)^2}{6\delta}\right)$$

at scheduled decision times. For method m with population rate Γ_m^{pop},

$$n_{\mathrm{pred},m}(\delta) = \min\{n \in \mathcal{T}_m : n\Gamma_m^{\mathrm{pop}} \geq b_{\mathrm{prac}}(n, \delta)\}.$$

This predictor is an implementation-scale diagnostic, not a certified theorem bound.

Calibration summary. For

$$\mathrm{ARE} = \left|\frac{\overline{\tau}_\delta}{n_{\mathrm{pred}}(\delta)} - 1\right|,$$

the archived 45-setting summaries are:

Method	Mean ARE	Median ARE	Max ARE
E full Top–Two	4.24%	3.52%	12.41%
E one-bit Top–Two	2.95%	2.56%	8.63%
E full companion (exploratory)	8.66%	10.06%	16.43%
E one-bit companion	12.19%	13.69%	21.57%
U nonparametric full	3.63%	3.06%	11.20%
U raw current-status	2.83%	2.61%	6.91%

The full-companion row is an exploratory numerical analogue only; no separate full-feedback second-order theorem is inferred from it.

E.3 Uncertainty summaries and pairing

For a scalar statistic Z over R independent replication units, we use the empirical standard error

$$\mathrm{se}(\overline{Z}) = \left[\frac{1}{R(R-1)}\sum_{r=1}^{R}(Z_r - \overline{Z})^2\right]^{1/2}.$$

Bootstrap percentile 95% intervals in the legacy paired benchmark use replication IDs as the resampling unit.

For Figure 1, if $(\tau_{\mathrm{full},r}, \tau_{1\mathrm{bit},r})$ denotes one explicitly coupled replication, every bootstrap resample recomputes

$$\frac{\overline{\tau}^*_{1\mathrm{bit}}/\overline{\tau}^*_{\mathrm{full}}}{T^\star_{1\mathrm{bit}}/T^\star_{\mathrm{full}}}.$$

Thus the statistic is a *ratio of resampled means*; it is not the average of the run-wise ratios $\tau_{1\mathrm{bit},r}/\tau_{\mathrm{full},r}$.

The new Q1/Q3 files are analyzed according to their recorded simulation units. Equal integer replication labels alone are not treated as evidence of a paired common-random-number construction.

E.4 Direct first-order diagnostic (Q1)

The deep-confidence Q1 runs use the clustered five-arm instance

$$\mu = (.70, .64, .63, .62, .61), \qquad \beta = .5,$$

and

$$\delta \in \{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-10}\},$$

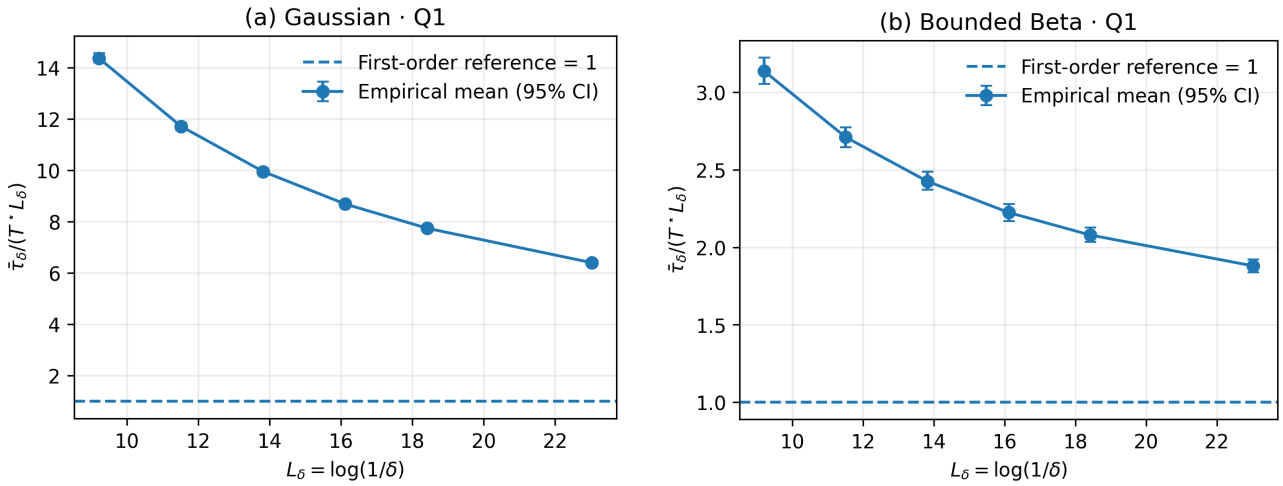


Figure S1: **Supplementary direct first-order diagnostic (Q1).** Each panel reports $\overline{\tau}_\delta/(T^\star \log(1/\delta))$ for the corresponding Q1 configuration. Left: Gaussian Model E. Right: raw-current-status Model U. The horizontal reference at one is asymptotic and is not a finite-δ upper bound. Confidence intervals summarize Monte-Carlo uncertainty in the empirical mean.

with 200 replications per confidence level. All supplied Q1 rows stop before their caps and contain no observed misidentification.

The direct normalization is

$$R_{\mathrm{FO}}(\delta) = \frac{\overline{\tau}_\delta}{T^\star \log(1/\delta)} = \frac{\overline{\tau}_\delta \Gamma^\star}{\log(1/\delta)}.$$

The population $\Gamma^\star$ used here must be read from the oracle record of the *same Q1 run*; values from the older representative benchmark must not be substituted.

Important separation from Q3. The E-Q1 and E-Q3 files have different stopping-time distributions and are not pooled. For Model U, the raw-current-status Q3 outcomes coincide with the Q1 raw trajectories at the four common confidence levels; they are therefore the same observations reused for a different comparison, not an independent replication set.

E.5 Deep-confidence full-versus-one-bit comparison (Q3)

Q3 uses

$$\delta \in \{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$$

and 200 replications per method and confidence level. All supplied trajectories stop and contain zero observed misidentifications.

The raw Q3 means are:

Model	δ	Full mean	One-bit mean
E	10^{-4}	4271.585	6887.180
E	10^{-6}	5044.205	8099.755
E	10^{-8}	5754.300	9325.650
E	10^{-10}	6475.910	10443.215
U	10^{-4}	1806.425	8512.050
U	10^{-6}	2147.375	9875.075
U	10^{-8}	2486.250	11283.175
U	10^{-10}	2819.925	12749.800

These are finite-confidence empirical means and are not substituted for the population characteristic-time ratios.

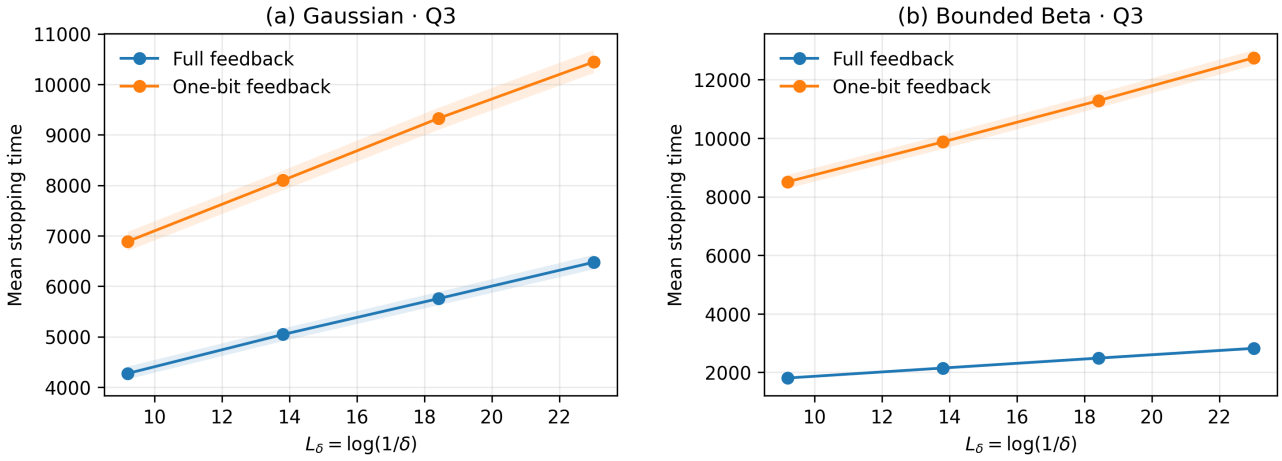


Figure S2: **Supplementary deep-confidence full-versus-one-bit comparison (Q3).** Curves show empirical mean stopping times with bootstrap uncertainty. Left: Gaussian full feedback versus one-bit Top–Two. Right: bounded nonparametric full feedback versus raw current-status one-bit. The methods share the same confidence grid, but no paired-bootstrap claim is made unless the saved simulation record explicitly establishes a common-random-number coupling.

E.6 Monte-Carlo correctness checks

The coarser correctness diagnostics use

$$\delta \in \{.10, .05, .02, .01\}.$$

The supplied Model-E Top–Two file contains 600 replications per confidence level. The supplied Model-U file contains 600 raw-current-status and 600 marginal-bit replications per confidence level. Every supplied row stops and no wrong recommendation is observed.

Method	Levels	Reps/level	Stop fraction	Observed errors
E Top–Two	4	600	1.000	0
U raw current-status	4	600	1.000	0
U marginal-bit diagnostic	4	600	1.000	0

These zero counts are implementation diagnostics only. With hundreds of Monte-Carlo replications they do not empirically certify rare error probabilities at the deep confidence levels used elsewhere in the paper.

E.7 Model-E finite-confidence companion

The separate Model-E identification-schedule companion is evaluated at

$$\delta \in \{10^{-2}, 10^{-3}, 10^{-4}\}$$

with 400 replications per method and level. Both the companion and the corresponding Top–Two trajectories stop in all supplied rows with zero observed misidentifications.

The corresponding empirical means are:

δ	E Top–Two	E companion
10^{-2}	31415.5525	35065.1425
10^{-3}	32025.4275	35751.0350
10^{-4}	32632.2500	36402.9100

E.8 Certified Model-U computational stress diagnostic

To examine the computational cost of the proof-level stopping calibration, we additionally run the theorem-certified Model-U boundary on the separated five-arm Beta instance

$$\mu = (.90, .50, .30, .20, .10), \qquad \beta = .5,$$

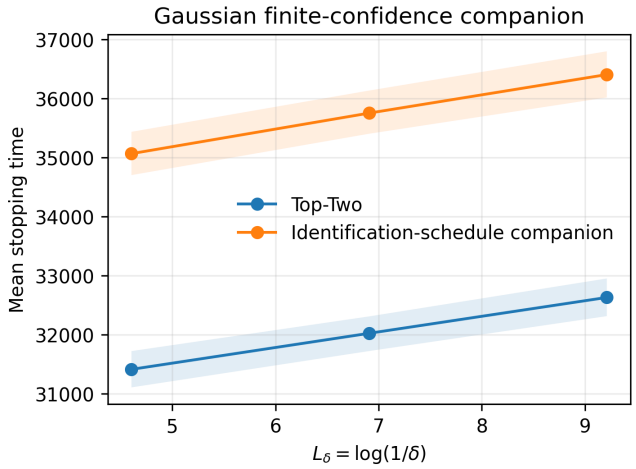


Figure S3: **Supplementary Model-E finite-confidence companion.** Empirical mean stopping times for the identification-schedule companion and the corresponding Top–Two run over the three finite-confidence levels. This diagnostic concerns the separate companion construction; it is not a direct test of the first-order Top–Two limit.

at

$$\delta \in \{.10, .05, .02, .01\}.$$

The certified sieve penalty is retained, raw-profile and stopping checks are performed every 50 rounds, and the cap is 2×10^5. The final run contains 200 replications.

δ	Reps	Stops before cap	Stop fraction
.10	200	0	0
.05	200	0	0
.02	200	0	0
.01	200	0	0

The sentinel recommendation -1 in these censored rows means "no recommendation was issued before the cap" and is not an arm index. Accordingly, `error=0` on a censored row is not interpreted as a correct recommendation. Because every trajectory is censored, this experiment does not yield a meaningful mean stopping-time curve. Its role is instead to document the large finite-sample conservatism and computational cost of the certified sieve calibration. These trajectories are never pooled with the practical raw-current-status experiment.

E.9 Implementation and reproducibility notes

Model-E numerical oracle. The theoretical design optimizes over probability measures on a compact threshold set. The simulation uses a finite query grid and separator grid and periodically refreshes the numerical oracle. Such refresh periods are computational implementation choices; the theorem is stated for the exact predictable design rule.

Model-U raw-profile solver. The reported practical Model-U method retains the distinct observed thresholds rather than replacing the observation model by a fixed binning. Profile likelihoods are computed by the monotone raw-profile solver; the bounded full-feedback comparator uses the nonparametric $K_{\inf}^{\pm}$ duals. Practical GLRs are checked periodically.

Certified versus practical boundary. The practical current-status plots use $b_{\mathrm{prac}}(n, \delta)$. The proof-certified rule includes the substantially larger direct-profile sieve penalty. The two rules therefore answer different empirical questions and their trajectories must not be combined.

Data accounting. Replication IDs, not individual confidence-level rows, are the independent Monte-Carlo units whenever several stopping levels are evaluated on one trajectory. The two old representative paired jobs have been re-audited from raw files. The new Q1, Q3, correctness, and Model-E companion files have likewise been read at the raw-row level. Final released artifacts should retain the corresponding configurations and population-oracle files beside every `trials.csv`.

Compute. The new large Model-U raw-profile experiments are CPU-only. The Linux server used for the latest runs has 44 physical CPU cores (88 logical CPUs) and approximately 121 GiB of installed RAM; worker counts are limited according to available memory. The final certified-U stress run used 16 worker processes. No GPU is required.

Figure generation. All final plots should be regenerated from raw CSV files with one plotting script, shared typography, shared panel dimensions, and the uncertainty rules stated above. Reformatting old and new datasets into a common visual style is harmless; pooling statistically different configurations is not.